

COPENHAGEN BUSINESS SCHOOL
MSc Thesis in Business Administration and Mathematical Business Economics
Department of Finance

Neural Additive and Factor-Augmented Extensions of MIDAS Regression for High-Dimensional Corporate Survival Forecasting

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Abstract

This thesis develops and evaluates a family of neural additive extensions to the MIDAS regression of Miao et al. (2025) for high-dimensional corporate survival forecasting, addressing structural restrictions through architectural non-linearities: a learned per-covariate aggregation replaces the fixed polynomial dictionary with a non-linear lag-to-feature mapping; learned shape functions replace each covariate’s linear contribution to prediction; and a factor channel supplements the additive predictor with cross-covariate interactions through a low-dimensional latent representation.

Complementary ablation designs isolate the contribution of each, with all variants sharing the IPCW-weighted logistic objective of the baseline. The full protocol is applied to two panels of very different size: a panel of Chinese manufacturing firms with a low Special Treatment event rate, and a much larger US panel with a substantially higher event rate, each evaluated across three forecast horizons.

On the Chinese panel, none of the neural variants produce a statistically meaningful AUC gain over the baseline at any horizon, and the per-covariate variants pay an iteration-level variance premium. An ablation that isolates the factor block on the linear baseline identifies its role as a regulariser of the per-covariate shape functions rather than as a carrier of new non-linear signal, and after Procrustes alignment removes the rotational ambiguity of the parameterisation, the population-aligned per-covariate shape functions are statistically indistinguishable from horizontal lines.

On the larger US panel the picture changes: the factor-augmented variant outperforms the linear baseline at every horizon. At the shortest horizon the gain is carried by the factor channel rather than by the per-covariate shapes, which remain flat; at longer horizons a sparse subset of per-covariate shape functions begins to capture genuine structure. The two panels together indicate that the absence of a gain on the Chinese panel reflects a finite-sample limitation rather than a structural ceiling of the additive neural family, and that the architecture’s flexibility becomes productive once the sample is large enough to support it.

Resumé

Dette speciale udvikler og evaluerer en familie af neurale additive udvidelser til MIDAS-regressionen fra Miao et al. (2025) til højdimensionel prognose af virksomheders overlevelse, og adresserer strukturelle restriktioner gennem arkitektur-niveau ikke-lineariteter: en lært per-kovariat aggregering erstatter den fastlåste polynomielle basis med en ikke-lineær mapping fra rå lag til aggregerede features; lærte shape-funktioner erstatter hver kovariats lineære bidrag til prædiktions; og en faktor-kanal supplerer den additive predictor med cross-kovariat interaktioner gennem en lavdimensional latent repræsentation.

Komplementære ablations-designs isolerer bidraget fra hver, mens alle varianter deler det IPCW-vægtede logistiske objektiv fra baselinen. Hele protokollen anvendes på to paneler af meget forskellig størrelse: et panel af kinesiske produktionsvirksomheder med en lav Special Treatment-event-rate, og et væsentligt større amerikansk panel med en markant højere event-rate, hver evalueret på tre prognose-horisonter.

På det kinesiske panel producerer ingen af de neurale varianter en statistisk meningsfuld AUC-gevinst over baselinen ved nogen horisont, og per-kovariat-varianterne har en højere iterations-niveau-varians. En ablation der isolerer faktor-blokken på den lineære baseline identificerer dens rolle som en regulariser af de per-kovariat shape-funktioner snarere end som bærer af nyt ikke-lineært signal, og efter Procrustes-justering, der fjerner parametriseringens rotationelle tvetydighed, er de populations-justerede per-kovariat shape-funktioner statistisk uskelnelige fra vandrette linjer.

På det større amerikanske panel ændrer billedet sig: den faktor-augmenterede variant overgår den lineære baseline ved alle horisonter. Ved den korteste horisont bæres gevinsten af faktor-kanalen snarere end af per-kovariat shape-funktionerne, der forbliver flade; ved længere horisonter begynder en sparsom delmængde af per-kovariat shape-funktioner at indfange ægte struktur. De to paneler indikerer tilsammen at fraværet af en gevinst på det kinesiske panel afspejler en begrænsning i stikprøvestørrelse snarere end et strukturelt loft for den additive neurale familie, og at arkitekturens fleksibilitet bliver produktiv når stikprøven er stor nok til at understøtte den.

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Chapter 1

Introduction

1.1 Motivation

Predicting corporate financial distress is a long-standing problem in financial econometrics. Its quantitative tradition begins with the discriminant- and ratio-based default-prediction models of [Altman \(1968\)](#), is reformulated as a logistic classification of accounting variables by [Ohlson \(1980\)](#), and is extended into a hazard-based survival framework by [Shumway \(2001\)](#). The practical relevance has only grown: the post-crisis Basel III and IV regulatory frameworks place firm-level probability-of-default modelling at the centre of internal-ratings-based capital requirements, banking supervision uses forward-looking default probabilities in stress-test exercises, and credit investors price corporate exposures using the same probability-of-default inputs. Under the internal-ratings-based approach, banks compute their own estimates of probability of default (PD), loss given default, and exposure at default, which feed the regulatory formula that determines risk-weighted assets and minimum capital. PD is the probability that the counterparty defaults within a fixed horizon, the firm-level survival-analytic quantity that all distress-prediction methods in this literature target. More precisely: a binary distress event must be predicted from each firm's covariate history at a fixed forecast horizon, with the practical complication that most firms do not experience the event during the observation window.

Three structural features of corporate-default panels compound the difficulty of prediction. First, each firm is observed as a multi-year panel of quarterly financial ratios, so each firm's predictor is itself a matrix of lagged values in which several ratios are each observed at multiple quarterly dates within the look-back window; the dimensionality of the lag-feature predictor easily exceeds the count of resolved-event firms. Second, distress events are rare in the calendar window, leaving the resolved-event count a small fraction of the firm count even on multi-year horizons. Third, at long prediction horizons a handful of firms with long follow-up necessarily carry most of the inferential weight, shrinking the effective sample size

below the nominal event count. Standard logistic regression on the raw lag-feature matrix is infeasible at this dimension, and naive aggregation that collapses each ratio to its mean or last observation discards precisely the lag-profile information that distinguishes a deteriorating firm from a stable one.

Mixed Data Sampling (MIDAS) regression, formalised by Ghysels et al. (2007), addresses the high-dimensional lag structure by projecting each covariate’s lagged values onto a low-dimensional dictionary of basis functions, replacing the raw lags with a small number of dictionary-weighted features and rendering standard linear estimation feasible. Among recent applications of MIDAS to corporate distress forecasting (Audrino et al., 2019; Jiang et al., 2021), the sg-LASSO-MIDAS framework of Miao et al. (2025) is the most general, combining MIDAS aggregation with inverse-probability-of-censoring weighting for the survival objective and sparse-group LASSO regularisation for variable selection at both the covariate level and the within-covariate dictionary level. sg-LASSO-MIDAS is the empirical benchmark against which this thesis evaluates its neural extensions, and is developed in Chapter 4. Two structural restrictions remain in the framework: each per-covariate contribution is linear in the aggregated features, and the predictor is additive across covariates with no cross-product terms (Section 4.4). This thesis asks whether those restrictions hold on a panel where financial ratios may interact across covariates and respond non-linearly to threshold or accelerating effects.

The recent machine-learning literature has produced architectures that directly relax these restrictions while preserving the additive structure of the linear baseline. Neural Additive Models (Agarwal et al., 2021) replace each linear per-covariate contribution with a learned shape function while retaining the sum-of-functions form, inheriting the dimension-reduction property of additive structure, and admitting per-covariate visualisation of the marginal effect. Factor-Augmented Neural Networks (Fan and Gu, 2024) introduce a low-dimensional factor channel on top of a per-covariate predictor, capturing controlled cross-covariate interactions through a small set of latent factors without abandoning the additive scaffolding. Neither building block has, to our knowledge, been combined with the MIDAS-IPCW protocol for corporate distress survival forecasting. This thesis fills that gap with a coordinated family of neural additive extensions of sg-LASSO-MIDAS.

1.2 Research Question

The thesis addresses the following primary research question: under the IPCW-weighted MIDAS protocol of Miao et al. (2025), can neural additive extensions of the sg-LASSO-MIDAS baseline that relax the linearity restriction, the additivity restriction, or both, improve out-of-sample default prediction or yield interpretive output that the linear baseline cannot provide

on a corporate panel with high censoring and low event frequency? The primary question decomposes into four operational sub-questions, each mapped to a specific architectural variant or variant family developed in Chapter 5 and evaluated in Chapter 6:

- (a) **NAM-MIDAS** Does per-covariate non-linearity alone improve performance over the linear baseline?
- (b) **FA-NAM-MIDAS** Does factor-augmented cross-covariate interaction on top of the additive predictor provide additional gain?
- (c) **Deep-MIDAS** Does dropping additivity altogether in favour of a joint feedforward predictor recover what the additive constraint costs?
- (d) **FA-MIDAS** Does the factor block alone on top of the fixed linear baseline inject non-linear signal that the baseline is missing?

Each sub-question is answerable on the empirical panel through a specific comparison between estimators reported in Table 6.2.

Beyond the four prediction questions, the thesis asks two complementary diagnostic questions: what non-linear signal do the learned shape functions encode at the population level once the rotational ambiguity of the per-covariate aggregations has been resolved, and whether the Residual variant's learned mixing weights γ_k expose any covariate-specific inadequacy in the Gegenbauer dictionary? Finally, the thesis asks whether the answers obtained on this panel persist when the full protocol is applied to a substantially larger panel from a different market.

1.3 Outline of the Thesis

The remainder of the thesis is organised as follows. Chapter 2 positions the contribution within the prior literature. Chapter 3 develops the survival-analytic setup, the IPCW correction for right censoring, the MIDAS lag transformation, and the time-dependent AUC metric. Chapter 4 develops the linear sg-LASSO-MIDAS baseline of Miao et al. (2025), and Chapter 5 introduces the family of neural additive extensions with their regularisation, training, and identifiability analysis. Chapter 6 reports the empirical application and the accompanying diagnostic analyses. Chapter 7 draws together the findings and outlines future research. Three appendices report additional empirical results, the covariate definitions, and the singular value decomposition underlying the Procrustes alignment.

1.4 Declaration of GenAI Use

Generative AI was used as a supplementary aid for language assistance, proofreading, search engine, and code debugging, in accordance with Copenhagen Business School's guidelines.

Chapter 2

Literature Review

This chapter positions the thesis’s contribution within three literatures: corporate distress prediction, MIDAS-based mixed-frequency regression, and neural additive and factor-augmented architectures.

2.1 Corporate Distress Prediction

The transition catalogued in Section 1.1 from [Altman \(1968\)](#)’s discriminant Z-score, through [Ohlson \(1980\)](#)’s logistic generalisation, to [Shumway \(2001\)](#)’s hazard-based survival framework reflects a sequence of methodological responses to specific limitations of each predecessor. Discriminant analysis presupposes multivariate normality of the covariates and equal covariance matrices across distressed and non-distressed firms; the logistic specification relaxes both assumptions by modelling the conditional default probability directly through a parametric link, at the cost of discarding any explicit time-to-event structure. The hazard reformulation in turn models the firm-period panel structure that the cross-sectional logistic treats as independent observations, recasting default as a survival process with a time-varying risk and a censoring mechanism. The IPCW-corrected logistic specification of [Miao et al. \(2025\)](#) that this thesis inherits sits in the survival branch, retaining the binary-classification objective at a fixed forecast horizon while handling censoring through the inverse-probability weighting developed in Section 3.5.

Beyond model form, the modern corporate-default literature is shaped by a second methodological pressure: predictor dimensionality. The natural response in the recent literature has been to regularise the linear estimator through structured penalties that exploit the covariate-by-lag organisation of the predictor. Section 2.2 reviews this strategy and the MIDAS aggregation it combines with; Section 2.3 reviews the neural additive and factor-augmented architectures the thesis builds on.

2.2 Mixed-Frequency Aggregation in High-Dimensional Settings

Mixed Data Sampling (MIDAS) regression of Ghysels et al. (2007) is the standard econometric tool for aggregating high-dimensional lag structures into a low-dimensional set of dictionary-weighted features; the formal setup, including the Gegenbauer polynomial dictionary used by the baseline of Chapter 4, is developed in Section 3.6. MIDAS has been widely applied to mixed-frequency macroeconomic forecasting (Clements and Galvão, 2008; Marcellino and Schumacher, 2010), where a target observed at low frequency is predicted from regressors observed at higher frequency. The framework is general for any setting where many lagged values of a few covariates compete for signal under a small sample.

Sparse-group LASSO regularisation, introduced by Simon et al. (2013), combines an ℓ_1 penalty at the within-group level with a group ℓ_2 penalty across groups, and is the natural regulariser when the predictors come with a known group structure. The Residual NAM-MIDAS variant developed in Chapter 5 was inspired by this convex-combination structure. The combination of MIDAS aggregation with sparse-group LASSO penalisation has been developed for time-series nowcasting (Babii et al., 2022), and Miao et al. (2025) extends it to corporate distress forecasting by adding the IPCW correction for right censoring. The specific configuration used as baseline in this thesis is developed in Section 4.3.

2.3 Neural Additive and Factor-Augmented Architectures

Generalised additive models, introduced by Hastie and Tibshirani (1990), write the predictor as a sum of smooth per-covariate functions, each separately inspectable, while allowing every contribution to be non-linear. Neural Additive Models (Agarwal et al., 2021) replace each smooth f_k with a small feedforward neural network, generalising the GAM functional class without abandoning the additive scaffold. NAM is the building block this thesis uses to relax the linearity restriction of sg-LASSO-MIDAS, and the per-covariate predictor architecture is developed in Section 5.5.

Factor-Augmented Sparse Throughput Deep ReLU Neural Networks of Fan and Gu (2024) target a complementary problem: introducing non-additive cross-covariate structure into a high-dimensional neural regression. The architecture decomposes the predictor into a non-additive factor channel built on a fixed PCA projection and an additive per-covariate channel. This thesis combines this architecture with the NAM predictor on the MIDAS-aggregated covariate space (Section 5.6) to relax both linearity and additivity in sg-LASSO-MIDAS.

Chapter 3

Methodological Foundations

This chapter establishes the common mathematical framework used throughout the thesis: the survival-analytic setup, the IPCW correction for right-censoring, the MIDAS lag aggregation, and the evaluation metrics for censored outcomes. These elements provide the shared notation and statistical foundation on which the baseline of Chapter 4 and the neural extensions of Chapter 5 both rely.

3.1 Basic Setup and Notation

Let $T \geq 0$ denote the *survival time* of a firm, defined in the corporate context studied in this thesis, the time from initial public offering (IPO) to the first occurrence of financial distress. We associate with each firm a vector of covariates that captures its financial condition over time. When covariates are observed at discrete quarterly time points, the covariate history for firm i consists of K financial ratios, each observed at d lagged quarterly dates over the s -year window prior to the prediction date. The precise covariate structure is formalised in Section 3.6.

Throughout the thesis, we consider i.i.d. samples of size N . For each firm $i \in [N] := \{1, \dots, N\}$, we observe a possibly censored version of the survival time, a censoring indicator, and covariates. The precise observation structure is described in Section 3.3.

The prediction target shared by all methods in this thesis is the conditional probability

$$\mathbb{P}(T \leq t \mid Z, T \geq s), \tag{3.1}$$

that a firm experiences distress within t years of listing, given that it has survived at least s years and given its covariate vector Z . The conditioning on $\{T \geq s\}$ reflects a practical constraint: the covariates used for prediction consist of quarterly financial data observed over the first s years after listing, so a meaningful prediction can only be formed for firms that have survived long enough to generate this covariate history. The choice of s and the prediction

horizon $t > s$ are design parameters specified in the empirical application (Chapter 6).

3.2 The Survival Function

The central object in the survival-analytic setup of this thesis is the survival function, which characterises the distribution of T directly in terms of the probability of being alive at time t .

The *survival function* gives the probability that a firm survives beyond time t :

$$S(t) = \mathbb{P}(T > t), \quad t \geq 0. \quad (3.2)$$

At the endpoints, $S(0) = 1$ and $S(t) \rightarrow 0$ as $t \rightarrow \infty$, reflecting that all firms are alive at listing and eventually experience the event.

When conditioning on covariates, we write the *conditional survival function*

$$S(t | Z) = \mathbb{P}(T > t | Z). \quad (3.3)$$

The prediction target (3.1) relates to the conditional survival function through

$$\mathbb{P}(T \leq t | Z, T \geq s) = 1 - \frac{S(t | Z)}{S(s | Z)}, \quad (3.4)$$

for $t > s$: the ratio $S(t | Z)/S(s | Z)$ is the probability of still being alive at t given survival to s , and one minus that is the default probability over $(s, t]$. The methods in this thesis model the conditional probability (3.1) directly through a logistic specification (Chapters 4 and 5), rather than estimating the full conditional survival function $S(t | Z)$ and recovering the target via (3.4). The direct approach is natural when the quantity of primary interest is a failure probability at a fixed horizon, rather than the full survival curve.

3.3 Right Censoring

In practice, the survival time T is not always fully observed. A key feature of survival data is *censoring*: for some firms, we only know that T exceeds a certain time, not its exact value. This section formalises the censoring mechanism that all methods in the thesis must handle.

3.3.1 The Observation Structure

Let $C \geq 0$ denote the *censoring time*. In the corporate default risk setting, C is determined by the end of the observation period: if the follow-up ends on December 31, 2020, then C_i for firm i is the number of years from its IPO to that date. What we actually observe is not (T, C)

separately, but the *observed time*

$$\tilde{T} = \min(T, C), \quad (3.5)$$

together with the *event indicator*

$$\delta = \mathbf{1}\{T \leq C\}, \quad (3.6)$$

which equals 1 if the event is observed and 0 if the observation is censored. The observed data for firm i thus consist of the triplet $(\tilde{T}_i, \delta_i, Z_i)$.

Example 3.1 (Censoring in the empirical dataset). In the Chinese manufacturing firm dataset studied in Chapter 6, firms are observed from their IPO through December 31, 2020. A firm whose IPO occurred in 2014 and which never experienced distress has $C = 6$ years. If the firm entered distress in 2017, then $T = 3$, and since $T < C$, we have $\delta = 1$ and $\tilde{T} = 3$. If instead the firm remains healthy, we have $\delta = 0$ and $\tilde{T} = C = 6$. In the empirical dataset, approximately 92.6% of observations are censored.

3.3.2 Independent Censoring

To recover default probabilities from censored data consistently, we need an assumption about how the censoring mechanism relates to the failure time. Following Miao et al. (2025), we assume:

Assumption 3.1 (Independent censoring). C is independent of T and Z .

This assumption states that the censoring mechanism carries no information about the failure time or the covariates. In the corporate setting, Assumption 3.1 is natural because the censoring time is determined solely by the observation period: it is a calendar date rather than a firm-specific event. No firms are selectively removed from the study before the end date.

Under Assumption 3.1, the Kaplan–Meier estimator of Section 3.4 and the inverse probability of censoring weights of Section 3.5 remain consistent. If the assumption were violated, for instance if firms at higher risk of distress were more likely to delist and thus be censored, the methods would produce biased estimates.

3.3.3 Sufficient Follow-Up

In addition to independent censoring, the IPCW approach of Section 3.5 requires that the follow-up period is long enough to observe events at the prediction horizon. Following Miao et al. (2025), we assume:

Assumption 3.2 (Sufficient follow-up). $\mathbb{P}(\tilde{T} \geq t) \geq C_r$, where $C_r > 0$ is a fixed constant.

This ensures that the probability of remaining uncensored and distress-free up to time t is bounded away from zero, which is necessary for estimating the censoring survival function at the prediction horizon. Note that C_r is a regularity constant unrelated to the censoring times C_i . The assumption guarantees that the IPCW weights remain bounded, preventing any single observation from dominating the weighted objective.

3.4 The Kaplan–Meier Estimator

The Kaplan–Meier (KM) estimator (Kaplan and Meier, 1958) is a nonparametric estimator of the survival function from right-censored data. In its general form, given D distinct event times $t_{(1)} < t_{(2)} < \dots < t_{(D)}$, it estimates the probability of surviving beyond time t as the product-limit

$$\hat{S}(t) = \prod_{j: t_{(j)} \leq t} \left(1 - \frac{d_j}{n_j}\right), \quad (3.7)$$

where d_j is the number of events at $t_{(j)}$ and n_j is the number at risk just before $t_{(j)}$.

In this thesis, the KM estimator is not used to estimate the event survival function $S(t) = \mathbb{P}(T > t)$ itself; default probabilities are instead modelled directly through the logistic specification of Chapters 4 and 5. The KM estimator is used exclusively to estimate the *censoring* survival function $H(t) = \mathbb{P}(C \geq t \mid C \geq s)$, which enters the IPCW weights of Section 3.5. To do so, the roles of events and censoring are reversed: observations with $\delta_i = 0$, namely the censored firms, are treated as “events,” and observations with $\delta_i = 1$, namely the defaults, are treated as “censored.” Applying (3.7) to this reversed sample yields the estimate

$$\hat{H}(t) = \hat{\mathbb{P}}(C \geq t \mid C \geq s), \quad (3.8)$$

which is consistent under Assumption 3.1, since the independence of C and T ensures that the censoring times form a valid survival sample when the failure times play the role of censoring.

3.5 Inverse Probability of Censoring Weighting

Throughout this thesis, censoring is handled by *inverse probability of censoring weighting* (IPCW). The idea is to reweight observations for which the outcome at horizon t is known so that they are collectively representative of the full population.

3.5.1 Motivation

The outcome at horizon t is unresolved for firms with $C_i < t$: we observe neither the event nor a survival time beyond t , so for these firms we cannot determine whether distress would

have occurred before the horizon. The missing information is asymmetric, though. A firm censored at year five may still have defaulted at year seven, whereas a firm observed to default at year three cannot retroactively become a non-default. Censoring therefore hides a fraction of would-be defaults but never would-be survivors, and since the censored firms are systematically the youngest, simply dropping them and counting observed defaults over the remainder underestimates the true default probability.

IPCW resolves this by up-weighting each observed default so that it stands in for the defaults hidden by censoring. If only a fraction $H(T_i)$ of firms are typically followed long enough to witness an event at time T_i , then each observed default is counted as $1/H(T_i)$ would-be defaults. Firms with unresolved outcomes at t receive zero event-weight, and firms that survived to t without distress are already correctly recorded as non-events.

3.5.2 The Observation Indicator and Weighting Scheme

Under Assumptions 3.1 and 3.2, the IPCW correction is built from two components. Define the *observation indicator*

$$\delta_i(t) = \mathbf{1}\{C_i \geq t \wedge T_i\}, \quad (3.9)$$

which equals one precisely when the outcome at horizon t is known for firm i : either because the firm defaulted before t , or because it survived past t while still under observation. The IPCW weight for firm i is then

$$w_i = \frac{\delta_i(t) \mathbf{1}\{\tilde{T}_i \leq t\}}{H(t \wedge \tilde{T}_i)}, \quad (3.10)$$

where $H(u) = \mathbb{P}(C \geq u \mid C \geq s)$ is the censoring survival function. The numerator picks out the firms that actually defaulted before the horizon; for such firms $\tilde{T}_i = T_i$, so the denominator is $H(T_i)$, the probability of being followed long enough to witness the event. The weight can therefore be read as *observed default* divided by *probability of being followed that long*, and each observed default is counted as $1/H(T_i)$ would-be defaults, in line with the motivation above.

In practice, H is estimated by the Kaplan–Meier estimator $\hat{H}$ from (3.8). The weights w_i are computed once from the training data and treated as fixed constants throughout estimation, applying unchanged to the linear sg-LASSO-MIDAS baseline of Chapter 4 and the neural extensions of Chapter 5. This horizon-specific form is the outcome-weighted IPCW used by Miao et al. (2025), tailored to logistic estimation at a fixed horizon.

Example 3.2 (IPCW weights on the empirical panel). In the empirical panel of Chapter 6, the firms with positive IPCW weight at each horizon are those that defaulted before it: 23 firms at $t = 8$, 28 at $t = 8.5$, and 34 at $t = 9$, out of 901 in total. The remaining firms carry zero IPCW weight, but they still enter the estimation through the unweighted normalisation term of the logistic objective in Chapter 4.

3.5.3 Validity Beyond Linear Models

An important property of the IPCW correction is that its validity does not depend on the functional form of the prediction model. The equivalence between the population objectives with and without censoring (Lemma A.2 of Miao et al. 2025) rests on Assumption 3.1 and the law of iterated expectations, neither of which involves the linearity of the predictor. The IPCW weights therefore serve the same role in the neural extensions of Chapter 5 as in the sg-LASSO-MIDAS baseline, and the censoring treatment is retained unchanged across architectures.

3.6 The MIDAS Transformation

The prediction target (3.1) requires a covariate vector Z that summarises each firm's financial condition. The setup is mixed-frequency in a specific sense: a single binary outcome per firm, observed at horizon t , is predicted from K financial ratios sampled at quarterly frequency over s years, yielding $K \times d$ lagged predictor values per firm. The many quarterly observations per firm carry quarter-to-quarter variation that simple aggregation to an annual frequency would discard, but also produce a lag structure too high-dimensional to estimate without restrictions. This section formalises the covariate construction and the Mixed Data Sampling (MIDAS) dimensionality reduction that all methods in the thesis build upon.

3.6.1 The High-Dimensional Lag Structure

Consider an i.i.d. sample $\{(\tilde{T}_i, \delta_i, Z_i)\}_{i=1}^N$ where all firms have survived at least s years. For each of K financial covariates, we observe d lagged quarterly values covering the s -year window prior to the conditioning threshold, where $d = s \times m - 1$ and $m = 4$ is the annual sampling frequency of the quarterly data used throughout this thesis. For covariate k , the lag vector is

$$Z_{i,k} = (z_{i,s-\frac{1}{m},k}, z_{i,s-\frac{2}{m},k}, \dots, z_{i,s-\frac{d}{m},k})^\top \in \mathbb{R}^d. \quad (3.11)$$

Stacking all lags and an intercept, the full covariate vector is

$$Z_i = \left(1, Z_{i,1}^\top, Z_{i,2}^\top, \dots, Z_{i,K}^\top\right)^\top \in \mathbb{R}^{K_z}, \quad K_z = K \times d + 1. \quad (3.12)$$

The parameter count K_z grows linearly with the number of lags d , and even moderate values of K and s produce a high-dimensional regression where direct estimation is infeasible. A naive remedy, applying the LASSO to all K_z parameters individually, treats every lag as if it were unrelated to the others, even though nearby quarters of the same financial ratio typically carry similar predictive information.

3.6.2 From Raw Lags to Dictionary-Weighted Features

The Mixed Data Sampling (MIDAS) approach of Ghysels et al. (2007) offers a structured solution by approximating each lag profile with a low-dimensional dictionary of basis functions. The linear predictor involving the raw covariates can be decomposed as

$$Z_i^\top \theta_0 = \theta_{0,1} + \frac{1}{d} \sum_{k=1}^K \sum_{j=1}^d \omega_k \left(\frac{j-1}{d} \right) z_{i,s-\frac{j}{m},k'} \quad (3.13)$$

where $\omega_k : [0,1] \rightarrow \mathbb{R}$ is a weight function that describes the shape of how covariate k 's influence is distributed across its lags: whether the most recent quarters dominate, whether the weight peaks in the middle of the window, whether it is roughly flat, and so on. Rather than treating the d lag coefficients of each covariate as unrelated free parameters, as the standard LASSO would, the MIDAS parametrisation views them as d samples of one underlying smooth function ω_k . The key idea is then to approximate each ω_k using a finite dictionary $\{w_1, \dots, w_L\}$ of basis functions:

$$\omega_k(u) \approx \sum_{l=1}^L \beta_{0,k,l}^* w_l(u), \quad u \in [0,1]. \quad (3.14)$$

This replaces the d free coefficients per covariate with only L coefficients, where typically $L \ll d$. In MIDAS terminology, this is the *restricted* specification, as opposed to the *unrestricted* variant that retains all d lag coefficients.

We define the MIDAS-weighted covariate vector:

$$X_i = \left(1, Z_{i,1}^\top W, Z_{i,2}^\top W, \dots, Z_{i,K}^\top W \right)^\top \in \mathbb{R}^p, \quad (3.15)$$

where $W = (w_l(\frac{j-1}{d})/d)_{j \in [d], l \in [L]} \in \mathbb{R}^{d \times L}$ is the weighting matrix and $p = K \times L + 1$. For each covariate k , the aggregated feature vector $h_{i,k} = Z_{i,k}^\top W \in \mathbb{R}^L$ summarises the d lags into L components, each corresponding to one basis function in the dictionary. The same notation $h_{i,k}$ is used throughout the remainder of the thesis, including in the neural extensions of Chapter 5, where the aggregation is performed by a learned network rather than by the fixed dictionary.

3.6.3 The Gegenbauer Polynomial Dictionary

The choice of dictionary matters for both numerical stability and estimation quality. Following Miao et al. (2025), we use a Gegenbauer polynomial dictionary: the power polynomials $\{1, x, x^2\}$ are orthogonalised via Gram–Schmidt with respect to the Chebyshev measure

$d\mu(x) = (1 - x^2)^{-1/2} dx$ on $[-1, 1]$, and shifted to $[0, 1]$ via $x = 2u - 1$ so that the basis lives on the same interval as the normalised lag positions. The resulting three functions w_1, w_2, w_3 capture a constant level, a linear trend, and a quadratic curvature of the lag profile. A covariate whose influence is roughly uniform across the window loads mostly on w_1 ; one whose influence decays or grows monotonically from recent to distant lags loads on w_2 ; and one with a peak or trough in the middle of the window loads on w_3 .

Evaluated at the d lag positions, the raw power polynomials u and u^2 produce nearly parallel columns in the design matrix: both are monotone in u , both are non-negative on $[0, 1]$, and their correlation over the $d = 23$ quarter-points is high. The two features then carry nearly the same information across firms, leaving LASSO with no stable way to decide which of them should hold the coefficient. The Gegenbauer basis decouples the three components so that each dictionary function contributes complementary rather than overlapping information, leaving the regularised estimator with a well-conditioned problem and stable variable selection across random subsamples.

More generally, the dictionary size L is a design choice that controls a bias–variance trade-off. A larger L enriches the space of representable weight functions, reducing the approximation error, but increases the number of parameters. In the linear baseline (Chapter 4), L is fixed at 3. In the neural extensions of Chapter 5, L is treated as a hyperparameter selected via cross-validation.

3.6.4 The MIDAS Approximation Error

The dictionary approximation (3.14) introduces an approximation error whenever the true weight functions ω_k do not lie exactly in the span of $\{w_1, \dots, w_L\}$. Denoting the best linear predictor under the MIDAS parametrisation by $X_i^\top \beta_0$, the MIDAS approximation error is

$$E_i = Z_i^\top \theta_0 - X_i^\top \beta_0. \quad (3.16)$$

This additional source of error does not arise in standard high-dimensional regression, where predictors enter the model directly. The theoretical analysis of Miao et al. (2025) explicitly accounts for E_i , providing finite-sample estimation bounds that decompose into a statistical error term and an approximation bias term.

The approximation error motivates a central question addressed by the neural extensions in Chapter 5: can the fixed dictionary W be replaced by a *learned* aggregation function that adapts to the data? When the true lag patterns decay non-monotonically, contain threshold effects, or take shapes that differ meaningfully across covariates, the polynomial basis may impose a binding constraint. A neural aggregation network can, in principle, approximate arbitrary continuous weight functions, potentially eliminating the approximation error

entirely. The residual architecture introduced in Section 5.4 provides a controlled way to test this hypothesis: it anchors the aggregation at the Gegenbauer basis and allows a per-covariate mixing parameter γ_k to determine whether the neural component improves upon the polynomial fit.

3.7 Evaluation Metrics for Censored Data

Standard classification metrics must be adapted for right censoring, since the true event status $\mathbf{1}\{T_i \leq t\}$ is not observed for firms censored before the horizon. This section describes how the time-dependent AUC is estimated and how its uncertainty is quantified in the empirical study.

3.7.1 Time-Dependent ROC Curves and AUC

Each method in this thesis produces a predicted risk score Y_i for firm i , taken as the estimated default probability or any monotone transformation thereof. A Receiver Operating Characteristic (ROC) curve plots the true positive rate against the false positive rate across all classification thresholds, and in the survival context both rates depend on the time horizon t :

$$\text{Se}(c, t) = \mathbb{P}(Y_i > c \mid T_i \leq t), \quad (3.17)$$

$$\text{Sp}(c, t) = \mathbb{P}(Y_i \leq c \mid T_i > t), \quad (3.18)$$

where c is the classification threshold. Because a large share of firms are censored before t , the indicators $\mathbf{1}\{T_i \leq t\}$ and $\mathbf{1}\{T_i > t\}$ cannot be formed for the censored subsample, so a plain empirical estimate of Se and Sp would discard or misclassify a substantial portion of the panel.

We therefore use the nearest-neighbour estimator (NNE) of [Heagerty et al. \(2000\)](#), which guarantees that the resulting estimates of Se and Sp are monotone in the marker. The estimator replaces the missing event indicator with a local kernel-smoothed estimate of $\mathbb{P}(T \leq t \mid Y)$, using firms with similar risk scores as donors: for each candidate threshold c , every firm contributes to the weighted estimates of Se and Sp according to how close its risk score is to c , with kernel bandwidth $\kappa_n = 0.25 \cdot n^{-1/2}$, with n the size of the evaluation sample, matching the implementation of [Miao et al. \(2025\)](#).

The area under the ROC curve at time t is

$$\text{AUC}(t) = \int_0^1 \text{Se}(c, t) d(1 - \text{Sp}(c, t)), \quad (3.19)$$

with the probabilistic interpretation $\text{AUC}(t) = \mathbb{P}(Y_i > Y_j \mid T_i \leq t, T_j > t)$: the probability that a firm that defaults before t receives a higher risk score than one that does not.

3.7.2 Uncertainty Quantification and Reporting

Out-of-sample AUC is a noisy summary of any single train/test split. Chapter 6 therefore repeats the estimation over ten independent train/test iterations with fixed seeds, stratified on the event indicator, so that each iteration yields one value of $\text{AUC}(t)$. Reported headline figures are the means across iterations, and the pointwise 95% confidence interval is constructed as the empirical 2.5th and 97.5th percentiles of the ten iteration-level AUCs.

Chapter 4

The sg-LASSO-MIDAS Framework

This chapter presents the baseline method: the sg-LASSO-MIDAS framework of Miao et al. (2025). It combines the IPCW censoring treatment and MIDAS dimensionality reduction of Chapter 3 with sparse-group LASSO regularisation, yielding a structured linear approach to corporate distress prediction. The baseline serves both as the empirical benchmark against which the neural extensions of Chapter 5 are evaluated and as the source of design choices that the extensions inherit.

4.1 Logistic Regression for Default Prediction

The starting point of Miao et al. (2025) is to model the conditional default probability (3.1) directly through a logistic specification. For a firm with covariate vector $Z_i \in \mathbb{R}^{K_z}$, the model assumes

$$\mathbb{P}(T_i \leq t \mid Z_i, T_i \geq s) = \sigma(Z_i^\top \theta_0(t, s)), \quad (4.1)$$

where $\sigma(u) = \exp(u)/(1 + \exp(u))$ is the logistic function and $\theta_0(t, s) \in \mathbb{R}^{K_z}$ is the true parameter vector, specific to the horizons t and s ; we write θ_0 for short when these are fixed.

The parameter θ_0 maximises the population conditional log-likelihood:

$$\theta_0 = \arg \max_{\theta \in \mathbb{R}^{K_z}} \mathbb{E} \left[\mathbf{1}\{T \leq t\} Z^\top \theta - \log(1 + \exp(Z^\top \theta)) \mid T \geq s \right]. \quad (4.2)$$

This is the standard logistic log-likelihood for the binary outcome $\mathbf{1}\{T \leq t\}$ among firms with $T \geq s$; see Lemma A.1 of Miao et al. (2025). However, equation (4.2) cannot be applied as written, because the event indicator $\mathbf{1}\{T_i \leq t\}$ is not observed for censored firms. The IPCW correction from Section 3.5 resolves this, as described in the next section.

4.2 The IPCW-Weighted Objective

Applying the OIPCW framework of Section 3.5 to the logistic model (4.1), Miao et al. (2025) obtain an equivalent characterisation of θ_0 that depends only on observed quantities. Under Assumptions 3.1 and 3.2,

$$\theta_0 = \arg \max_{\theta \in \mathbb{R}^{K_z}} \mathbb{E} \left[\frac{\delta_i(t) \mathbf{1}\{\tilde{T}_i \leq t\}}{H(t \wedge \tilde{T}_i)} Z_i^\top \theta - \log(1 + \exp(Z_i^\top \theta)) \mid \tilde{T}_i \geq s \right], \quad (4.3)$$

where $\delta_i(t) = \mathbf{1}\{C_i \geq t \wedge T_i\}$ is the observation indicator and $H(u) = \mathbb{P}(C \geq u \mid C \geq s)$ is the censoring survival function introduced in Section 3.5. The proof is given in Lemma A.2 of Miao et al. (2025).

The MIDAS transformation of Section 3.6 replaces the raw covariate vector $Z_i \in \mathbb{R}^{K_z}$ with the lower-dimensional MIDAS-weighted vector $X_i \in \mathbb{R}^p$, where $p = K \times L + 1$. The population parameter of interest becomes $\beta_0 \in \mathbb{R}^p$. Given the i.i.d. sample $\{(\tilde{T}_i, \delta_i, X_i)\}_{i=1}^N$, the empirical OIPCW risk function is

$$R_N(\beta) = \frac{1}{N} \sum_{i=1}^N \left[-\frac{\delta_i(t) \mathbf{1}\{\tilde{T}_i \leq t\}}{\hat{H}(t \wedge \tilde{T}_i)} X_i^\top \beta + \log(1 + \exp(X_i^\top \beta)) \right], \quad (4.4)$$

where $\hat{H}$ is the Kaplan–Meier estimate of the censoring survival function from (3.8). The two terms inside the sum form the standard logistic negative log-likelihood: the IPCW-weighted first term rewards a high logit $X_i^\top \beta$ on the firms that default before t , and $\log(1 + \exp(X_i^\top \beta))$ is the softplus normalisation that penalises a high logit on every observation regardless of outcome. Together they push the estimator toward separating defaulting firms from non-defaulting firms on the logit scale.

4.3 Sparse-Group LASSO Estimation

Even after the MIDAS dimensionality reduction, the number of parameters p can remain large relative to the effective sample size. To address this, Miao et al. (2025) apply the sparse-group LASSO penalty of Simon et al. (2013). For $\beta \in \mathbb{R}^p$, define

$$\Omega(\beta) = \alpha \|\beta\|_1 + (1 - \alpha) \|\beta\|_{2,1}, \quad \|\beta\|_{2,1} = \sum_{G \in \mathcal{G}} \|\beta_G\|_2, \quad (4.5)$$

where $\alpha \in [0, 1]$ balances the LASSO (ℓ_1) and group LASSO ($\ell_{2,1}$) components, and $\mathcal{G}$ is a pre-specified collection of groups. The group structure follows directly from the MIDAS parametrisation: each covariate $k \in \{1, \dots, K\}$ contributes L dictionary-weighted features,

forming a natural group $G_k = \{2 + (k - 1)L, \dots, 1 + kL\}$. The intercept forms its own group $G_0 = \{1\}$.

The sparse-group LASSO estimator minimises the penalised empirical risk:

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} R_N(\beta) + \lambda \Omega(\beta), \quad (4.6)$$

where $\lambda \geq 0$ controls the overall regularisation strength. The penalty (4.5) encourages sparsity at two levels simultaneously: the ℓ_1 component can zero out individual parameters within a group, while the $\ell_{2,1}$ component can zero out entire groups. This is well suited to the MIDAS setting. Group-level sparsity drops an uninformative covariate entirely, while within-group sparsity zeroes an individual dictionary component when the covariate's lag profile does not require it: for example, the quadratic component can be dropped if a constant level and linear trend already capture the shape.

The tuning parameters (λ, α) are selected by cross-validation against the time-dependent AUC, with implementation details given in Section 6.2.2.

4.4 Structural Limitations

Beyond the MIDAS approximation error established in Section 3.6.4, two further structural restrictions limit the flexibility of the sg-LASSO-MIDAS framework. First, each per-covariate contribution $h_{i,k}^\top \beta_k$ is linear in the firm-level aggregated features. Second, the predictor $X_i^\top \beta = \beta_0 + \sum_{k=1}^K h_{i,k}^\top \beta_k$ is additive across covariates and admits no non-additive cross-product terms. Both restrictions, and how they motivate the neural extensions, are unpacked in Section 5.1.

The neural additive extensions of Chapter 5 relax the linearity restriction through per-covariate shape functions f_k , while learned aggregation networks address the dictionary's approximation error. The factor-augmented variant additionally captures cross-covariate structure that the additive predictor cannot represent.

Chapter 5

Neural Additive MIDAS Extensions

This chapter develops the central methodological contribution of the thesis: a family of neural additive extensions of the sg-LASSO-MIDAS framework. The extensions retain the additive structure of the baseline but introduce two distinct sources of nonlinearity. First, the per-covariate aggregation is, depending on the variant, either retained from the Gegenbauer dictionary or replaced by a learned aggregation network. Second, the linear per-covariate contribution is replaced by a learned shape function. The IPCW censoring treatment and group structure of the baseline are retained throughout.

5.1 From the Linear Baseline to a Neural Additive Model

The sg-LASSO-MIDAS framework of Chapter 4 writes the default-event logit as a sum of covariate contributions, $\text{logit}_i = \beta_0 + \sum_{k=1}^K (\tilde{Z}_{i,k}^\top W) \beta_k$: each covariate's contribution depends only on itself, not on any other. What the baseline imposes beyond additivity is that the contribution is *linear* in the dictionary-aggregated features $\tilde{Z}_{i,k}^\top W$: the marginal effect of a unit change in any aggregated feature is the constant $\beta_{k,\ell}$, regardless of the firm's level of that or any other aggregated feature. For financial ratios whose effect on distress risk is plausibly threshold-like, accelerating, or state-dependent, linearity is an economically restrictive assumption, and one that no choice of regularisation tuning can relax within the baseline.

The neural additive model of Agarwal et al. (2021) relaxes the linearity of each per-covariate contribution. Let $h_{i,k} \in \mathbb{R}^L$ denote the aggregated feature vector for covariate k : in the baseline this equals $\tilde{Z}_{i,k}^\top W$, while in the variants developed in Section 5.3 it is replaced by a learned aggregation $g_k(\tilde{Z}_{i,k})$. The linear contribution $h_{i,k}^\top \beta_k$ is then replaced by a nonlinear shape function $f_k(h_{i,k})$ learned by a small feedforward network, yielding

$$\text{logit}_i = \beta_0 + \sum_{k=1}^K f_k(h_{i,k}), \quad (5.1)$$

where each $f_k : \mathbb{R}^L \rightarrow \mathbb{R}$ maps the L -dimensional aggregated feature vector for covariate k to a contribution on the logit scale. Architectural details of f_k are given in Section 5.5.

Two observations motivate the choice to preserve additivity rather than move immediately to a non-additive predictor $f_\theta(h_i)$ on the feature vector $h_i = (h_{i,1}^\top, \dots, h_{i,K}^\top)^\top$. First, additive models inherit the convergence rate of L -dimensional nonparametric problems rather than KL -dimensional ones (Stone, 1985). This structural reduction is a statistical advantage when N is small relative to KL , as in corporate default panels. Neural networks can in principle circumvent the curse of dimensionality (Barron, 1993; Schmidt-Hieber, 2020), but additivity provides this dimension reduction unconditionally.

Second, additivity gives a form of interpretability that a non-additive predictor cannot. The mapping $h_{i,k} \mapsto f_k(h_{i,k})$ depends only on covariate k 's aggregated features, so each financial ratio's learned effect can be analysed in isolation, for instance through the marginal-effect and shape-function plots developed in Section 5.5.2.

5.2 Model Overview and Variants

The NAM-MIDAS family relaxes the two structural restrictions of the sg-LASSO-MIDAS baseline (Section 4.4), the fixed Gegenbauer aggregation and the linear predictor, so that any performance gain or loss can be attributed to a specific component. The variants compared in the empirical study comprise three NAM-MIDAS configurations that relax the two restrictions in different combinations, plus three further variants that capture cross-covariate effects in different ways: a factor-augmented additive model that adds a low-dimensional factor term to the additive predictor, a non-additive neural predictor that maps the joint feature vector directly to the logit, and a factor-only ablation that places that factor term on the linear baseline alone.

The Ablation NAM-MIDAS holds the Gegenbauer aggregation fixed while replacing the linear predictor with $\sum_k f_k(h_{i,k})$ (Section 5.5); the Full NAM-MIDAS additionally replaces the dictionary with per-covariate aggregation networks g_k (Section 5.3). The Residual NAM-MIDAS (Section 5.4) interpolates between the two through a per-covariate mixing weight $\gamma_k \in [0, 1]$, which serves as a diagnostic for each covariate's departure from the dictionary. The FA-NAM-MIDAS (Section 5.6) supplements the additive structure with a low-dimensional factor term $g_0(F_i)$ extracted via principal components, capturing dominant cross-covariate comovements. The Deep-MIDAS predictor (Section 5.7) drops additivity in favour of an unrestricted feedforward network with shared aggregation. A final ablation, FA-MIDAS, places the factor block of FA-NAM-MIDAS directly on the fixed linear baseline, omitting the per-covariate shape functions, to isolate the standalone contribution of the factor channel. Table 5.1 summarises the variants.

Table 5.1. Model variants compared in the empirical study. The columns report the aggregation, predictor, factor block, and source section for each variant.

Variant	Aggregation $h_{i,k}$	Predictor	Factor	Source
sg-LASSO-MIDAS	$\tilde{Z}_{i,k}^\top W$	$\sum_k h_{i,k}^\top \beta_k$	—	Ch. 4
Ablation NAM-MIDAS	$\tilde{Z}_{i,k}^\top W$	$\sum_k f_k(h_{i,k})$	—	Sec. 5.5
Full NAM-MIDAS	$g_k(\tilde{Z}_{i,k})$	$\sum_k f_k(h_{i,k})$	—	Sec. 5.3
Residual NAM-MIDAS	$\gamma_k g_k(\tilde{Z}_{i,k}) + (1 - \gamma_k) \tilde{Z}_{i,k}^\top W$	$\sum_k f_k(h_{i,k})$	—	Sec. 5.4
FA-NAM-MIDAS	$g_k(\tilde{Z}_{i,k})$	$g_0(F_i) + \sum_k f_k(h_{i,k})$	PCA	Sec. 5.6
Deep-MIDAS	$g_\phi(\tilde{Z}_{i,k})$	$f_\theta(h_i)$	—	Sec. 5.7
FA-MIDAS	$\tilde{Z}_{i,k}^\top W$	$g_0(F_i) + \sum_k h_{i,k}^\top \beta_k$	PCA	Sec. 5.6

5.3 Per-Covariate Aggregation Networks

The NAM-MIDAS variants replace the fixed Gegenbauer dictionary W of the baseline by a family of K per-covariate networks,

$$g_k : \mathbb{R}^d \rightarrow \mathbb{R}^L, \quad h_{i,k} = g_k(\tilde{Z}_{i,k}), \quad k = 1, \dots, K. \quad (5.2)$$

The per-covariate design recognises that different financial ratios are heterogeneous in how their lags inform distress risk: short-term profitability measures, slow-moving leverage ratios, and liquidity buffers plausibly require different weightings of recent versus distant lags. Several variants share an aggregation across covariates: the baseline and Ablation NAM-MIDAS via the Gegenbauer dictionary, and Deep-MIDAS via a shared aggregation network. In all of these, the aggregation function itself is identical for every covariate, so any covariate-specific adaptation enters only at the predictor stage, through the linear coefficients β_k for the linear predictors or through the joint predictor in Deep-MIDAS. Per-covariate networks g_k instead provide flexibility at the aggregation stage, letting each covariate have its own possibly nonlinear lag mapping and capturing the heterogeneity in lag dynamics that the data may show. Figure 5.1 shows this aggregation step within the per-covariate pipeline, with the baseline’s fixed Gegenbauer projection alongside the NAM-MIDAS learned g_k .

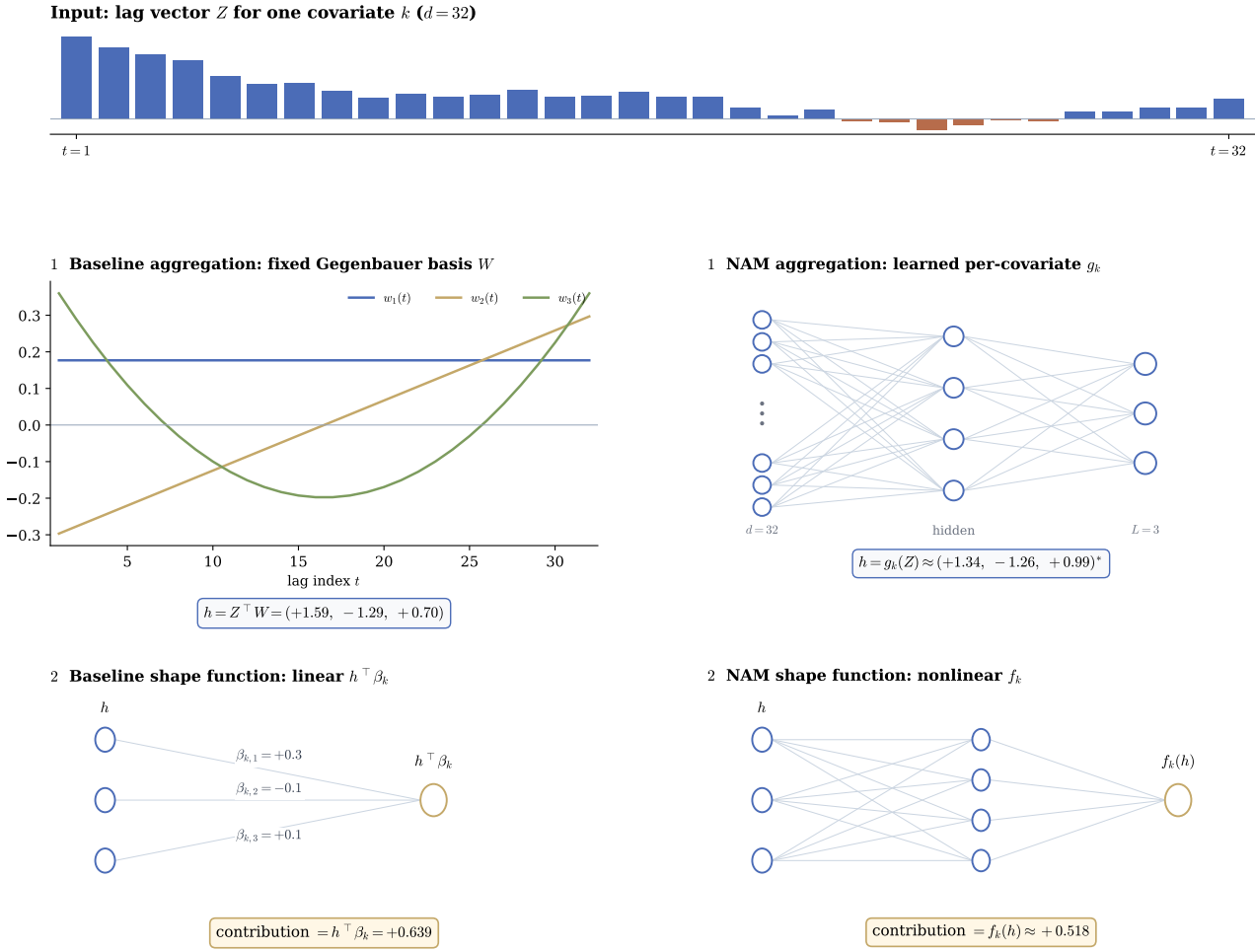


Figure 5.1. Per-covariate pipeline for a single covariate k , comparing the sg-LASSO-MIDAS baseline (left) and NAM-MIDAS (right). Values are illustrative.

5.3.1 Architecture

Each aggregation network g_k is a feedforward neural network with $n_{\text{agg}} \in \{1, 2, 3, 4\}$ hidden layers, first hidden width H_{agg} , and a linear output layer mapping to the L aggregated features. Denoting the hidden widths by $H_1 = H_{\text{agg}}$ and $H_\ell = \max(\lfloor H_{\ell-1}/2 \rfloor, \max(4, L))$ for $\ell = 2, \dots, n_{\text{agg}}$, the network computes

$$g_k(z) = W_k^{(\text{out})} a_{n_{\text{agg}}}(z) + b_k^{(\text{out})}, \quad a_\ell(z) = \text{ELU}\left(W_k^{(\ell)} a_{\ell-1}(z) + b_k^{(\ell)}\right), \quad (5.3)$$

with $a_0(z) = z$, and with the Exponential Linear Unit activation $\text{ELU}(u) = u$ for $u \geq 0$ and $\alpha(e^u - 1)$ for $u < 0$. The matrices $W_k^{(\ell)}$, $W_k^{(\text{out})}$ and vectors $b_k^{(\ell)}$, $b_k^{(\text{out})}$ are the layer- ℓ weights and biases of network g_k . Halving the hidden width at each successive layer, subject to a floor of $\max(4, L)$, keeps both the parameter count and the per-layer forward and backward pass cost proportional to the small output dimension. For L between two and five in the empirical application, uniformly wide hidden layers would carry capacity that the L -dimensional

output cannot directly use. The progressive compression follows the standard funnel-shape heuristic in feedforward design. The floor of $\max(4, L)$ ensures the final hidden width is no smaller than the output dimension, so that the output projection $W_k^{(\text{out})}$ is always full rank. ELU is preferred over ReLU in the aggregation network because hidden units stay active even when their pre-activation is negative, avoiding the dead units that ReLU can develop in a small-width feedforward stack. No dropout is applied to the hidden layers of g_k . Regularisation of the aggregation stage comes instead from weight decay on the parameters of g_k and from the group dropout of Section 5.9, which zeroes the output $h_{i,k}$ of randomly selected covariates during training rather than acting inside any individual g_k .

The parameter count of a single aggregation network with depth n_{agg} is

$$p_{\text{agg}} = d H_1 + \sum_{\ell=2}^{n_{\text{agg}}} H_{\ell-1} H_{\ell} + H_{n_{\text{agg}}} L + \sum_{\ell=1}^{n_{\text{agg}}} H_{\ell} + L, \quad (5.4)$$

where the first two terms are weights between successive layers, $H_{n_{\text{agg}}} L + L$ count the output-layer weights and biases, and the rest are hidden-layer biases. The aggregation stage as a whole therefore contributes $K p_{\text{agg}}$ parameters to the model, scaling linearly in the number of covariates.

5.4 Residual Aggregation with Per-Covariate Mixing

The per-covariate aggregation networks of Section 5.3 replace the Gegenbauer dictionary W entirely. In cases where the dictionary is already a close approximation to the true lag structure, this full replacement discards a strong prior at the cost of additional parameters that must be fitted from data. The residual aggregation mechanism introduced in this section sits between the baseline and the full neural aggregation: each aggregated feature is a weighted average of the Gegenbauer feature and the neural feature,

$$h_{i,k} = (1 - \gamma_k) \tilde{Z}_{i,k}^{\top} W + \gamma_k g_k(\tilde{Z}_{i,k}), \quad k = 1, \dots, K, \quad (5.5)$$

where $\gamma_k \in [0, 1]$ is a covariate-specific mixing parameter learned jointly with the other model components. The mechanism preserves the Gegenbauer dictionary as a soft anchor and allows the neural aggregation to enter only to the extent warranted by the data for each covariate.

5.4.1 Per-Covariate Mixing Parameters

Each mixing weight γ_k is obtained from an unconstrained scalar parameter $\gamma_{k,\text{raw}} \in \mathbb{R}$ through the sigmoid transform, which maps any real number to $(0, 1)$:

$$\gamma_k = \sigma(\gamma_{k,\text{raw}}) = \frac{1}{1 + e^{-\gamma_{k,\text{raw}}}} \in (0, 1), \quad (5.6)$$

so that the weighted average in (5.5) is automatically well-defined throughout training without explicit constraints on γ_k . The raw parameters $\{\gamma_{1,\text{raw}}, \dots, \gamma_{K,\text{raw}}\}$ are trained by stochastic gradient descent jointly with the aggregation networks g_k and the downstream predictor, and no parameter sharing is imposed across covariates, so each γ_k evolves independently.

The raw parameters are initialised at $\gamma_{k,\text{raw}}^{(0)} = -2$ for all k , which through (5.6) gives $\gamma_k^{(0)} = \sigma(-2) \approx 0.119$. At initialisation the aggregation therefore resembles the Gegenbauer dictionary with a small neural correction: roughly 88% of the feature weight lies on the fixed dictionary and 12% on the untrained neural network. This starting point has two advantages. It places the model close to the baseline at the start of training, so the early gradient signal is informative about departures from the baseline rather than about recovering it from random initialisation. It also avoids vanishing gradients in $\gamma_{k,\text{raw}}$: the sigmoid derivative $\sigma'(x) = \sigma(x)(1 - \sigma(x))$ falls toward zero as $\sigma(x)$ approaches 0 or 1. At the chosen initialisation $\gamma_k^{(0)} = 0.119$ the derivative is ≈ 0.105 , well away from saturation. An initialisation closer to $\gamma_k = 0$ or $\gamma_k = 1$ would give a derivative near zero, blocking gradient updates to $\gamma_{k,\text{raw}}$ and freezing the mixing weight. A uniform initialisation of $\gamma_k^{(0)}$ across covariates also ensures that any heterogeneity in the fitted γ_k values reflects what the data require during training, rather than a structural assumption built into the starting point that some covariates already deserve more neural flexibility than others.

5.4.2 Diagnostic Interpretation

Once training has converged, the vector $(\gamma_1, \dots, \gamma_K)$ admits a direct reading as a covariate-level diagnostic of dictionary adequacy. For covariate k , a small value of γ_k indicates that the model has chosen to place most of the aggregation weight on the Gegenbauer dictionary, which is an indirect statement that the polynomial basis already provides a satisfactory representation of the relevant lag structure for that particular financial ratio. A large value of γ_k carries the opposite reading: the model has found the fixed dictionary inadequate and prefers to route most of the aggregation through the learned network g_k . Intermediate values indicate partial disagreement between the basis and the data.

γ_k is a relative weighting. A small γ_k does not assert that the Gegenbauer dictionary is

objectively correct for covariate k , only that in the presence of the candidate neural aggregation g_k and the associated regularisation, the model prefers the dictionary. The diagnostic should therefore be read alongside the shape functions f_k (Section 6.3.3) and the predictive performance reported in Section 6.3.1, rather than as an isolated summary.

5.5 The NAM Predictor

The aggregation stage of Sections 5.3 and 5.4 produces, for each firm i and each covariate k , an aggregated feature vector $h_{i,k} \in \mathbb{R}^L$. The NAM-MIDAS predictor combines these additively,

$$\eta_i = \beta_0 + \sum_{k=1}^K f_k(h_{i,k}), \quad f_k : \mathbb{R}^L \rightarrow \mathbb{R}, \quad (5.7)$$

where η_i denotes the logit entering the IPCW-weighted logistic loss of Section 5.8.

5.5.1 Architecture

Each predictor network f_k is a feedforward neural network with $n_{\text{pred}} \in \{1, 2, 3, 4\}$ hidden layers, first hidden width H_{pred} , and a linear output layer mapping to a single scalar. Denoting the hidden widths by $\tilde{H}_1 = H_{\text{pred}}$ and $\tilde{H}_\ell = \max(\lfloor \tilde{H}_{\ell-1}/2 \rfloor, 4)$ for $\ell = 2, \dots, n_{\text{pred}}$, the network computes

$$\begin{aligned} f_k(h) &= v_k^{(\text{out})\top} \tilde{a}_{n_{\text{pred}}}(h) + c_k^{(\text{out})}, \\ \tilde{a}_\ell(h) &= \text{Dropout}_p \left(\text{ReLU} \left(V_k^{(\ell)} \tilde{a}_{\ell-1}(h) + c_k^{(\ell)} \right) \right), \end{aligned} \quad (5.8)$$

with $\tilde{a}_0(h) = h$, rectified linear activation $\text{ReLU}(u) = \max(u, 0)$, and independent Bernoulli dropout with rate p applied to the post-activation vectors at each hidden layer during training (Section 5.9). The matrices $V_k^{(\ell)}$ and vectors $c_k^{(\ell)}$, $v_k^{(\text{out})}$, $c_k^{(\text{out})}$ are the layer- ℓ weights and biases of network f_k .

Unlike the aggregation networks (Section 5.3.1), the predictor uses ReLU activations and applies dropout at each hidden layer during training. ReLU is standard in feedforward classifier heads and interacts naturally with dropout. Dropout is confined to the predictor because the shape functions f_k have more scope to overfit than the lag aggregations g_k , so stochastic regularisation is targeted at the component most at risk.

The parameter count of a single predictor network with depth n_{pred} is

$$p_{\text{pred}} = L \tilde{H}_1 + \sum_{\ell=2}^{n_{\text{pred}}} \tilde{H}_{\ell-1} \tilde{H}_\ell + \tilde{H}_{n_{\text{pred}}} + \sum_{\ell=1}^{n_{\text{pred}}} \tilde{H}_\ell + 1, \quad (5.9)$$

where the first two terms are weights between successive layers, $\tilde{H}_{n_{\text{pred}}} + 1$ are the output-layer weight and bias, and the rest are hidden-layer biases. The predictor stage as a whole therefore contributes $Kp_{\text{pred}} + 1$ parameters, the additive 1 being the shared intercept β_0 .

5.5.2 Interpretability via Shape Functions

Equation (5.1) is an additive model in the sense of [Hastie and Tibshirani \(1990\)](#) because each shape function f_k depends only on $h_{i,k}$. For any fixed covariate k , the mapping $h \mapsto f_k(h)$ is a function of L arguments. Two questions must be settled before the visual reading developed in Section 6.3.3 can be evaluated: what the L axes of $h_{i,k}$ represent, and what a finite-dimensional plot of f_k actually shows.

The interpretation of the L axes depends on the variant. For the sg-LASSO-MIDAS baseline and the Ablation NAM-MIDAS variant the aggregation is the fixed Gegenbauer projection $h_{i,k} = \tilde{Z}_{i,k}^\top W$, so the components inherit the polynomial-order interpretation of W : $h_{k,1}$ approximates the mean of the lag profile, $h_{k,2}$ a linear trend, $h_{k,3}$ a curvature. For the Residual NAM-MIDAS variant the small mixing weight $\tilde{\gamma}_k \approx 0.13$ documented in Section 6.3.2 keeps that decoding approximately valid. For the Full NAM-MIDAS and FA-NAM-MIDAS variants, where the aggregation is fully learned, $h_{i,k} = g_k(\tilde{Z}_{i,k})$ has no predetermined economic label, and each of the L components is whatever the per-covariate aggregation network has chosen to extract from the lag window. An economic reading must then be reconstructed *a posteriori*, either formally, by regressing $h_{k,\ell}$ on a panel of candidate raw-lag summaries within an iteration, or informally, by inspecting the shape-function visualisations of f_k and reading off how each component $h_{k,\ell}$ shifts the predicted contribution relative to the others; the known economic meaning of the underlying ratio k then suggests what each component appears to capture. The latter route is the one pursued in Section 6.3.3 for two illustrative FA-NAM-MIDAS iterations.

When $L > 1$, the natural mapping $h \mapsto f_k(h)$ cannot be rendered directly on two axes, and the plots in Section 6.3.3 take two complementary forms. The first is a single-coordinate scatter $(h_{i,k,\ell}, f_k(h_{i,k}))$ for a chosen slice index $\ell \in \{1, \dots, L\}$, pooled across observations and training iterations. This is the standard NAM shape plot, but it is not a slice in which the other $L - 1$ coordinates of $h_{i,k}$ are held fixed: those coordinates vary freely across the plotted observations and contribute to the visible vertical spread. The plotted curve is therefore the projection of an L -dimensional surface onto one axis, not a marginal-effect plot in the additive-model sense. The second form is direct multi-dimensional rendering: for $L = 2$ a 2D contour or 3D height surface displays the full f_k without compression, and for $L = 3$ a 3D scatter or pairwise 2D projection retains all of the input information up to colour encoding. Because the random-search grid for L includes $\{2, 3\}$ alongside higher values, several fitted iterations admit this direct reading without slicing.

Both visualisation strategies are conditional on the identifiability arguments developed in Section 5.11: the level-shift and slope-redistribution ambiguities displace and reshape fitted surfaces between iterations of the same fold without changing the prediction rule, so within-iteration shapes are identified only up to such level shifts and slope redistribution.

For comparison, the linear baseline’s shape function $f_k^{\text{lin}}(h) = h^\top \beta_k$ admits a more compact visualisation in lag space. Composing f_k^{lin} with the Gegenbauer aggregation gives a single weighted sum over raw lags, contribution = $\sum_t w_{\text{eff}}(t) Z_t$, with per-lag effective weight $w_{\text{eff}}(t) = \sum_{\ell=1}^L \beta_{k,\ell} w_\ell(t)$, which by construction lies in the L -dimensional span of the Gegenbauer dictionary. The NAM-MIDAS shape function relaxes this restriction: its dependence on h is nonlinear, so the composition $f_k(g_k(Z))$ is not a single weighted sum over lags and is not constrained to any polynomial span. Figure 5.2 illustrates the consequence on a hypothetical bimodal target lag-weight pattern, with a peak on the most recent lags and a smaller bump in the mid-range. The baseline’s best fit within the Gegenbauer span cannot reproduce the bimodal pattern and smooths the peak and the bump into a single unimodal curve, while the learned shape function recovers both without overfitting individual points.

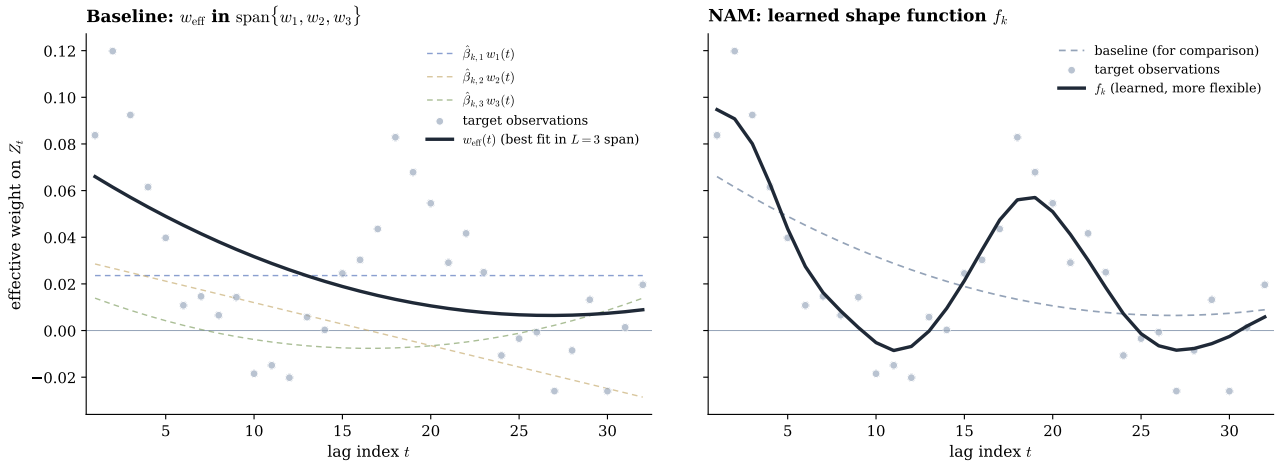


Figure 5.2. Side-by-side fit of a hypothetical bimodal lag-weight target by the linear baseline (left) and the NAM-MIDAS shape function (right). *Left:* thin dashed lines are the three scaled Gegenbauer basis components $\hat{\beta}_{k,\ell} w_\ell(t)$, and the thick solid line is their sum, the baseline’s best w_{eff} , which is confined to the L -dimensional Gegenbauer span. *Right:* the thick solid line is the learned shape function f_k , which is unconstrained by that span and recovers both the recent-lag peak and the mid-range bump; the faint dashed line repeats the baseline fit for comparison.

5.6 Factor-Augmented NAM-MIDAS

The three NAM-MIDAS variants of Sections 5.3 to 5.5 share a purely additive predictor: each covariate’s contribution depends only on its own aggregated features. The FA-NAM-MIDAS variant relaxes this restriction in a controlled way, following the factor-augmented neural network framework of [Fan and Gu \(2024\)](#): a small number of latent factors is extracted from

the aggregated feature matrix and passed through a dedicated factor network, while the per-covariate shape functions are retained alongside. The resulting predictor is additive over $K + 1$ channels: K covariate-specific channels plus one joint channel that captures variation shared across covariates.

5.6.1 Motivation: Beyond Additivity

The additive predictor $\beta_0 + \sum_k f_k(h_{i,k})$ cannot capture interactions between covariates because each shape function f_k takes only its own aggregated feature $h_{i,k}$ as input. The effect of $h_{i,k}$ on the logit is therefore independent of the values of $h_{i,k'}$ for any other covariate k' . In corporate distress prediction the absence of interactions is a substantive limitation, because financial ratios are driven by shared latent forces such as macroeconomic cycles, sectoral shocks, or funding conditions; the effect of a given leverage ratio on distress risk plausibly differs across regimes in which profitability and liquidity ratios are jointly high or jointly depressed. The factor-augmented decomposition follows Fan and Gu (2024) in treating a small number of common components as the source of shared variation across covariates, modelling the residuals separately and preserving covariate-level interpretability while adding controlled cross-covariate interactions.

5.6.2 Factor Estimation via PCA

After the aggregation stage produces the per-covariate feature vectors $h_{i,k} \in \mathbb{R}^L$, the stacked feature vector

$$H_i = (h_{i,1}^\top, \dots, h_{i,K}^\top)^\top \in \mathbb{R}^{KL} \quad (5.10)$$

summarises the aggregated information for firm i at the current horizon. Factors are obtained by the linear projection $F_i = P^\top H_i \in \mathbb{R}^r$, where $P \in \mathbb{R}^{KL \times r}$ is a projection matrix estimated by principal component analysis on the training sample and the factor dimension r is a small hyperparameter relative to KL .

Following Fan and Gu (2024), the projection matrix P is estimated by principal component analysis (PCA): the leading r orthogonal directions that capture the most variance in a fixed reference set of stacked features become the columns of P . Concretely, let $\bar{H} \in \mathbb{R}^{n \times KL}$ denote the matrix whose rows are the stacked feature vectors H_i for the n training firms, computed under the sg-LASSO-MIDAS Gegenbauer aggregation $h_{i,k} = \tilde{Z}_{i,k}^\top W$. Column-standardising $\bar{H}$ to $\bar{H}_{\text{std}}$ by subtracting column means and dividing by column standard deviations, the sample covariance $\Sigma = \bar{H}_{\text{std}}^\top \bar{H}_{\text{std}} / n$ is eigendecomposed, and P is set to the matrix of its leading r eigenvectors, scaled column-wise by the inverse column standard deviations so that P operates directly on unstandardised stacked features. The projection matrix is estimated once at the start of training and held fixed thereafter, so no gradient flows through P during

neural network training. The factors seen by the model at each forward pass are $F_i = P^\top H_i$, with H_i assembled from whichever aggregated features the model is currently producing.

Estimating P from the Gegenbauer-aggregated training features anchors the factor space to a stable reference that does not depend on the neural network parameters; re-estimating P during training would create a moving target for g_0 . The eigendecomposition is performed on the standardised covariance, so the leading factors are directional components of variation across firms rather than directions of large raw magnitude.

5.6.3 The Factor Network

The factors $F_i \in \mathbb{R}^r$ enter the predictor through a dedicated feedforward network $g_0 : \mathbb{R}^r \rightarrow \mathbb{R}$, yielding the augmented logit

$$\eta_i = \beta_0 + g_0(F_i) + \sum_{k=1}^K f_k(h_{i,k}). \quad (5.11)$$

The predictor decomposes the logit into three structurally distinct components: a scalar intercept β_0 , a factor contribution $g_0(F_i)$ that depends on all covariates through the linear projection P , and the K covariate-specific contributions $f_k(h_{i,k})$ familiar from the NAM-MIDAS predictor. Holding the covariate-specific terms fixed at any configuration, g_0 is free to capture nonlinear, regime-dependent effects of the factor summary on the logit, while the per-covariate shape functions continue to capture the variation specific to each covariate, after the factor channel has absorbed the shared cross-covariate variation.

5.6.4 Architecture and Design Choices

The factor network g_0 is a feedforward neural network with $n_{\text{fac}} \in \{1, 2\}$ hidden layers, first hidden width H_{fac} , ReLU activation, and dropout at each hidden layer. With $n_{\text{fac}} = 2$ the second hidden width is halved to $\max(\lfloor H_{\text{fac}}/2 \rfloor, 4)$, matching the halving rule used in the aggregation and predictor stages. The depth ceiling of two hidden layers reflects the low dimensionality of the factor input together with the limited training sample: deeper networks would risk overfitting without meaningfully improving expressive power for the smooth mapping the factor block must learn.

The factor dimension r is treated as a hyperparameter in the random search of Section 5.10, with $r \in \{2, 3, 4, 5\}$ reflecting the expectation that the number of common components is much smaller than KL .

Two design choices distinguish FA-NAM-MIDAS. First, the factor network g_0 is a network

rather than a linear inner product. A linear factor contribution would reduce to

$$g_0^{\text{lin}}(F_i) = \alpha^\top F_i = \alpha^\top P^\top H_i = (P\alpha)^\top H_i = \beta^\top H_i,$$

where $\beta = P\alpha$, a linear function of H_i that is already representable by the per-covariate channels. Without nonlinearity in g_0 , factor augmentation would add nothing beyond what the per-covariate channels can already represent. Second, the PCA projection is pre-trained and fixed rather than estimated jointly with the network parameters, following [Fan and Gu \(2024\)](#). This separates the factor-structure problem from the function-approximation problem and avoids the substantial training complexity of jointly optimising a $KL \times r$ projection matrix.

5.7 The Deep-MIDAS Predictor

The Deep-MIDAS predictor reverses the architectural choices of the NAM-MIDAS variants: a single shared aggregation network g_ϕ replaces the per-covariate g_k , and a feedforward predictor without additive restriction replaces the additive sum $\sum_k f_k(h_{i,k})$. Neither specification is a special case of the other, so including Deep-MIDAS tests which architecture is more useful on corporate-default data.

5.7.1 Shared Aggregation and Predictor

First, the aggregation stage uses a single shared network $g_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^L$, applied identically to all K covariates,

$$h_{i,k} = g_\phi(\tilde{Z}_{i,k}), \quad k = 1, \dots, K, \quad (5.12)$$

rather than a collection of K per-covariate networks g_k as in Equation (5.2). The shared aggregator imposes a single lag-to-feature mapping across all financial ratios, sacrificing the covariate-specific flexibility that motivates the separate design (Section 5.3).

Second, the predictor is a feedforward network $f_\theta : \mathbb{R}^{KL+1} \rightarrow \mathbb{R}$ taking all K covariates' aggregated features and an intercept as input,

$$\eta_i = f_\theta(1, h_{i,1}^\top, \dots, h_{i,K}^\top), \quad (5.13)$$

in contrast to the additive composition $\beta_0 + \sum_k f_k(h_{i,k})$ of Equation (5.1). The predictor is a feedforward neural network with $n_{\text{pred}} \in \{1, 2, 3, 4\}$ hidden layers, first hidden width $H_{\text{pred}}^{\text{deep}}$, ReLU activations, and dropout at each hidden layer. Hidden widths halve as $H_1 = H_{\text{pred}}^{\text{deep}}$ and $H_\ell = \max(\lfloor H_{\ell-1}/2 \rfloor, 8)$ for $\ell \geq 2$, with a floor of eight reflecting the higher input dimension $KL + 1$ relative to the per-covariate predictors of Section 5.5. The first hidden width is chosen

to be substantially larger than the NAM predictor width, so that the Deep-MIDAS predictor has the capacity to represent nonlinear combinations of aggregated features across covariates.

The hidden layers of f_θ mix aggregated features across all K covariates from the first hidden layer onward, so the model admits arbitrary nonlinear interactions between any subset of raw lags, limited only by capacity and regularisation. The only structural restriction retained from the sg-LASSO-MIDAS framework is the MIDAS feature-extraction step itself: raw lags enter only through the aggregated features $h_{i,k}$ of dimension L .

Whether NAM-MIDAS or Deep-MIDAS has more parameters in aggregate depends on the chosen hidden widths: NAM-MIDAS' K separate aggregation networks contribute a factor of K at the aggregation stage, while Deep-MIDAS' predictor on $\mathbb{R}^{KL+1}$ adds parameters at the predictor stage.

5.8 IPCW-Weighted Loss Function

The estimation target for all neural variants of this chapter is the same population parameter identified for the sg-LASSO-MIDAS baseline in Section 4.2, namely the logistic default-probability coefficients under the IPCW characterisation of Equation (4.3). The only difference in this loss between the baseline and the neural variants is how the logit η_i is computed: the linear score $X_i^\top \beta$ of the baseline is replaced by the neural logit $\eta_i(\Phi)$ produced by the architecture of the corresponding variant (Equation (5.1) for the NAM-MIDAS family, Equation (5.11) for the FA-NAM-MIDAS variant, and Equation (5.13) for the Deep-MIDAS predictor). The IPCW-weighted empirical risk used for training is the direct analogue of Equation (4.4),

$$\mathcal{L}(\Phi) = \frac{1}{N} \sum_{i=1}^N [-w_i \eta_i(\Phi) + \log(1 + \exp(\eta_i(\Phi)))] , \quad (5.14)$$

where Φ collects all trainable parameters of the variant and $w_i = \delta_i(t) \mathbf{1}\{\tilde{T}_i \leq t\} / \hat{H}(t \wedge \tilde{T}_i)$ are the OIPCW weights of Equation (3.10), computed once from the training sample and treated as fixed constants throughout estimation. The structural reading of the two terms inside the bracket is the same as for the baseline empirical risk of Equation (4.4), with $X_i^\top \beta$ replaced by $\eta_i(\Phi)$.

The regularisation terms developed in Section 5.9 are added to this loss on a per-variant basis to form the total training objective.

5.9 Regularisation

The IPCW-weighted loss of Equation (5.14) controls the model's fit to the data but places no restriction on the magnitude or roughness of the trainable parameters Φ . In a high-

dimensional setting where the total parameter count of the NAM-MIDAS family substantially exceeds the sample size (Equation (5.9) and Equation (5.4)), the unregularised minimiser is not statistically well defined: many configurations of weights attain near-identical training loss and the choice among them is determined by initialisation and optimiser noise rather than by the data.

Four distinct devices are combined. Group dropout (Section 5.9.1) randomly zeros entire shape-function outputs $f_k(h_{i,k})$, implementing a stochastic analogue of the group sparsity that the sg-LASSO penalty enforces in the baseline. An additive structured penalty (Section 5.9.2) applies ℓ_1 and ℓ_2 norms to each predictor subnetwork f_k 's first-layer weights, mirroring the sg-LASSO group structure of the baseline. An output penalty (Section 5.9.3) shrinks the magnitude of the shape-function outputs $f_k(h_{i,k})$ directly rather than through the weights. Weight decay (Section 5.9.4) applied through the Adam optimiser, the adaptive stochastic gradient method of Kingma and Ba (2015), acts as a background ℓ_2 on all parameters. Collecting the loss and the explicitly additive penalty terms, the total training objective is

$$\mathcal{L}_{\text{tot}}(\Phi) = \mathcal{L}(\Phi) + \lambda_1 \sum_{k=1}^K \|V_k^{(1)}\|_1 + \lambda_2 \sum_{k=1}^K \|V_k^{(1)}\|_2 + \lambda_{\text{out}} \frac{1}{K} \sum_{k=1}^K \frac{1}{N} \sum_{i=1}^N f_k(h_{i,k})^2, \quad (5.15)$$

with $V_k^{(1)}$ the first-layer weight matrix of predictor f_k . The four penalty strengths $(\lambda_1, \lambda_2, \lambda_{\text{out}}, \lambda_{\text{wd}})$ and the group-dropout probability p_{group} are selected alongside the architectural hyperparameters via the random search described in Section 5.10.3.

The four devices are described here for the additive NAM-MIDAS variants, whose predictor is a sum of per-covariate shape functions f_k . Two variants of Table 5.1 do not share that structure and are regularised by the subset of devices that still applies. The Deep-MIDAS predictor has no per-covariate subnetworks, so group dropout, the structured penalty, and the output penalty, all defined on the individual f_k , have no counterpart; it is regularised by weight decay on all parameters together with the hidden-layer dropout built into the joint predictor f_θ (Equation (5.13)). The FA-MIDAS ablation holds the linear baseline of Chapter 4 fixed at its fitted coefficients and trains only the factor network g_0 , which is regularised by weight decay and hidden-layer dropout exactly as in FA-NAM-MIDAS.

5.9.1 Group Dropout

During training, group dropout draws an independent Bernoulli mask $M_i \in \{0, 1\}^K$ with $P(M_{i,k} = 0) = p_{\text{group}}$ for each observation i and each covariate k , and replaces the aggregate logit of Equation (5.1) by

$$\eta_i^{\text{drop}} = \beta_0 + \frac{1}{1 - p_{\text{group}}} \sum_{k=1}^K M_{i,k} f_k(h_{i,k}), \quad (5.16)$$

where the factor $(1 - p_{\text{group}})^{-1}$ preserves the expected logit under the mask. Without this scaling, the random masking would shrink the logit in expectation by a factor of $1 - p_{\text{group}}$, since $E[M_{i,k}] = 1 - p_{\text{group}}$. At evaluation time, the mask is disabled and the logit takes the deterministic form η_i of Equation (5.1). The inverted scaling cancels the shrinkage so that the training-time logit matches this in expectation:

$$E[\eta_i^{\text{drop}}] = \beta_0 + \frac{1}{1 - p_{\text{group}}} \cdot (1 - p_{\text{group}}) \sum_{k=1}^K f_k(h_{i,k}) = \eta_i.$$

The masking introduces stochasticity at the level of entire shape functions: on a given training observation, each f_k is either fully present or fully absent. Two consequences follow. First, each f_k receives gradient signal on roughly a fraction $1 - p_{\text{group}}$ of training observations rather than on all of them, so the optimiser cannot route the entire logit through one dominant covariate while leaving the remaining subnetworks underfitted: when the dominant covariate is masked, the loss on that observation must be reduced using the surviving f_j , which forces gradient flow into the remaining subnetworks. Second, the model implicitly trains over the 2^K subsets of active covariates and is evaluated on the full set, which acts as an ensemble averaging across covariate subsets and encourages each f_k to contribute a signal that remains useful in combination with arbitrary subsets of the other shape functions. The probability p_{group} is treated as a hyperparameter, with the search grid specified in Section 6.2.3.

5.9.2 Additive Structured Penalty

The additive structured penalty places an ℓ_1 and an ℓ_2 norm on the first-layer weight matrix $V_k^{(1)} \in \mathbb{R}^{H_{\text{pred}} \times L}$ of each predictor subnetwork f_k , summed across the K covariates as in Equation (5.15). The restriction to the first layer is deliberate: in the feedforward map of Equation (5.8), $V_k^{(1)}$ is the only matrix that carries the identity of the input covariate, since all subsequent layers operate on the already mixed hidden representation. Shrinking $V_k^{(1)}$ therefore controls how strongly covariate k is allowed to enter the shape function, which is the quantity the baseline sg-LASSO penalty of Equation (4.5) also controls through the group norm $\|\beta_k\|_2$ on the linear coefficients.

Writing out the penalty component of Equation (5.15),

$$\mathcal{R}_{\text{struct}}(\Phi) = \lambda_1 \sum_{k=1}^K \sum_{j,\ell} |V_{k,j\ell}^{(1)}| + \lambda_2 \sum_{k=1}^K \left(\sum_{j,\ell} (V_{k,j\ell}^{(1)})^2 \right)^{1/2}, \quad (5.17)$$

the ℓ_1 term sums the absolute values of every entry of every first-layer matrix, while the ℓ_2 term takes a per-covariate ℓ_2 norm over all entries of $V_k^{(1)}$ and sums these across k , reproducing at the level of weight matrices the sparse-group geometry of the baseline penalty

(Equation (4.5)). Setting $\lambda_1 = 0$ leaves Equation (5.17) as a pure group penalty on $\{V_k^{(1)}\}_{k=1}^K$ with the same group-sparsity behaviour as the sg-LASSO group term, which drives entire matrices $V_k^{(1)}$ to zero rather than producing element-wise shrinkage. The combination (λ_1, λ_2) is treated as a two-dimensional hyperparameter pair, with the search grid (specified in Section 6.2.3) including zero on each axis so that pure ℓ_1 , pure group ℓ_2 , and their combinations are all covered as special cases.

Two features of Equation (5.17) are worth noting. First, the penalty is smooth almost everywhere but nondifferentiable at zero for both norms. Adam (Section 5.9.4) responds to this kink by taking smooth gradient steps; it has no threshold rule that would set weights below a cutoff to exactly zero, so exact sparsity at the weight level is not produced. Weights are instead pushed toward zero by the magnitude of the gradient near the origin. The group-level sparsity behaviour of the baseline is therefore approximated rather than reproduced exactly. Second, deeper layers of each f_k are not structured-penalised; they are instead regularised by the weight decay of Section 5.9.4, which reflects the interpretation that first-layer weights carry covariate identity while deeper layers carry only the shape of the shared nonlinearity.

5.9.3 Output Penalty

The output penalty shrinks the magnitude of the shape-function outputs $f_k(h_{i,k})$ directly rather than indirectly through the weights that produce them. Its form, already displayed in Equation (5.15), is the empirical second moment of the stacked outputs averaged over observations and covariates,

$$\mathcal{R}_{\text{out}}(\Phi) = \lambda_{\text{out}} \frac{1}{K} \sum_{k=1}^K \frac{1}{N} \sum_{i=1}^N f_k(h_{i,k})^2. \quad (5.18)$$

This expression is computed by passing the training observations through the network at each optimiser step and averaging the squared outputs $f_k(h_{i,k})^2$, so unlike the weight-level penalty of Equation (5.17), its value depends on the training inputs as well as on the parameters.

The output penalty addresses a specific problem of the additive specification: two shape functions can cancel one another to produce a small aggregate logit while each taking on arbitrarily large magnitudes in opposite directions, and the weight-level penalties of Equation (5.17) and Section 5.9.4 do not rule out this configuration if the contributions happen to cancel across the support of $h_{i,k}$. The output penalty removes this possibility by penalising each f_k on its own empirical scale before the additive aggregation, which has the further effect of possibly stabilising the shape functions displayed in Section 6.3.3: without Equation (5.18), the ensemble averaging of Section 5.10.2 produces shape functions whose variance at any

fixed input grows across restarts with the magnitude of the individual f_k , and Equation (5.18) directly caps that magnitude. The search grid for λ_{out} (specified in Section 6.2.3) contains zero, so the penalty can be switched off entirely if the additive cancellation problem is not active for a given horizon, with the random search picking $\lambda_{\text{out}} = 0$ in that case.

5.9.4 Weight Decay

The weight decay term applies a single ℓ_2 coefficient λ_{wd} to every trainable parameter in Φ through the Adam update rule. For each parameter w with loss-gradient g_t at optimiser step t , Adam maintains exponential moving averages of the gradient and its squared values,

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (5.19)$$

and updates the parameter by the bias-corrected first moment divided by the square root of the bias-corrected second moment,

$$w_{t+1} = w_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \quad (5.20)$$

with $\hat{m}_t = m_t / (1 - \beta_1^t)$, $\hat{v}_t = v_t / (1 - \beta_2^t)$, learning rate η , and ϵ a small constant for numerical stability. The denominator $\sqrt{\hat{v}_t}$ makes the effective step size adaptive: parameters with historically small gradients receive larger steps. The implementation uses decoupled weight decay: rather than adding an ℓ_2 term to the gradient g_t before Equation (5.20), the optimiser multiplies each parameter by $(1 - \eta \lambda_{\text{wd}})$ at every step, modifying Equation (5.20) to

$$w_{t+1} = (1 - \eta \lambda_{\text{wd}}) w_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}. \quad (5.21)$$

The decoupled form is numerically preferable because it keeps the shrinkage strength independent of the per-parameter variance estimates $\hat{v}_t$ that Adam uses. On the aggregation networks of Section 5.3 and the factor network of Section 5.6, which carry no structured penalty, the weight decay is the only parameter-level regularisation in effect, with the factor network additionally applying within-network dropout at each hidden layer. On the predictor subnetworks of Section 5.5, the weight decay acts on all layers jointly, supplementing the first-layer-only penalty of Equation (5.17) with a light background shrinkage on deeper layers.

The search grid for λ_{wd} (specified in Section 6.2.3) is centred on small values relative to the λ_1 and λ_2 ranges, reflecting that the structured and output penalties carry the covariate- and function-level work, leaving weight decay to provide only background ℓ_2 shrinkage on the remaining parameters. The three penalty terms are therefore not interchangeable: replacing

one with a larger value of another produces observably worse validation loss, because each addresses a problem specific to a different layer of the architecture.

5.10 Training Procedure

The training procedure implements the minimisation of the penalised objective Equation (5.15) under the iterated evaluation protocol of Chapter 6. Four design choices, motivated by the IPCW-weighted survival setting, warrant description: a per-lag, per-covariate standardisation applied before the first forward pass, a two-phase ensemble scheme that separates epoch selection from final fitting, a random search over the extended hyperparameter space of the NAM and FA-NAM variants (Section 6.2.3), and a gradient-clipping rule that bounds the update norm under the IPCW weighting. Each choice addresses a concrete failure mode of neural optimisation in this setting, and the subsections below describe them in order.

5.10.1 Covariate Standardisation

Before any forward pass, each of the $K \cdot d$ raw lag columns in $\{Z_{i,k,j}\}$ is centred and scaled by its training-set mean and standard deviation, computed separately for every (k, j) pair rather than pooled across covariates or across lags. Let $\bar{Z}_{k,j}^{\text{train}}$ and $s_{k,j}^{\text{train}}$ denote the sample mean and standard deviation of the j -th lag of covariate k on the training sample; each observation is then replaced by

$$\tilde{Z}_{i,k,j} = \frac{Z_{i,k,j} - \bar{Z}_{k,j}^{\text{train}}}{\max(s_{k,j}^{\text{train}}, 10^{-8})}, \quad (5.22)$$

with the floor of 10^{-8} in the denominator protecting against columns that are numerically constant on the training sample. The test sample is standardised using the same training-set statistics, not its own, which preserves the boundary between training and evaluation and prevents the test sample from leaking into the standardisation.

The per-lag choice matters specifically for the MIDAS construction. The aggregation networks of Section 5.3 and the fixed Gegenbauer dictionary both operate on the standardised lag vector $\tilde{Z}_{i,k}$ as a whole, but the d lags of a single covariate have very different marginal distributions in practice, with recent lags typically more volatile than distant ones. Pooling across lags would mix these scales, blurring the contrast that the aggregation networks and the Gegenbauer dictionary act on. The per-lag rescaling therefore serves the MIDAS structure rather than the neural network, and is applied uniformly to the baseline and to the neural variants of Table 5.1 so the comparison across variants is not affected by the preprocessing step.

5.10.2 Two-Phase Ensemble Training

For each combination of hyperparameters and each train/test iteration, the model is fitted in two phases and the procedure is replicated n_{restarts} times with independent random initialisations and independent validation splits. The predictions from the restarts are averaged to form the ensemble prediction used in the final evaluation.

The first phase performs a stratified partition of the training sample into an internal training fold of 85% and an internal validation fold of 15%, stratified on the outcome so that defaults and non-defaults appear in both folds in proportion to their population frequency. The stratification is essential given the low default rate of the sample: a simple random split with fraction 0.15 would leave some folds with too few positives to yield a stable IPCW-weighted validation loss. The model is trained on the internal training fold using the Adam optimiser applied to Equation (5.15), with the IPCW-weighted validation loss (Equation (5.14)) evaluated on the internal validation fold at every epoch. Training continues until the validation loss has not improved for a number of epochs equal to a patience hyperparameter (search grid in Section 6.2.3), at which point the epoch index at which the validation loss attained its minimum is recorded as the optimal stopping time E^* .

The second phase retrains the same architecture on the full training sample for exactly E^* epochs, without early stopping and without any held-out fold. This phase is the fit that produces the predictions used on the test set. The two-phase scheme recovers, at the cost of an extra optimisation pass, the 15% of training observations otherwise lost to the internal validation split: the first phase estimates the generalisation-optimal stopping time, and the second phase fits the parameters at that stopping time on the full sample. The retrain matters because the IPCW weighting concentrates information in the small number of observed defaults, and discarding a stratified 15% of those positives would noticeably degrade the fit on the full training distribution.

The restart dimension averages over two sources of variation: the random weight initialisation of the aggregation, predictor, and factor subnetworks, and the random stratified split drawn at the start of each first phase. The averaged prediction is

$$\hat{\pi}_i = \frac{1}{n_{\text{restarts}}} \sum_{b=1}^{n_{\text{restarts}}} \sigma(\eta_i^{(b)}(\hat{\Phi}^{(b)})), \quad (5.23)$$

with $\hat{\Phi}^{(b)}$ the full parameter vector fitted in restart b and σ the sigmoid link. The ensemble size is subject to a trade-off between prediction variance and compute cost; the specific value used in this thesis is given in Section 6.2.3.

5.10.3 Hyperparameter Selection via Random Search

Hyperparameter selection uses random search rather than exhaustive grid search; see [Bergstra and Bengio \(2012\)](#) for the standard efficiency argument. The search grid for each variant, the number of sampled configurations, the cross-validation protocol, and the selection criterion are specified in Sections 6.2.3 and 6.2.4.

Two methodological features merit comment. First, the MIDAS output dimension L is searched jointly with the other hyperparameters, and the Gegenbauer dictionary W is regenerated at every draw to match the sampled L ; this differs from the baseline, in which L is fixed, and reflects the fact that the aggregation networks of Section 5.3 can exploit a richer or coarser MIDAS target basis depending on the effective signal dimensionality of each covariate. Second, the FA-NAM-MIDAS search is conducted separately from the NAM-MIDAS search. The optimal configurations need not coincide on the shared hyperparameters, since the presence of the factor block reduces the predictive load on the per-covariate shape functions and may therefore shift the optimum on H_{pred} and on the output-regularisation strength. A separate search ensures that each variant is tuned to its own optimum rather than being constrained to share the other's configuration.

5.10.4 Gradient Clipping

At every optimiser step, the global ℓ_2 norm of the gradient of Equation (5.15) with respect to the parameter vector Φ is bounded above by a threshold c_{clip} treated as a hyperparameter (search grid in Section 6.2.3). Denoting by $g_t = \nabla_{\Phi} \mathcal{L}_{\text{tot}}(\Phi_t)$ the raw gradient at step t , the clipped gradient used in the Adam update is

$$\tilde{g}_t = g_t \cdot \min\left(1, \frac{c_{\text{clip}}}{\|g_t\|_2}\right), \quad (5.24)$$

so that $\|\tilde{g}_t\|_2 \leq c_{\text{clip}}$ uniformly across steps. When the raw norm is below the threshold, Equation (5.24) returns g_t unchanged; when the raw norm exceeds the threshold, the gradient is rescaled to the threshold while preserving its direction.

Gradient clipping is a standard stability technique in neural training, but its role here is structural: it responds directly to the IPCW weighting of Equation (5.14), where a small number of observations can carry disproportionately large weights. The weight $w_i = \delta_i(t) \mathbf{1}\{\tilde{T}_i \leq t\} / \hat{H}(t \wedge \tilde{T}_i)$ is bounded below by $1 / \hat{H}(t)$ for defaulters that have not been censored, and unbounded above in finite samples when $\hat{H}$ approaches zero for observations with long follow-up. The per-observation contribution to g_t is therefore heavy-tailed even when the raw residual of the logistic loss is well behaved, and the heavy-tailedness concentrates at precisely the observations whose signal the IPCW correction most needs to preserve. Without

clipping, a single extreme weight can drive the full-batch gradient in a direction dictated by one observation, and the Adam moment estimates accumulate the effect of that spike over subsequent steps through their exponentially weighted averages. Equation (5.24) bounds the per-step contribution of any one observation without discarding it, which preserves the consistency of the IPCW estimator: the empirical risk is still computed with the full weight w_i , and clipping only rescales the optimiser's per-step movement, not the loss surface itself, so the set of stationary points of the empirical risk is preserved and the IPCW-consistent target of the optimisation is unaffected.

5.11 Identifiability and the Interpretation of Shape Functions

This section formalises a structural feature of the additive specification: the individual shape functions f_k are not uniquely identified by the data, even though their sum is. Three distinct sources of non-identification jointly determine how the shape-function plots of Section 6.3.3 should be read: a level-shift transformation, a slope-redistribution transformation between correlated covariates, and a within-covariate collinearity ambiguity for multi-input shape functions.

The additive decomposition of Equation (5.1) identifies the aggregate logit $\sum_k f_k(h_{i,k})$ but not the individual components f_k . An observation that [Hastie and Tibshirani \(1990\)](#) establish for generalised additive models carries over to the neural case without alteration: the likelihood depends on the f_k only through their sum, so any transformation of the form

$$f_k(h) \mapsto f_k(h) + c_k, \quad \sum_{k=1}^K c_k = 0, \quad (5.25)$$

leaves the logit unchanged for every observation and therefore attains the same value of the IPCW-weighted empirical risk. The intercept β_0 absorbs the aggregate shift in Equation (5.25) and fixes the predictive scale, but how the level is split across covariates is not pinned down by the objective. The practical consequence is that the shape functions f_k are identified only up to *prediction equivalence*, not up to *shape equivalence*, and any interpretation of a single f_k is conditional on the particular additive decomposition returned by the optimisation procedure, not on a population-level function object.

The set of prediction-equivalent solutions described by Equation (5.25) is only the simplest source of non-identification. When f_k is approximately linear in $h_{i,k}$ over large portions of the covariate support, and when two aggregated features $h_{i,k}$ and $h_{i,k'}$ happen to be near-collinear on the training distribution, additional transformations of the form

$$f_k(h) \mapsto f_k(h) + \alpha h, \quad f_{k'}(h) \mapsto f_{k'}(h) - \alpha h \quad (5.26)$$

shift slope between the two covariates while leaving the fitted logit numerically unchanged on the training sample. These transformations characterise a family of numerically equivalent solutions; they are not applied during training but describe the directions along which different random initialisations are free to drift without paying a cost in training loss. Such near equivalences are not exact, but on finite samples the gradient of the objective at the fitted parameters is small along these near-invariant directions, so different random initialisations reach solutions that differ along them. The stacked shape-function output $\sum_k f_k$ is consequently well behaved across restarts, while the individual f_k are not.

A third source of non-identification, specific to multi-input shape functions ($L > 1$), arises within a single covariate. When the components $h_{k,1}, \dots, h_{k,L}$ are themselves near-collinear on the training distribution, as documented for the FA-NAM-MIDAS variant in Section 6.3.3, the function f_k is constrained by the data only along the dominant eigendirection of the scatter $(h_{i,k,1}, \dots, h_{i,k,L})$; behaviour along the orthogonal directions is essentially unconstrained and can drift freely between iterations without affecting the prediction. This is a within-covariate analogue of Equation (5.26) and is not present in classical generalised additive models, where each shape function f_k maps a real-valued covariate $X_{i,k} \in \mathbb{R}$ to a real-valued contribution, leaving no internal coordinates that could be collinear.

Three consequences organise how the shape functions of Section 5.5.2 should be read. First, a single f_k from a single restart is one of many equivalent solutions, not an estimate of a population quantity, and the ensemble averaging of Section 5.10.2 averages across members of that family rather than reducing a sampling variance. Second, the output penalty of Section 5.9.3 partially resolves the level shift in Equation (5.25) by shrinking each f_k toward zero on the training empirical measure; the returned $\{f_k\}$ are therefore the smallest-norm members of this set, but the set of equivalent solutions itself is unchanged. Third, the cross-restart instability of shape functions grows with both the collinearity of the aggregated features $h_{i,k}$ and the approximate linearity of the fitted f_k ; when the predictor is close to the linear baseline, the ensemble spread of each f_k is therefore expected to be large even when the aggregate prediction is stable.

Chapter 6

Empirical Application

This chapter applies the NAM-MIDAS family developed in Chapter 5 to a panel of Chinese manufacturing firms, the same dataset used by Miao et al. (2025) for the sg-LASSO-MIDAS baseline. The neural variants are evaluated on predictive performance against the baseline and on the interpretive outputs the architectural extensions are designed to deliver. The headline result is that the neural variants achieve at most parity with the baseline at the shortest horizon and lag at longer horizons, with generally higher iteration variance. The diagnostic analyses produce a mixed picture: the learned mixing weights γ_k provide a per-covariate verdict on whether the Gegenbauer dictionary is sufficient or whether the data prefer a neural aggregator, and the PCA factor block decomposes cross-covariate dependence into a small set of latent factors, while the per-covariate shape functions that motivated the NAM architecture vary substantially across iterations at this sample size.

6.1 Data

The empirical analysis is based on a panel of Chinese manufacturing firms listed on the Shanghai and Shenzhen stock exchanges, with IPO dates spanning 2002 to 2014 and follow-up through December 31, 2020. The dataset is the same as that used by Miao et al. (2025) and is sourced from the iFinD database (Hithink RoyalFlush Information Network, 2025). It contains $N = 901$ firms.

6.1.1 Distress Event and Censoring

The event of interest is corporate financial distress as operationalised through the Special Treatment (ST) designation imposed by the Chinese securities regulator. A firm is designated ST if it meets any of several solvency triggers: two consecutive years of negative earnings, the combination of one recent year of negative earnings with negative equity in the most recent year, an audit opinion expressing substantial doubt or high delisting risk, or other

situations identified by the stock exchange as abnormal activities or high delisting risk (Miao et al., 2025). The failure time T_i is the number of years from the firm's IPO to its first ST designation; firms that are never designated are right-censored at the end of the observation window (Section 3.3).

Of the 901 firms, 67 are designated ST ($\delta_i = 1$) and 834 remain non-distressed ($\delta_i = 0$), giving a censoring rate of approximately 92.6% that makes the IPCW correction of Section 3.5 essential. The empirical Kaplan–Meier survival curve, shown in Figure 6.1, reflects this structure: the survival probability remains above 0.9 for horizons beyond ten years and decays gradually.

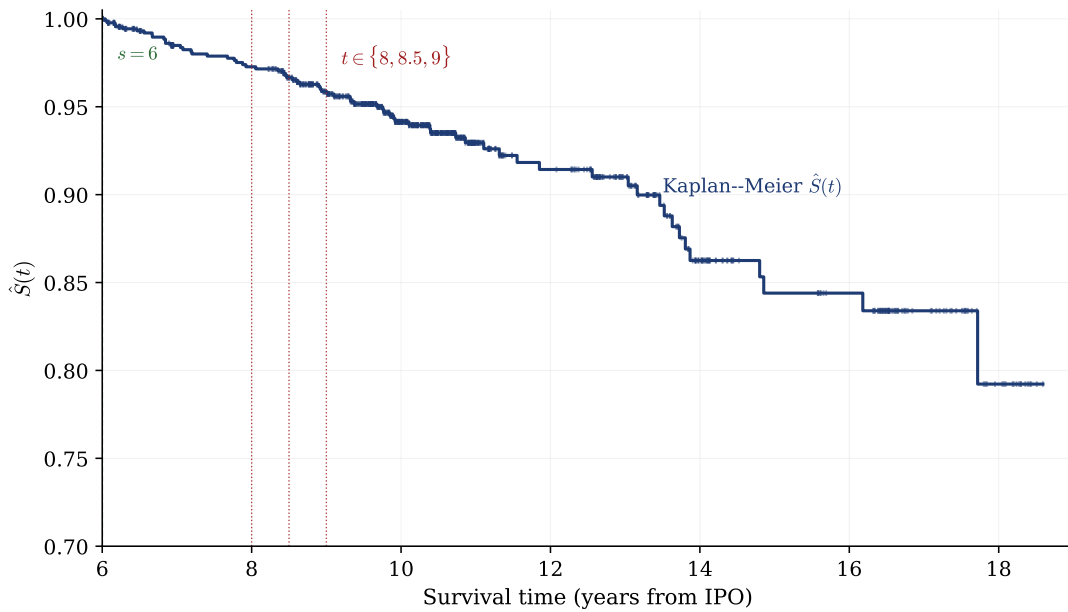


Figure 6.1. Kaplan–Meier estimate of the non-distress survival function $\Pr(T > t)$ for the Chinese manufacturing panel ($N = 901$, 67 ST designations). Vertical lines mark the conditioning year $s = 6$ and the three evaluation horizons $t \in \{8, 8.5, 9\}$ used throughout the empirical application.

6.1.2 Covariates

Each firm is characterised by $K = 32$ quarterly financial covariates. The covariate set spans the standard categories used in financial-distress prediction: profitability ratios, cash-flow ratios, leverage and capital-structure ratios, short-term liquidity ratios, efficiency ratios, and per-share measures. The full list of covariate definitions is provided in Appendix B.

Conditional on a chosen evaluation horizon t and a look-back window of s years, the MIDAS design matrix is constructed by collecting the $m \cdot s - 1 = 4s - 1$ most recent quarterly observations of each covariate for each firm, with the current-period quarter excluded as in Section 3.6.1. With $s = 6$ this corresponds to a $901 \times 23 \times 32$ three-dimensional covariate array, which the sg-LASSO-MIDAS baseline reduces to a $901 \times L \cdot 32$ Gegenbauer-expanded

matrix, and which the NAM variants pass through per-covariate aggregation networks. All 901 firms supply the required $4s - 1$ quarterly observations at each reference quarter, so the full panel enters the estimation sample at every horizon.

6.1.3 Estimation Sample and Horizons

The empirical study evaluates prediction performance at three forecast horizons $t \in \{8, 8.5, 9\}$ years from IPO, with a fixed look-back window of $s = 6$ years. The three horizons are the windows with the largest available event counts in the panel: 23, 28, and 34 firms have entered ST status by $t = 8, 8.5, 9$ respectively. Earlier horizons concentrate too few events for time-dependent AUC estimation to be informative, while horizons beyond nine years fall into the low-density right tail of the failure distribution. At each horizon the full panel of 901 firms is split 80/20 into a training sample of 720 and a test sample of 181 firms, stratified by event status.

More events at longer horizons might be expected to yield better predictive performance, but two structural factors push in the opposite direction. First, the look-back window pulls covariates from increasingly distant past as the horizon lengthens, so the predictive signal in the financial ratios becomes staler relative to the event. Second, by design, the IPCW correction inflates the observation weights $1/\hat{H}(t \wedge \tilde{T}_i)$ as the horizon lengthens, to compensate for censoring bias: $\hat{H}$ shrinks at later times, so a few long-followed firms carry a growing share of the total weight. The estimator's effective sample size therefore declines with t , and its variance rises even as the nominal event count grows.

6.2 Implementation Details

6.2.1 Common Setup

All seven methods are estimated on the same horizon-specific samples and evaluated under the iterated 80/20 train/test protocol with stratified splits and pointwise 95% percentile confidence intervals described in Section 3.7.2. The OIPCW weights are computed from a horizon-specific Kaplan–Meier estimate of the censoring survival function as set out in Section 3.5.

6.2.2 sg-LASSO-MIDAS Baseline

The linear baseline is the sparse-group LASSO MIDAS estimator of Miao et al. (2025), implemented using the `sparsegl` package for the penalised regression step and the `midasml` package for the Gegenbauer dictionary construction. The Gegenbauer dictionary is fixed at $L = 3$ basis functions per covariate, giving the baseline $LK + 1 = 97$ estimated coefficients.

The sparse-group LASSO tuning parameters λ and α are selected jointly by 5-fold stratified cross-validation, with α taken from a grid of $\{0, 0.1, 0.3, 0.5, 0.7, 0.9, 1\}$ controlling the within-group versus between-group sparsity, and λ selected from 200 candidate values between $\lambda_{\max}$, the smallest value for which the penalised optimum is identically zero, and a small fraction of $\lambda_{\max}$. The baseline is refit on the full training sample at the cross-validated (λ, α) pair before evaluation.

6.2.3 NAM-MIDAS Variants

The three NAM-MIDAS variants, Ablation, Full, and Residual, share a common R implementation based on the `torch` package and a common predictor architecture (Section 5.5), differing only in the per-covariate aggregation: Ablation uses the fixed Gegenbauer dictionary of the baseline, Full uses the learned aggregation networks of Section 5.3, and Residual interpolates via the mixing weights of Section 5.4. See Table 5.1 for the formal summary. Each variant is trained with the IPCW-weighted loss of Section 5.8 and the regularisation machinery of Section 5.9.

The full hyperparameter search grid for the NAM-MIDAS family is given in Table 6.1. An exhaustive search would require $4^9 \times 5^5 = 8.19 \times 10^8$ configurations; hyperparameters are therefore selected by random search over $n_{\text{random}} = 50$ configurations drawn uniformly from the grid, following the protocol of Section 5.10.3.

Each configuration is evaluated by three-fold stratified cross-validation on the training sample, with patience-based early stopping under a hard cap of 300 epochs at the cross-validation stage to bound the training time per configuration. The cross-validation is conducted with the Full NAM-MIDAS variant as the scoring architecture, and the resulting configuration is then reused by the Ablation and Residual variants when their final ensembles are trained, so that all three variants are compared at a common hyperparameter context that isolates the effect of the aggregation choice. The selection criterion is the time-dependent AUC at the iteration's evaluation horizon, computed using the nearest-neighbour estimator (Section 3.7.1) with smoothing bandwidth $0.25 \cdot n_{\text{val}}^{-1/2}$. The mean AUC across the three folds is the per-configuration score, and the configuration attaining the highest mean score is retained. A configuration is discarded if fewer than two folds complete without numerical failure.

After cross-validation, the selected configuration is retrained on the full training sample with $n_{\text{restarts}} = 2$ and the predictions of the two restarts are averaged to form the final ensemble, following the two-phase training procedure of Section 5.10.2.

Table 6.1. Hyperparameter search space for the NAM-MIDAS variants. FA-NAM-specific parameters are below the second rule.

Parameter	Symbol	Search values
Learning rate	η	$\{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$
MIDAS output dim	L	$\{2, 3, 4, 5\}$
Agg hidden dim	H_{agg}	$\{8, 16, 32, 64, 128\}$
Agg layers	n_{agg}	$\{1, 2, 3, 4\}$
Pred hidden dim	H_{pred}	$\{4, 8, 16, 32\}$
Pred layers	n_{pred}	$\{1, 2, 3, 4\}$
ℓ_1 penalty	λ_1	$\{0, 0.001, 0.01, 0.1, 0.5\}$
ℓ_2 penalty	λ_2	$\{0, 0.001, 0.01, 0.1, 0.5\}$
Dropout	p_{drop}	$\{0, 0.1, 0.2, 0.4, 0.5\}$
Group dropout	p_{group}	$\{0, 0.1, 0.2, 0.3, 0.5\}$
Weight decay	λ_{wd}	$\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$
Output penalty	λ_{out}	$\{0, 0.001, 0.01, 0.1\}$
Patience	—	$\{20, 30, 40, 50\}$
Gradient clip	—	$\{0.5, 1.0, 2.0, 5.0\}$
<i>FA-NAM only</i>		
Number of factors	r	$\{2, 3, 4, 5\}$
Factor hidden dim	H_{factor}	$\{8, 16, 32, 64\}$
Factor layers	n_{factor}	$\{1, 2\}$

6.2.4 FA-NAM-MIDAS

The factor-augmented variant extends the Full NAM-MIDAS estimator by introducing a PCA factor block that operates on the Gegenbauer-aggregated representation of the training sample, as specified in Section 5.6. The PCA projection is computed from an eigen decomposition of the training covariance and is held fixed during gradient training; only the downstream factor network g_0 and the per-covariate shape functions f_k are updated.

The factor block adds three hyperparameters: the number of factors r , the factor-network width H_{factor} , and the factor-network depth n_{factor} , all listed in Table 6.1. For the reason given in Section 5.10.3, the FA-NAM random search is conducted separately from the NAM search, with the same $n_{\text{random}} = 50$ configurations and 3-fold stratified cross-validation scheme. All other settings, including standardisation, regularisation, ensembling with $n_{\text{restarts}} = 2$, and early stopping, are inherited from the shared infrastructure.

6.2.5 Deep-MIDAS

The Deep-MIDAS comparison model is the non-additive predictor described in Section 5.7. It consists of a shared neural aggregation network g_ϕ that maps each covariate's standardised lag vector $\tilde{Z}_{i,k}$ to a common L -dimensional feature representation $h_{i,k}$, followed by a single feedforward predictor f_θ on the stacked features that outputs the logit. The hyperparameter

grid and training infrastructure are otherwise held identical to the NAM-MIDAS family, with the MIDAS output dimension fixed at $L = 3$ rather than searched, so that differences in test performance can be attributed to the change in architectural form rather than to differences in the estimation protocol.

6.2.6 FA-MIDAS

The FA-MIDAS ablation keeps the sg-LASSO-MIDAS baseline fixed as an offset and trains only the PCA factor block g_0 of Section 5.6. The linear part is locked at the baseline’s $L = 3$, so the cross-validation search covers only the number of factors r , the factor-network width and depth, and the standard regularisation hyperparameters; all other settings are inherited from the shared infrastructure.

6.3 Results

6.3.1 Prediction Performance

Table 6.2. Distress prediction performance. Reported values are the average out-of-sample time-dependent AUC across ten stratified 80/20 splits; the iteration standard deviation is shown in parentheses. Pointwise 95% confidence intervals, computed as the 2.5th and 97.5th percentiles of the iteration-level AUCs, are reported below each mean.

Model	Prediction horizon t		
	$t = 8$	$t = 8.5$	$t = 9$
sg-LASSO-MIDAS	0.818 (0.069) [0.775, 0.861]	0.812 (0.053) [0.779, 0.845]	0.805 (0.067) [0.764, 0.846]
NAM-MIDAS (ablation)	0.715 (0.117) [0.642, 0.787]	0.657 (0.233) [0.513, 0.802]	0.565 (0.205) [0.437, 0.692]
NAM-MIDAS (full)	0.823 (0.074) [0.777, 0.868]	0.792 (0.107) [0.725, 0.858]	0.728 (0.117) [0.656, 0.801]
NAM-MIDAS (residual)	0.794 (0.145) [0.705, 0.884]	0.725 (0.123) [0.649, 0.801]	0.706 (0.100) [0.644, 0.768]
FA-NAM-MIDAS	0.818 (0.082) [0.767, 0.869]	0.793 (0.078) [0.745, 0.842]	0.792 (0.091) [0.736, 0.848]
FA-NAM-MIDAS ($L = 3$)	0.737 (0.122) [0.704, 0.784]	0.807 (0.089) [0.770, 0.843]	0.763 (0.084) [0.731, 0.799]
Deep-MIDAS	0.759 (0.101) [0.718, 0.807]	0.810 (0.074) [0.777, 0.845]	0.750 (0.110) [0.719, 0.787]
FA-MIDAS	0.780 (0.075) [0.741, 0.823]	0.805 (0.065) [0.767, 0.848]	0.793 (0.066) [0.762, 0.837]

Table 6.2 reports the headline prediction performance of the baseline and the neural

variants at the three evaluation horizons. Two patterns dominate the comparison of the baseline with the four NAM-family variants and organise the subsequent discussion.

First, on point estimates, the sg-LASSO-MIDAS baseline is never dominated by the neural variants. At the shortest horizon, $t = 8$, the Full NAM-MIDAS and FA-NAM-MIDAS estimators are statistically indistinguishable from the baseline, with mean AUCs of 0.823 and 0.818 against a baseline of 0.818. These differences are small relative to the iteration-level standard deviations and well within the 95% confidence intervals. At $t = 8.5$ and $t = 9$, however, the baseline preserves its point-estimate lead across all NAM variants. FA-NAM-MIDAS is the closest competitor at $t = 9$ with a mean AUC of 0.792 against the baseline's 0.805, and its confidence interval $[0.736, 0.848]$ overlaps the baseline's $[0.764, 0.846]$. The Ablation variant underperforms dramatically, with mean AUC collapsing to 0.565 at $t = 9$.

Second, the neural variants carry a substantially larger iteration-level variance than the baseline. Excluding the Ablation variant, the NAM-family standard deviation ratio ranges from 1.07 (Full NAM at $t = 8$) to 2.32 (Residual at $t = 8.5$). FA-NAM-MIDAS reduces this spread (ratio 1.19 at $t = 8$, 1.47 at $t = 8.5$, 1.36 at $t = 9$) but does not eliminate it, consistent with the pooling interpretation of the factor block (Section 5.6.1): by summarising the covariates into a small number of common factors, it averages out some of the per-covariate noise that drives the other NAM variants' volatility. The reduction is clearest against Ablation and Residual; against Full NAM it holds only at the longer horizons, where FA-NAM is tighter than Full NAM at $t = 8.5$ and $t = 9$ but slightly wider at $t = 8$. For some variants, a few iterations drive the high variance: AUCs of 0.182 and 0.257 for the Ablation variant at $t = 8.5$ (iteration 3) and $t = 9$ (iteration 5), both well below random, and 0.467 for the Residual variant at $t = 8$ (iteration 2); Table A.4 reports the full per-iteration AUC matrix across variants and horizons. The pattern is informative: the worst collapses occur in the Ablation variant and to a lesser extent in the Residual variant, whose aggregation remains partly tied to the Gegenbauer dictionary. The Full and FA-NAM-MIDAS variants, with fully learned aggregation, do not produce comparable collapses on the same iterations, indicating an architecture-specific failure rather than a data-specific one. The two-phase ensemble with $n_{\text{restarts}} = 2$ partially mitigates these collapses but does not eliminate them.

Together, these two observations show that the NAM family does not provide a statistically meaningful improvement in mean predictive performance over the sg-LASSO-MIDAS baseline at any of the three horizons, and pays a 7–132% variance premium relative to the linear model. The pattern of no AUC improvement combined with higher variance suggests that the additional flexibility of f_k over β_k fits noise rather than adding signal.

The joint-predictor variant Deep-MIDAS of Section 5.7 replaces the K separate aggregators and predictors of the additive NAM family with a single shared aggregator g_ϕ and a joint multi-layer predictor on the concatenated outputs, dropping the per-covariate decomposition

that underwrites the interpretability claim of FA-NAM-MIDAS. Trained under the same protocol as the $L = 3$ variant of FA-NAM-MIDAS, Deep-MIDAS attains mean AUCs of 0.759 (0.101), 0.810 (0.074) and 0.750 (0.110) at $t = 8$, $t = 8.5$ and $t = 9$, with 95% confidence intervals $[0.718, 0.807]$, $[0.777, 0.845]$ and $[0.719, 0.787]$ respectively. The architectural simplification therefore does not recover the predictive performance lost by abandoning the linear specification, and the case for choosing Deep-MIDAS over FA-NAM-MIDAS on this sample is empirically weak.

The factor-augmented linear variant FA-MIDAS of Section 5.6 keeps the sg-LASSO-MIDAS baseline as a fixed offset and adds a trainable factor network g_0 acting on the first r principal components of the Gegenbauer-aggregated features, without the per-covariate NAM-MIDAS predictor that FA-NAM-MIDAS adds. The linear part is locked to the baseline ($L = 3$, matching degree + 1), so only r , the factor-network width and depth, and the standard regularisation hyperparameters are searched in cross-validation. FA-MIDAS attains mean AUCs of 0.780 (0.075), 0.805 (0.065) and 0.793 (0.066) at $t = 8$, $t = 8.5$ and $t = 9$, with 95% confidence intervals $[0.741, 0.823]$, $[0.767, 0.848]$ and $[0.762, 0.837]$ respectively. At every horizon the iteration-level standard deviation is closer to the baseline's than for any NAM-MIDAS variant, consistent with the fact that the linear part is held fixed and only a small low-dimensional g_0 is trained. The mean AUC sits within the baseline's 95% confidence interval at all three horizons, so the factor augmentation does not improve predictive performance, but it also does not pay the volatility premium of the per-covariate neural variants. FA-MIDAS therefore isolates the contribution of the factor block on top of the linear baseline, and the result is that the factor block on its own does not raise mean AUC at any horizon relative to sg-LASSO-MIDAS. This sharpens the interpretation of the FA-NAM-MIDAS row of Table 6.2: the AUC gap between FA-NAM-MIDAS and the bare NAM-MIDAS variants, which is clearest at $t = 9$ where FA-NAM-MIDAS reaches 0.792 against a Full NAM-MIDAS mean of 0.728, cannot be attributed to non-linear signal that the linear baseline is missing and the factor block is recovering. If that were the case, FA-MIDAS would already outperform sg-LASSO-MIDAS, which it does not. The factor block instead helps FA-NAM-MIDAS by absorbing some of the cross-iteration instability that the K per-covariate shape functions introduce, rather than by injecting non-linear information unavailable to the linear baseline. The regularisation of the additive predictor, not the discovery of a new non-linearity, is what the factor block contributes in this sample.

A common feature of the three $L = 3$ variants reported above (FA-NAM-MIDAS ($L = 3$), Deep-MIDAS, FA-MIDAS) is that they all peak at $t = 8.5$ rather than declining monotonically with the horizon, in contrast to the four NAM-family rows whose cross-validation searches over $L \in \{2, 3, 4, 5\}$. The explanation is mechanical: with L locked to a single value, the variant's effective complexity is fixed across horizons, so the chosen L matches the data well at exactly one horizon and is suboptimal at the others. The empirical pattern indicates that

$L = 3$ is close to optimal at $t = 8.5$ but that $t = 8$ and $t = 9$ would have selected a different L (plausibly $L = 2$ or $L = 4$) under variable- L cross-validation. The variable- L rows of Table 6.2 do not show this peak because their CV is free to pick a different L at each horizon. The $L = 3$ lock is retained for the variants used in the shape-stability analysis of Section 6.3.6, where having a common L across iterations is required for orthogonal Procrustes alignment to be meaningful.

6.3.2 Learned Mixing Parameters γ_k

The Residual NAM-MIDAS variant parameterises the aggregation as a per-covariate weighted average of the Gegenbauer linear and neural aggregations, with mixing weight $\gamma_k \in [0, 1]$ (Section 5.4). The empirical question is what values of γ_k the optimisation selects across covariates and horizons.

Across the thirty independent training iterations, the learned γ_k values are concentrated tightly near the Gegenbauer end of the interval. Pooled across covariates and iterations, individual γ_k values lie in $[0.118, 0.211]$ at $t = 8$, in $[0.122, 0.169]$ at $t = 8.5$, and in $[0.120, 0.148]$ at $t = 9$. Averaging first across iterations within each covariate and then across covariates, the mean $\bar{\gamma}_k$ is 0.137, 0.133, and 0.130 at the three horizons respectively. The distribution narrows as the horizon lengthens. At $t = 8$, the distribution of γ_k across covariates and iterations is mildly bimodal: most observations lie in $[0.12, 0.15]$, with a smaller secondary cluster near 0.20–0.21 traceable to a single training iteration in which the optimiser converged to uniformly elevated mixing across all covariates rather than to specific covariates being systematically high. At $t = 8.5$ and $t = 9$ the distribution is unimodal with all observations below 0.17. No covariate at any horizon reaches $\gamma_k = 0.5$, and 95% of covariates lie within 0.02 of the cross-covariate mean at $t = 9$. Figures A.1 and A.2 visualise this distribution across covariates and horizons.

Two observations follow from this distribution. First, the optimisation places roughly 87% of the aggregation weight on the fixed Gegenbauer dictionary and only 13% on the neural aggregation network, *uniformly across covariates and horizons*; the residual mixing settles on a small, roughly constant neural correction rather than discovering covariate-specific structure. This 13% solution is empirically suboptimal, since the Full NAM-MIDAS variant in Table 6.2 attains a higher mean AUC than the Residual variant. At $\gamma_k = 1$ the residual aggregation reduces exactly to the neural aggregation of Full NAM-MIDAS, so the residual architecture has access to that better solution; the gap therefore reflects an optimisation difficulty rather than an architectural constraint, with the joint optimisation of γ_k and the other parameters converging to a local minimum near $\gamma_k = 0.13$ instead of pushing γ_k upward.

Second, the absence of a pronounced cross-sectional tail of high- γ_k covariates is inconsistent with the narrative that some specific financial ratios benefit from neural aggregation

while others remain on the Gegenbauer dictionary. Within the residual architecture, the contribution of the neural aggregation is diffuse and small.

The training protocol itself shapes this result in identifiable ways. The γ_k parameter is initialised at approximately 0.12 following the construction in Section 5.4.1, where the choice was justified by two considerations: it places the model close to the baseline at the start of training so the early gradient signal is informative about departures from the dictionary, and it avoids the vanishing-gradient region of the sigmoid that an initialisation near $\gamma_k = 0$ or $\gamma_k = 1$ would imply. The same choice, however, places a strong prior on the Gegenbauer end of the mixture, and the observed end value of 0.13 represents only a small upward push from this starting point. The protocol provides no separate learning rate for γ_k and no mechanism pushing it toward either 0 or 1, and with one scalar per covariate against the much larger parameter count of the aggregation networks, the gradient signal on γ_k is small. The joint dynamics therefore tend to settle in the gradient-flat interior of $[0, 1]$ rather than committing to either end. A more neutral starting point at $\gamma_k = 0.5$ would likely have produced a more indicative reading of where the optimisation converges; the γ_k findings should be read as “no productive lift detected from the initialisation” rather than as “ $\gamma_k = 0.13$ is the data-implied optimum”.

6.3.3 Shape Function Visualisation

The NAM family is designed, as motivated in Sections 5.5 and 5.11, to make visible the per-covariate mapping $f_k: \mathbb{R}^L \rightarrow \mathbb{R}$ from the aggregated features $h_{i,k}$ back to the logit. The shape-scatter plot in Figure A.4 shows the five coordinate projections $(h_{i,k,\ell}, f_k(h_{i,k}))$ of this L -dimensional function (Section 5.5.2) for the six covariates with the highest NAM importance, with iterations pooled across horizons and the raw f_k values reported on the logit scale; columns $\ell = 4, 5$ pool only iterations with $L \geq \ell$. Within each colour, points cluster at iteration-specific levels rather than scatter freely, reflecting the cross-iteration level-shift ambiguity of Equation (5.25), which displaces the entire surface f_k by an iteration-specific constant without changing its shape. Under the additive structure and with adequate data, the resulting curves would be expected to converge to interpretable marginal-effect shapes whose form is stable across training iterations. Two effects temper this expectation at the available sample size. First, each panel is still a one-dimensional projection of a higher-dimensional surface, so part of the visible vertical spread at a given $h_{i,k,\ell}$ reflects variation in the suppressed coordinates rather than estimator noise. Second, the individual shape functions estimated here do not exhibit cross-iteration stability and vary substantially across iterations at any fixed input. An iteration-level scalar summary for the six covariates with the highest cross-horizon NAM importance is reported in Figure A.5.

Across the ten iterations of a fixed horizon, the per-iteration mean of f_k across test obser-

variations varies by approximately 0.54 to 1.15 logit units in range, with iteration-level standard deviations of approximately 0.19 to 0.36 logit units (Figure A.5). These magnitudes are at least comparable to and often substantially larger than the typical mean absolute contribution $\mathbb{E}[|f_k(h_{i,k})|]$ itself, which falls in the range 0.20 to 0.30 logit units. Every (covariate, horizon) cell of Figure A.5 contains iterations whose per-iteration mean is positive and iterations whose per-iteration mean is negative.

This variability cannot be empirically separated at the available sample size into its underlying sources. The additive decomposition with correlated covariates is non-identified up to level-shift (Equation (5.25)) and slope-redistribution (Equation (5.26)) ambiguities, which permit infinitely many equivalent shape configurations under the same prediction rule. On top of this theoretical non-identification, the estimator is statistically unstable across the random 80% training subsamples used in each iteration, and stochastic optimisation introduces additional iteration-to-iteration noise. The remainder of the chapter therefore relies on aggregate diagnostics that are invariant under the identifiability ambiguities, in particular the mean absolute contribution $\mathbb{E}[|f_k(h_{i,k})|]$ and the relative ranking of covariates by it.

Despite the cross-iteration instability documented above, the shape function estimated within a single iteration carries an internally coherent gradient that can be read directly from the data when the input dimension L is small. Such within-iteration readings characterise the fit on a given fold rather than a population-level economic effect, but they illustrate the kind of nonlinear structure that FA-NAM-MIDAS can express.

A natural way to mitigate this instability without abandoning the shape-function visualisation would be to average f_k across iterations that share a common L . For variants with a fixed aggregation, such as the Ablation NAM-MIDAS, this average is well defined on a common domain and can be plotted directly. For FA-NAM-MIDAS, the per-iteration aggregator g_k produces iteration-specific feature coordinates, so cross-iteration averaging requires aligning the learned representations to a common reference before pooling, which we leave for future work. We illustrate this with two iterations from FA-NAM-MIDAS, our most consistent variant (Table 6.2): the $L = 2$ iteration at $t = 8$ with the highest FA-NAM-MIDAS AUC of 0.930 on this fold, plotted for $k = 3$ (fixed assets turnover, the covariate with the largest mean absolute contribution at $\mathbb{E}|f_3| = 0.668$); and the $L = 3$ iteration at $t = 8.5$ with the highest $L = 3$ FA-NAM-MIDAS AUC of 0.776, plotted for $k = 15$ (EBIT over operating income). The per-iteration L and AUC values are reported in Tables A.3 and A.4.

The model writes the default-event logit as $\eta_i = \beta_0 + \sum_k f_k(h_{i,k}) + g_0(F_i)$, so $f_k > 0$ raises and $f_k < 0$ lowers the predicted default probability $\sigma(\eta_i)$. A strongly negative f_k at a given $h_{i,k}$ therefore corresponds to a high predicted survival probability $1 - \sigma(\eta_i)$ for that observation.

Unlike the Gegenbauer-anchored coordinates of the baseline, where each $h_{k,\ell}$ inherits a

fixed polynomial-order interpretation as documented in Section 5.5.2, FA-NAM-MIDAS draws its per-covariate features purely from the learned aggregator g_k with no fixed- W anchor, so the L components $h_{k,\ell}$ have no predetermined economic label such as “mean”, “trend”, or “curvature”. Each component must nonetheless represent some aspect of the lag-window profile of covariate k that is informative for the default decision, since otherwise the network has no incentive to allocate capacity to it. We can therefore use the visualisations of the shape function f_k together with the geometry of the $h_{k,\ell}$ scatter to suggest, ex post, what each learned component appears to capture for a given iteration.

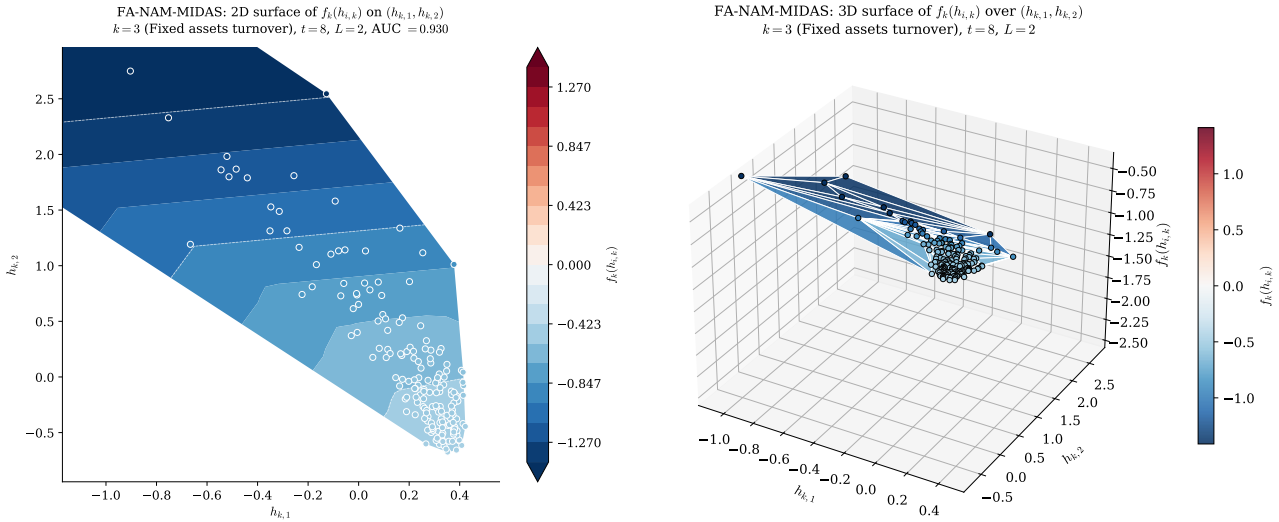


Figure 6.2. Within-iteration shape function f_3 for the fixed assets turnover ratio at $t = 8$, from one FA-NAM-MIDAS iteration with $L = 2$ and $AUC = 0.930$. *Left:* top-down 2D contour. *Right:* same surface as a 3D height field.

Reading Figure 6.2 for $k = 3$, f_3 is uniformly negative across the test sample, so fixed assets turnover acts as a default-suppressing factor for every firm in the sample. The strongest suppression is concentrated at the high- $h_{k,1}$, low- $h_{k,2}$ corner. The iso- f_3 contours are approximately parallel diagonal lines, which means that f_3 depends primarily on a single linear combination of $(h_{k,1}, h_{k,2})$ rather than on each component separately. The two learned components are themselves nearly collinear, with sample correlation $\rho(h_{k,1}, h_{k,2}) = -0.90$, so they act as correlated proxies of a single underlying summary of the turnover profile rather than as two independent features. The shape function f_3 then depends on the joint position along that single direction. We cannot label that summary as “mean” or “trend” *a priori* without the Gegenbauer scaffolding, but the visual gradient is unambiguous on the direction in which the network reads turnover information out of the lag window.

Reading Figure 6.3 for $k = 15$ (EBIT over operating income), f_{15} is again uniformly negative, and the dominant gradient runs along $h_{k,1}$, with the strongest negative contribution at high $h_{k,1}$. The pairwise sample correlations $(\rho_{12}, \rho_{13}, \rho_{23}) = (-0.64, -0.11, +0.84)$ confirm

FA-NAM-MIDAS: 3D scatter of $(h_{k,1}, h_{k,2}, h_{k,3})$ with colour = $f_k(h_{i,k})$
 $k=15$ (EBIT / op. income), $t=8.5$, $L=3$, AUC = 0.776

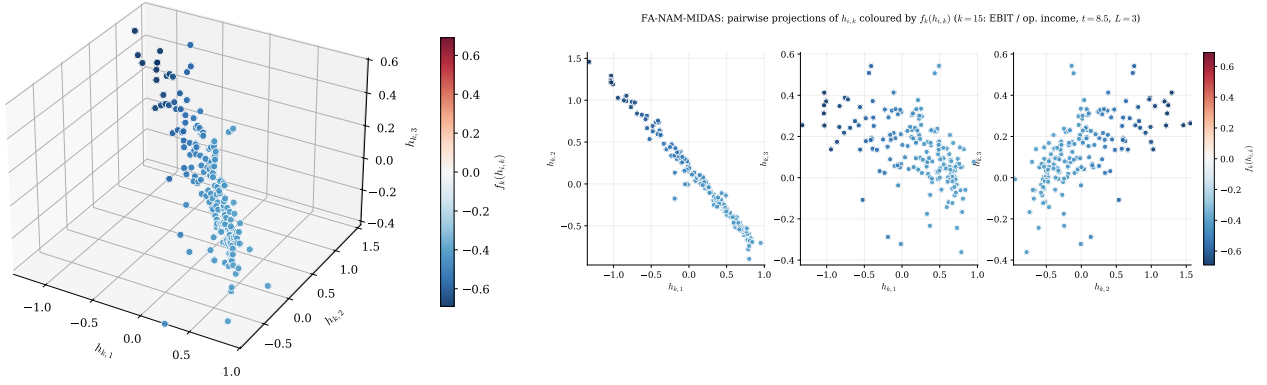


Figure 6.3. Within-iteration shape function f_{15} for the EBIT-over-operating-income ratio at $t = 8.5$, from one FA-NAM-MIDAS iteration with $L = 3$ and AUC = 0.776. *Left:* 3D scatter of test observations in $(h_{k,1}, h_{k,2}, h_{k,3})$ coloured by f_{15} . *Right:* the three pairwise 2D projections.

that the three learned components do not all collapse to a single direction. The $(h_{k,1}, h_{k,2})$ projection shows a clear monotone arc, suggesting that $h_{k,1}$ and $h_{k,2}$ together are tracking a low-dimensional summary of the profitability path. The third component $h_{k,3}$ moves nearly independently of $h_{k,1}$ but is strongly correlated with $h_{k,2}$, and inspection of the $(h_{k,2}, h_{k,3})$ panel shows a curve-like relationship rather than a diagonal scatter. $h_{k,1}$ appears to track the firm's overall EBIT margin level over the lag window, while $h_{k,2}$ and $h_{k,3}$ together appear to encode a non-linear feature of the trajectory's shape, since the network has chosen to spend two of its three components on this refinement rather than collapsing them onto $h_{k,1}$.

The same per-covariate contribution f_k can also be read directly in predicted-probability units. Composing the contributions from all covariates with the bias and the factor block gives the model's predicted default probability for each test observation, $\sigma(\beta_0 + \sum_k f_k(h_{i,k}) + g_0(F_i))$. The shift produced by removing covariate k alone, $\sigma(\eta_i) - \sigma(\eta_i - f_k)$, expresses the same logit contribution on the probability scale. Figure 6.4 plots the two predicted probabilities for the $L = 2$ fixed-assets-turnover iteration: covariate $k = 3$ pulls firms with high $h_{k,1}$ from a default probability as high as forty percent (when k 's contribution is set to zero) down to between one and thirty percent in the full model, with the median absolute shift across all 181 test observations being roughly 0.8 percentage points. The visible asymmetry, in which the shift is largest for firms whose without- k probability sits in the intermediate range, is a direct consequence of the sigmoid link.

Across the full set of FA-NAM-MIDAS iterations, the collinearity observed in the two examples above is the rule rather than the exception: 75.5% of all (iteration, k) pairs have at least one pairwise correlation $|\rho| > 0.9$ between $h_{k,\ell}$ components, with a median across pairs of 0.98. The L -dimensional learned representation therefore often collapses, in practice,

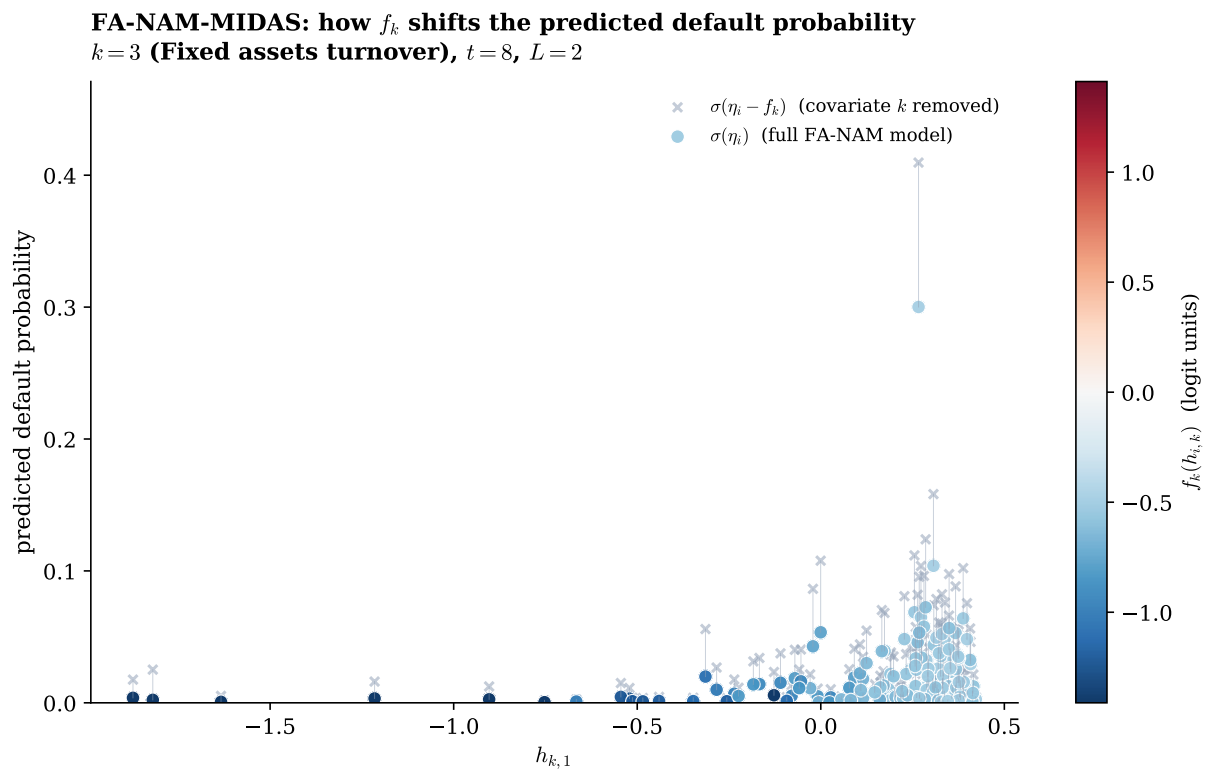


Figure 6.4. Per-observation contribution of covariate $k = 3$ (fixed assets turnover) to the predicted default probability for the FA-NAM-MIDAS iteration of Figure 6.2. Each test observation is shown twice: as a grey cross at the probability when k 's contribution is set to zero, and as a coloured dot at the full-model probability $\sigma(\eta_i)$.

to a one- or two-dimensional functional of the lag profile. This empirical pattern is the within-covariate collinearity flagged in Section 5.11 as a third source of non-identification specific to multi-input shape functions. The collapse is itself informative: the network discovers that a low-dimensional projection is sufficient at the available sample size. Together with the level-shift and slope-redistribution ambiguities of Section 5.11, the within-covariate collinearity means that reading the *level* of f_k at a single iteration remains conditional on the iteration shown, and only the sign of the gradient and the relative ranking of covariates by their mean absolute contribution $\mathbb{E}[|f_k(h_{i,k})|]$ are identified. The top-ranked covariates under this ranking, alongside the variance-weighted PCA loading mass that the factor block of FA-NAM-MIDAS relies on, are reported in Table A.1 of Appendix A.

6.3.4 Factor Analysis

The FA-NAM-MIDAS estimator pre-computes a $KL \times r$ PCA projection P from the Gegenbauer-aggregated training covariates and trains a factor network g_0 on the r -dimensional factor score $F_i = P^\top H_i$, where H_i is the stacked aggregated feature vector defined in Equation (5.10). The number of factors r is selected in cross-validation from $\{2, 3, 4, 5\}$; $r = 3$ is chosen most often (13 of 30 iterations), followed by $r = 2$ (8), $r = 5$ (6), and $r = 4$ (3); see Table A.3 for the full per-iteration listing. Averaged across iterations within each horizon, the per-factor variance shares decay gradually: F_1 captures roughly 14% at all horizons, F_2 about 10%, F_3 around 7%, F_4 declines from about 7% at $t = 8$ to under 6% at $t = 9$, and F_5 approximately 5%. The cumulative share across the five components is 0.444 at $t = 8$, 0.437 at $t = 8.5$, and 0.414 at $t = 9$. No single factor dominates the spectrum: the first factor explains roughly 14% across horizons against 5% for the fifth, a ratio of approximately 3:1. The spectrum is therefore slowly decaying, consistent with the heterogeneous, high-dimensional nature of the Chinese corporate financial panel, in which no small set of latent factors structurally dominates the covariance of the 32 ratios. Table 6.3 reports the per-(covariate, factor) loading magnitudes for a representative iteration with $L = 5$.

Table 6.3. Per-covariate PCA factor-loading magnitudes $M_{k,j} = \sqrt{\sum_{\ell} P_{k\ell,j}^2}$ for a representative iteration at $t = 8.5$ with $L = 5$. Per-factor in-sample variance shares are reported in the bottom row. Loadings below 0.3 are suppressed, and the largest values per factor (top 20%) are highlighted.

Covariate	F1	F2	F3	F4	F5
01 Acc. receivable turnover					
02 Current assets turnover	0.587	0.587	0.709	1.245	0.419
03 Fixed assets turnover		0.311		0.655	0.843
04 Total assets turnover	0.486	0.652	0.651	1.308	0.893
05 Current ratio	0.635	0.520	1.051	0.709	
06 Quick ratio	0.577	0.419	0.879	0.594	
07 Equity ratio	0.618	0.494	0.449		0.357
08 Tangible / liabilities	0.618	0.462	0.964	0.673	
09 Tangible / net debt					
10 EBITDA / liabilities	0.732	0.858	0.724	0.541	
11 Cash-flow / debt	0.330	0.506	0.561		0.563
12 ROA	0.360	0.983	0.706	1.033	
13 Net profit / assets	0.500	0.902	0.643	0.893	
14 Net profit / op. income	0.688	0.749	0.355	0.615	0.379
15 EBIT / op. income	0.629	0.701	0.364	0.653	0.491
16 Working capital / assets	0.699	0.608	0.792	0.301	0.583
17 Retained earnings / assets	0.362	0.797		0.468	
18 EBIT / assets		0.992	0.711	1.046	
19 Market value / liabilities	0.592	0.394	0.877	0.606	
20 Op. income / assets	0.548	0.619	0.579	1.276	0.796
21 Equity / liabilities	0.622	0.466	0.980	0.667	
22 Debt ratio	0.624	0.664	0.591	0.452	0.374
23 Interest-bearing debt ratio	0.433	0.407	0.447		0.550
24 Equity multiplier	0.558	0.473	0.319	0.325	0.429
25 Current / total assets	0.445	0.396	0.420	0.300	0.978
26 Current / total liabilities					0.594
27 Op. cash-flow / share		0.365	0.340		0.776
28 Op. income / share	0.331	0.396	0.526	0.727	0.666
29 Profit before tax / dividend		0.607	0.647	0.704	
30 Net assets / share			0.309		0.404
31 Op. cash-flow / op. income		0.429	0.363	0.375	0.829
32 Net op. cash-flow / op. income		0.430	0.364	0.376	0.829
Variance share	0.147	0.090	0.064	0.054	0.046

Reading Table 6.3, interpretable factor structure emerges. Factor 1, the largest single component at 14.7% of the in-sample variance, loads broadly on capital-structure ratios (EBITDA-over-liabilities (10), equity-over-liabilities (21), debt ratio (22)), short-term liquidity ratios (current ratio (5), working-capital-over-assets (16)), and profit-margin ratios (net-profit-over-operating-income (14), EBIT-over-operating-income (15)); it is best read as a composite financial-health factor that summarises solvency, liquidity, and profitability simultaneously

rather than isolating a single theme. Factor 2 has its largest loadings on the asset-based profitability ratios (return on assets (12), EBIT-over-assets (18), net profit over assets (13), retained earnings over assets (17)), on the income-margin ratios (14, 15), and on EBITDA-over-liabilities (10); it is a comprehensive profitability factor that pools per-asset returns, profit margins, and earnings coverage of liabilities into a single dimension. Factor 3 concentrates on the short-term liquidity ratios (current and quick ratios (5, 6), working-capital-over-assets (16)) and on the long-term solvency ratios (tangible-assets-over-liabilities (8), EBITDA-over-liabilities (10), market-value-over-liabilities (19), equity-over-liabilities (21)); it summarises the firm's overall ability to cover its liabilities and is best read as a creditworthiness factor that combines short-term liquidity with long-term solvency. Factor 4 loads jointly on the asset-turnover ratios (current assets (2) and total assets (4)) and on the asset-based profit ratios (return on assets (12), net profit over assets (13), EBIT-over-assets (18), operating-income-over-assets (20)); it is best read as an asset-utilisation factor in which the per-asset returns are pulled jointly by how much profit each asset generates and how rapidly assets turn over. Factor 5 has its largest loading on the current-to-total-assets ratio (25), with strong contributions from fixed-assets and total-assets turnover (3, 4), operating-income-over-assets (20), and the operating cash-flow ratios per share and per income (27, 31, 32); it captures asset-structure composition together with operating cash-flow conversion, an operating-efficiency factor that mixes how the asset base is allocated with how rapidly it is turned and how much of the resulting income becomes cash.

The trained factor network contributes a significant share of the overall logit signal. Decomposing the logit contribution by its mean absolute value across test observations, the factor component $g_0(F_i)$ carries roughly 55% at $t = 8$, 36% at $t = 8.5$, and 54% at $t = 9$, with the per-covariate terms $\sum_k f_k(h_{i,k})$ carrying the remainder. The factor share is non-monotonic across horizons, peaking at the boundary horizons rather than at the middle. The larger share at $t = 9$ is consistent with the observation that the FA-NAM-MIDAS variance is tighter than the Full NAM-MIDAS variance at $t = 8.5$ and $t = 9$: the pooled factor representation averages out the per-covariate iteration-level noise. Figure A.3 visualises the factor share across horizons.

6.3.5 Interpretation of Selected Hyperparameters

Beyond the headline AUCs, the hyperparameter configurations selected across the random-search iterations themselves carry information about the structure of the estimation problem. Two pools of selections are relevant: thirty NAM-side configurations and thirty FA-NAM-side configurations governing the factor block of FA-NAM-MIDAS. Table 6.4 summarises the most frequent choices in each pool, with the patterns interpreted below; the underlying per-iteration selections are reported in Tables A.2 and A.3.

Table 6.4. Most frequent hyperparameter selections across the random-search iterations. The NAM-HP pool ($n = 30$) governs the Full, Ablation, and Residual variants. The FA-NAM-HP pool ($n = 30$) governs the factor block of FA-NAM-MIDAS. The lag-dictionary size L is reported separately by horizon because preferences differ across both horizon and pool. Highlighted entries mark selections that saturate a grid boundary, identifying candidates for grid expansion.

Hyperparameter	Search grid	Most frequent selection	Frequency
<i>Regularisation (NAM-HP pool, $n = 30$)</i>			
ℓ_1 on weights λ_1	$\{0, 10^{-3}, 10^{-2}, 10^{-1}, 0.5\}$	$\leq 10^{-3}$	18/30
ℓ_2 on weights λ_2	$\{0, 10^{-3}, 10^{-2}, 10^{-1}, 0.5\}$	$\leq 10^{-3}$	21/30
Output penalty	$\{0, 10^{-3}, 10^{-2}, 10^{-1}\}$	$\{10^{-2}, 10^{-1}\}$	20/30
Predictor dropout	$\{0, 0.1, 0.2, 0.4, 0.5\}$	> 0 (most often 0.4)	25/30
Group dropout	$\{0, 0.1, 0.2, 0.3, 0.5\}$	0	14/30
<i>Architecture (NAM-HP pool, $n = 30$)</i>			
Predictor depth n_{pred}	$\{1, 2, 3, 4\}$	1	17/30
Aggregation depth n_{agg}	$\{1, 2, 3, 4\}$	spread (no clear preference)	—
Aggregation width H_{agg}	$\{8, 16, 32, 64, 128\}$	128	12/30
<i>Architecture (FA-NAM-HP pool, $n = 30$)</i>			
Factor count r	$\{2, 3, 4, 5\}$	3	13/30
Factor-net depth n_{factor}	$\{1, 2\}$	1	20/30
<i>Lag-dictionary size L (per horizon, $n = 10$ each)</i>			
NAM-HP, $t = 8$	$\{2, 3, 4, 5\}$	spread, $L \in \{2, 5\}$ tied	3/10 each
NAM-HP, $t = 8.5$	$\{2, 3, 4, 5\}$	$L = 5$	4/10
NAM-HP, $t = 9$	$\{2, 3, 4, 5\}$	$L = 3$	4/10
FA-NAM-HP, $t = 8$	$\{2, 3, 4, 5\}$	$L = 2$	5/10
FA-NAM-HP, $t = 8.5$	$\{2, 3, 4, 5\}$	$L \in \{2, 5\}$ tied	4/10 each
FA-NAM-HP, $t = 9$	$\{2, 3, 4, 5\}$	$L = 4$	5/10

The regularisation profile, summarised in the upper block of Table 6.4, concentrates at the *activation* level rather than at the *weight* level. The per-covariate ℓ_1 and ℓ_2 penalties on the predictor's first-layer weights are selected at or below 10^{-3} in roughly two-thirds of configurations, while the output penalty on the shape-function outputs $f_k(h_{i,k})^2$ is chosen at 10^{-2} or 10^{-1} in twenty of thirty. Predictor dropout is strictly positive in twenty-five of thirty, with 0.4 as the single most common value (ten of thirty), while group dropout, which operates at the covariate-block level, is more often set to zero than to any positive value. The pattern suggests that, with the data standardisation and the modest per-covariate widths already in place, penalising the magnitude of the learned effects is more informative than shrinking the weight matrices directly.

The architecture block of Table 6.4 reveals an asymmetry between the joint predictor and the per-covariate aggregation networks. The predictor depth n_{pred} is set at the lowest grid value of one in seventeen of thirty configurations, and deeper predictors are never strongly favoured, which is consistent with the additive decomposition leaving little useful work for stacked nonlinearities at the joint stage. The aggregation network is, by contrast, generously parameterised: its width H_{agg} takes the largest available value of 128 in twelve of thirty configurations, while its depth n_{agg} shows no clear concentration across the grid $\{1, 2, 3, 4\}$. The useful flexibility of the architecture therefore lives in the per-covariate aggregator, where the model is willing to spend parameters, rather than in the joint predictor, which it keeps shallow. The FA-NAM-MIDAS factor block reinforces this reading: the factor count r is selected at the small value of three in thirteen of thirty configurations, and the factor-network depth at one in twenty of thirty.

Horizon-specific preferences for the lag-dictionary size L differ between the two pools. The NAM-HP selections most often choose $L = 5$ at $t = 8.5$ but drop to $L = 3$ at $t = 9$. The FA-NAM-HP selections show a more striking pattern: at $t = 8.5$ the distribution is bimodal, with $L = 2$ and $L = 5$ each chosen in four of ten iterations, indicating two competing dictionary sizes that achieve comparable cross-validation AUC under the factor-augmented architecture. At $t = 9$ the distribution concentrates, with $L = 4$ leading in five of ten iterations and combining with a small factor count $r \in \{2, 3\}$ in nine of ten cases. The two pools therefore allocate temporal flexibility differently: the NAM-HP selection peaks in dictionary richness at the middle horizon, whereas the factor-augmented variant settles on a richer dictionary at the long horizon, where the added factor block appears to support rather than substitute for a richer lag basis.

Several entries in Table 6.4 saturate the upper end of their search grid and identify candidates for grid expansion. The aggregation width H_{agg} takes its grid maximum of 128 in twelve of thirty configurations; the output penalty its upper two grid values (10^{-2} and 10^{-1}) in twenty of thirty; and the NAM-HP L its grid maximum of five at $t = 8.5$, with

FA-NAM-HP also tying at $L = 5$ at the same horizon. Extending these search ranges upward would clarify whether the apparent preferences reflect a true optimum or a censored grid.

These patterns have two implications. First, they provide a validation of the regularisation design introduced in Section 5.9: the output penalty and dropout carry most of the regularisation load, while ℓ_1 and ℓ_2 on the weights act mainly as fallbacks. Second, the tendency of the random search to select shallow joint predictors with substantial output-level shrinkage is consistent with a flat region of the loss landscape near the chosen configurations, which motivates the two-phase ensemble training strategy described in Section 5.10.

6.3.6 Shape Stability via Procrustes Alignment

The shape figures of Section 6.3.3 are extracted from one fitted iteration of FA-NAM-MIDAS, and reading them across iterations at face value is complicated by an exact symmetry of the per-covariate term in the loss. For any orthogonal $Q \in O(L)$, replacing g_k with $Q g_k$ and f_k with $f_k \circ Q^\top$ gives

$$(f_k \circ Q^\top)(Q g_k(z_{i,k})) = f_k(Q^\top Q g_k(z_{i,k})) = f_k(g_k(z_{i,k})),$$

so the per-covariate contribution is unchanged. The IPCW-weighted log-loss of Section 5.8 is thus invariant under any L -dimensional rotation of g_k paired with the inverse rotation of f_k . Random initialisation selects one element of this equivalence class for each iteration, so two iterations that fit the same per-covariate contribution can present it in arbitrarily rotated parameterisations of the L -dimensional internal feature space. Direct visual comparison of f_k across iterations therefore confounds genuine disagreement about the underlying relationship with mere rotation of the coordinate system.

This rotational ambiguity is a structural, within-covariate source of non-identification, distinct from the two between-covariate sources catalogued in Section 5.11: the level-shift ambiguity of Equation (5.25) redistributes constants across the K shape functions while preserving their sum, and the slope-redistribution ambiguity of Equation (5.26) transfers an approximately linear contribution between two near-collinear covariates. Both operate *between* covariates and leave each f_k 's L -dimensional input geometry fixed. The rotation $(g_k, f_k) \mapsto (Q g_k, f_k \circ Q^\top)$ instead operates *within* a single covariate's L -dimensional internal feature space. It is the structural counterpart of the within-covariate collinearity flagged in Section 5.11: that ambiguity is data-dependent, arising on finite samples where the columns of H_k are near-collinear and f_k becomes essentially unconstrained along the low-variance directions, whereas the rotation symmetry is exact for any empirical distribution of H_k , regardless of whether its columns are well-spread or near-collinear, since it follows from the parameterisation of the per-covariate composition $f_k \circ g_k$ and not from the data. Procrustes

alignment removes only this rotational source; the level-shift and slope-redistribution ambiguities of Equations (5.25) and (5.26) remain present in the population-aligned shapes $\bar{f}_k$ reported below.

To check whether the underlying shape functions agree across iterations once the rotational ambiguity is removed, we apply orthogonal Procrustes alignment.

To produce comparable feature spaces across iterations we re-fit FA-NAM-MIDAS with $L = 3$ held constant, save the trained aggregation and predictor weights for every restart, and evaluate each $g_k^{(b)}$ on the full panel of $n = 901$ standardised lag matrices to obtain $H_k^{(b)} \in \mathbb{R}^{n \times L}$. We designate the iteration b^* with the highest cross-validated AUC as reference and, for each pair (b, k) , find the rotation

$$Q_k^{(b)} = \arg \min_{Q \in O(L)} \|H_k^{(b)} Q - H_k^{(b^*)}\|_F = U_{k,b} V_{k,b}^\top,$$

where $U_{k,b} S_{k,b} V_{k,b}^\top = (H_k^{(b)})^\top H_k^{(b^*)}$ is the singular value decomposition; the closed-form solution to the orthogonal Procrustes problem in this form is due to Schönemann (1966), with the construction of the SVD and the derivation of the rotation reviewed in Appendix C. The rotated shape function $\tilde{f}_k^{(b)}(h) = f_k^{(b)}((Q_k^{(b)})^\top h)$ is then comparable across b , and we report the population average $\bar{f}_k(h) = B^{-1} \sum_b \tilde{f}_k^{(b)}(h)$ together with the restart standard deviation. To summarise stability per covariate we project the reference-space points onto the linear direction along which $\bar{f}_k$ varies most across the panel, sample $\bar{f}_k$ on $n_g = 200$ grid points along that axis, and compute the shape signal-to-noise ratio

$$\text{SNR}_k = \frac{\max \bar{f}_k - \min \bar{f}_k}{\frac{1}{n_g} \sum_g 2 \text{sd}_b[\tilde{f}_k^{(b)}(h_g)]}.$$

The numerator measures the amplitude of the population shape; the denominator measures the typical restart spread along that axis. $\text{SNR}_k > 1$ would indicate that the population shape rises above the restart noise; $\text{SNR}_k \ll 1$ that the restart cloud absorbs the curve.

Figure 6.5 reports the six highest- SNR_k covariates at $t = 8.5$. Even for the top covariate the SNR is only 0.05, meaning the amplitude of $\bar{f}_k$ is twenty times smaller than the typical restart spread, and the remaining 26 covariates all sit at $\text{SNR}_k \leq 0.02$. The same pattern holds at $t = 8$ (best $\text{SNR}_k = 0.04$ for $k = 28$) and $t = 9$ (best $\text{SNR}_k = 0.01$ for $k = 22$). After orthogonal alignment removes the rotational ambiguity of g_k , the population per-covariate FA-NAM-MIDAS shape functions are statistically indistinguishable from a horizontal line.

Two consequences follow. First, the single-iteration shapes of Figures 6.2 and 6.3 are honest descriptions of one fitted model, not of *the* relationship between covariate k and default risk. Re-running the pipeline rotates each $g_k^{(b)}$ into a different parameterisation, and

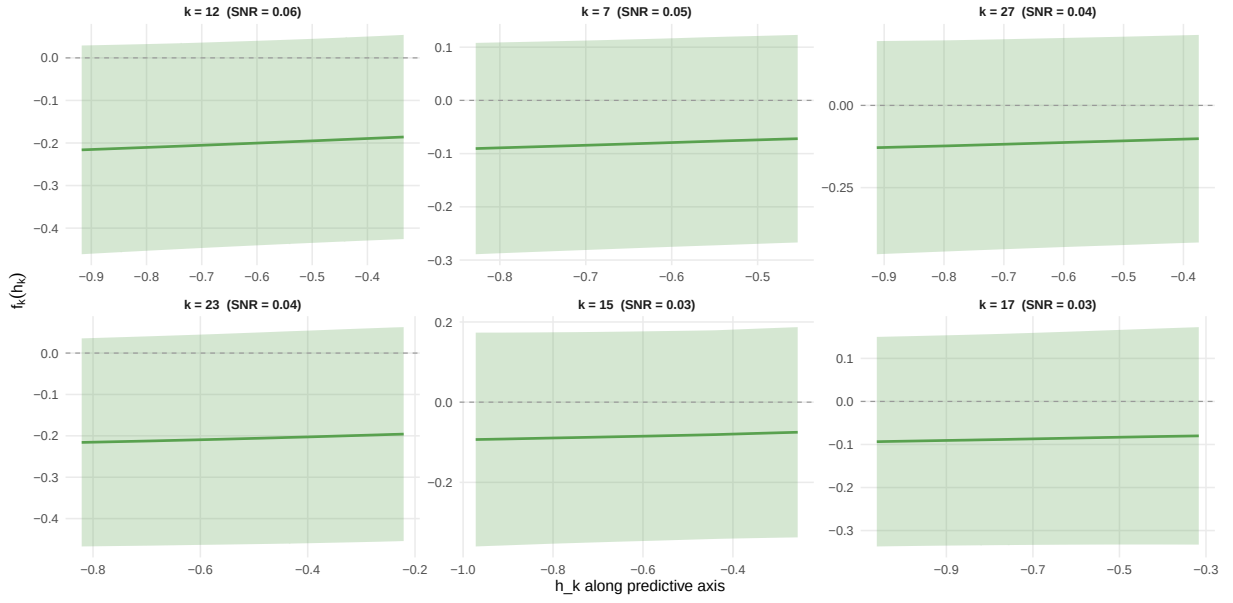


Figure 6.5. Procrustes-aligned population shape functions $\bar{f}_k$ for the six covariates with highest shape signal-to-noise ratio at $t = 8.5$. The reference iteration is the one with the highest cross-validated AUC ($b^* = 6$, CV-AUC = 0.842). Ribbons span ± 1 restart standard deviation around the population mean. SNR values are reported above each panel.

the only invariants that survive averaging are the level shift carried by the bias term and the sign of $\partial \bar{f}_k / \partial h_{k,1}$ where the SNR is large enough to read it. Second, the near-flatness of the population shapes is consistent with Section 6.3.1, where FA-NAM-MIDAS matches but does not exceed the sg-LASSO-MIDAS baseline: the additional flexibility of the per-covariate predictors does not produce population-scale shape information that the linear baseline is missing, and the factor network g_0 of Section 6.3.4 carries the non-linear contribution that distinguishes FA-NAM-MIDAS from the linear model.

6.4 Additional Results

6.4.1 US Panel Description and Retrieval

Section 6.3.6 showed that the per-covariate shape functions of FA-NAM-MIDAS on the Chinese panel are statistically indistinguishable from horizontal lines after Procrustes alignment across restarts. Whether this finding reflects a structural feature of the model or a finite-sample limitation of the panel motivates a further investigation on a larger panel from a different economy.

A parallel panel of US firms is therefore assembled from Wharton Research Data Services (Wharton Research Data Services, 2025). The construction mirrors that of the Chinese panel in Section 6.1 as closely as the available US data allow: the panel uses the same $K = 32$ financial

ratios, the same look-back window $s = 6$ years, and the same three evaluation horizons $t \in \{8, 8.5, 9\}$. The sample period spans 1985 to 2016 in firm entry dates, with follow-up through December 31, 2020 paralleling the Chinese panel of Section 6.1.1. The panel contains $N = 11,170$ firms with 2,961 distress events, for an event rate of approximately 26.5%.

The Chinese ST flag of Section 6.1.1 has no direct US counterpart. The closest analogue in CRSP (Center for Research in Security Prices, 2025) is the delisting code, which classifies the cause of a firm's removal from the exchange. Codes in the range 400–599 mark delistings for financial-distress reasons, covering insufficient capital (520–529), delinquent filing (550–559), bankruptcy (560–569), and listing-requirement violations (580–589). A firm enters the event set ($\delta_i = 1$) if it experiences any delisting in this range within the observation window, and is otherwise right-censored ($\delta_i = 0$) at December 31, 2020.

Compared with the Chinese ST flag, the US event captures a sharper, later-stage notion of distress: ST is a reversible warning that a firm can lose if its financial position recovers, whereas a CRSP financial delisting is terminal. The event rate of 26.5% is markedly higher than the 7.4% in the Chinese panel of Section 6.1.1. Despite the stricter event criterion, the higher rate reflects a broader sectoral scope: the panel spans firms across the economy rather than the manufacturing firms of the Chinese panel, capturing a more heterogeneous mix with a higher overall failure prevalence.

The 32 quarterly covariates are constructed from the Compustat North America Fundamentals Quarterly file (S&P Global Market Intelligence, 2025). Each ratio in the Chinese covariate set is mapped to its closest Compustat counterpart, spanning the same accounting categories: profitability ratios, cash-flow ratios, leverage and capital-structure ratios, short-term liquidity ratios, efficiency ratios, and per-share measures. The full mapping is reported in Table B.1.

6.4.2 Prediction Performance

Table 6.5. Distress prediction performance on the US panel. Reported values are the average out-of-sample time-dependent AUC across ten stratified 80/20 splits; the iteration standard deviation is shown in parentheses. Pointwise 95% confidence intervals, computed as the 2.5th and 97.5th percentiles of the iteration-level AUCs, are reported below each mean.

Model	Prediction horizon t		
	$t = 8$	$t = 8.5$	$t = 9$
sg-LASSO-MIDAS	0.803 (0.014) [0.786, 0.825]	0.809 (0.008) [0.798, 0.818]	0.802 (0.008) [0.790, 0.814]
FA-NAM-MIDAS	0.870 (0.015) [0.848, 0.888]	0.869 (0.013) [0.851, 0.886]	0.869 (0.014) [0.847, 0.887]

Table 6.5 reports the prediction performance of the linear baseline and FA-NAM-MIDAS. FA-NAM-MIDAS achieves mean AUCs of 0.870, 0.869, and 0.869 at $t = 8, 8.5$, and 9 , against baseline means of 0.803, 0.809, and 0.802. The improvement is approximately 6 to 7 percentage points at every horizon, with non-overlapping 95% confidence intervals throughout: the upper bound of the sg-LASSO-MIDAS interval lies strictly below the lower bound of the FA-NAM-MIDAS interval at each evaluation point.

This pattern reverses the headline finding of Section 6.3.1 on the Chinese panel, where FA-NAM-MIDAS matched but did not exceed the linear baseline. The US panel offers a more favourable setting in which the architectural flexibility of FA-NAM-MIDAS can pay off: a markedly larger sample, a higher event rate, and potentially richer non-linear structure in the panel itself. Under these conditions the per-covariate shape functions f_k and the factor channel g_0 capture non-linear signal that did not emerge on the smaller Chinese panel.

The FA-NAM-MIDAS mean AUCs are also remarkably stable across horizons, with the three values lying within 0.001 of each other. The FA-NAM-MIDAS iteration standard deviations of 0.015, 0.013, and 0.014 are modestly larger than the baseline values of 0.014, 0.008, and 0.008, a variance premium of approximately 1.1 to 1.8 across horizons. The premium is substantially smaller than the 1.1 to 2.3 documented for the NAM-MIDAS-family, excluding the ablation variant, on the Chinese panel in Section 6.3.1.

6.4.3 Shape Stability via Procrustes Alignment

Applying the Procrustes-aligned shape-stability analysis of Section 6.3.6 to the US-panel FA-NAM-MIDAS restarts at $L = 3$ yields a more nuanced pattern than the Chinese panel suggested. The maximal cross-covariate shape signal-to-noise ratio is 0.01 at $t = 8$, 0.37 at $t = 8.5$, and 0.70 at $t = 9$, compared with the Chinese values of 0.04, 0.05, and 0.01 at the same horizons.

At the shortest horizon $t = 8$ the US per-covariate shapes are essentially flat, and in fact slightly flatter than on the Chinese panel. The substantial AUC improvement at $t = 8$ (Table 6.5) therefore cannot be attributed to population-level non-linearity in the per-covariate shape functions f_k ; by elimination, it must instead be carried by the factor block g_0 . The factor block's role thus shifts from regulariser of the unstable per-covariate shapes on the small Chinese panel (Section 6.3.4) to signal-bearing channel on the larger US panel, while the per-covariate shape functions remain unable to capture a population-level non-linearity at the shortest horizon.

At the longer horizons $t = 8.5$ and $t = 9$, the per-covariate shape functions begin to capture non-trivial structure, but only for a sparse subset of covariates. The six highest-SNR shapes at $t = 9$ are shown in Figure 6.6; one covariate reaches SNR 0.37 at $t = 8.5$, and three clear SNR 0.4 at $t = 9$, while the rest stay flat (Figure 6.7, upper panel). Longer-horizon distress prediction therefore appears to draw on a different mix of channels than short-horizon

prediction.

The Procrustes residuals in the lower panel of Figure 6.7 give a complementary diagnostic. The residual for a covariate measures how far each restart's aggregation g_k remains from the reference after optimal rotation: a value near zero means two restarts produce the same aggregated features up to rotation, while a large value means they produce genuinely different ones. Most covariates align to small residuals, but several, including two of the three highest-SNR covariates, show large residuals and pronounced outliers, indicating that for some restarts the aggregation is not a rotated version of the reference but a substantively different solution. The rotational symmetry that Procrustes removes is therefore not the only source of per-covariate restart disagreement; a residual instability remains that the alignment cannot reach. The two diagnostics should therefore be read together. A large residual signals instability but not its source: it is consistent both with genuine structure that the model learns inconsistently and with noise that each restart fits differently. Only where a high residual coincides with a high SNR, as for covariate 13 in Figure 6.7, does the instability sit on top of genuine population-level structure; where a high residual coincides with a low SNR, as for covariate 30 in Figure 6.7, the simpler reading is restart-to-restart noise-fitting with no underlying signal.

6.4.4 Case Study: Distress Probability on Selected Firms

To illustrate how the AUC improvement of Section 6.4.2 translates into firm-level risk assessment, this section reports FA-NAM-MIDAS distress probabilities for six firms drawn from the US panel: four well-known bankruptcy cases (Enron, Eastman Kodak, Circuit City, and General Motors) and two long-standing survivors (Apple and Microsoft) used as controls.

The six firms are held out from the training panel before the train/test split is constructed in each iteration. This holdout, and the fixing of L at 3 across all iterations for compatibility with the Procrustes alignment, are the only differences from the variant trained for Table 6.5; the architecture, the hyperparameter grid, and the ten random seeds for the train/test split are otherwise identical. Holding the six firms out at panel-construction time means that each firm-level prediction is genuinely out-of-sample: the model has never seen these firms during training, validation, or ensemble retraining at any horizon or iteration.

For each held-out firm, a predicted distress probability $\hat{p}$ is produced by averaging the FA-NAM-MIDAS output across the ten iterations and the two ensemble restarts per iteration. The firm's 32×23 covariate matrix is standardised using the (k, j) -specific mean and standard deviation of the iteration's training portion, then passed through the trained g_k , f_k , and g_0 networks (Section 6.2.4); the resulting logit is mapped through the sigmoid, and the reported $\hat{p}$ is the mean across the 20 models.

Each firm's lag window ends at its own reference date (event date for failures, 2020-12-31

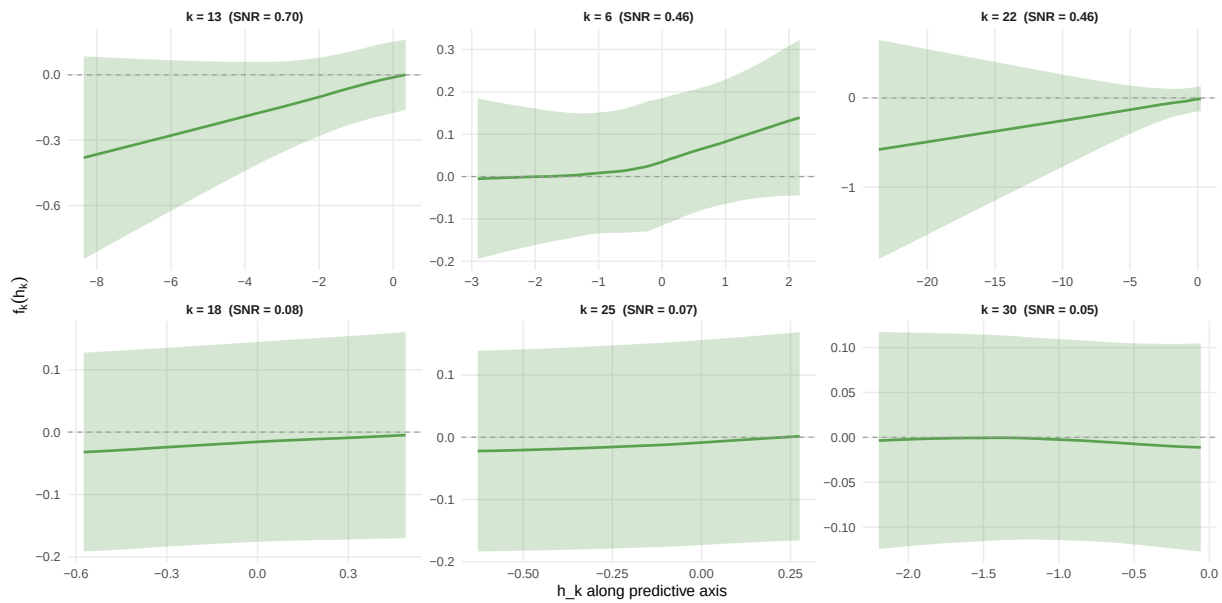


Figure 6.6. As Figure 6.5, but for the US panel at $t = 9$.

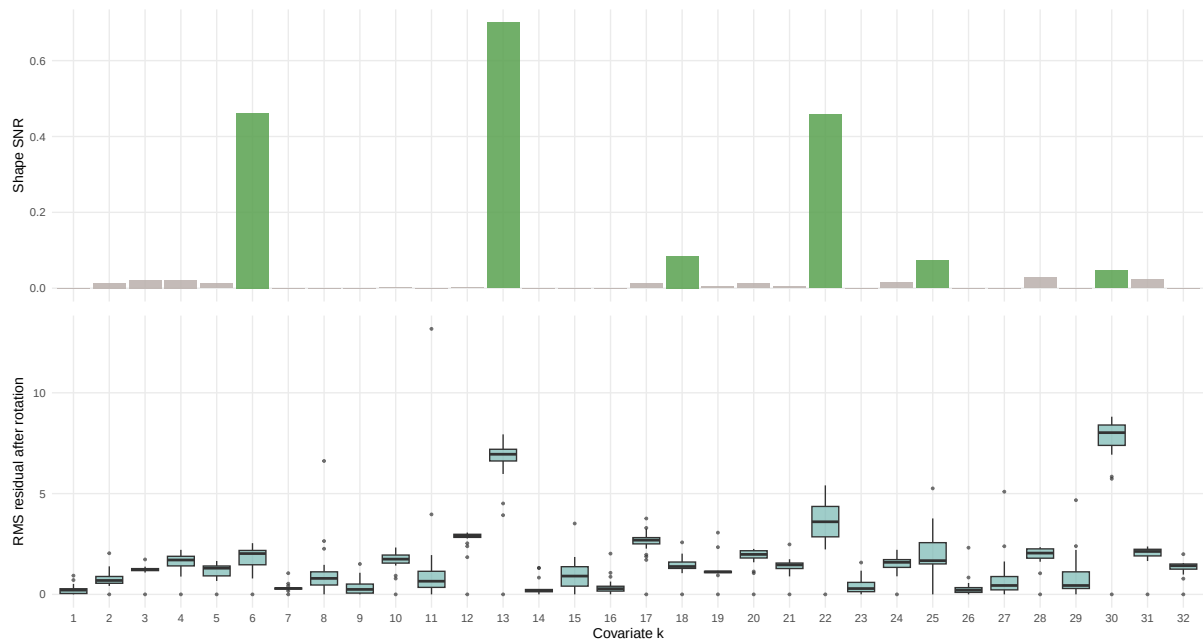


Figure 6.7. Per-covariate diagnostics on the US panel at $t = 9$. The upper panel reports the shape signal-to-noise ratio per covariate, with the six highest highlighted; the lower panel reports the distribution of the Procrustes RMS residual after rotation across restart pairs, per covariate.

for survivors), so the chronological position of the prediction varies firm by firm. Table 6.6 reports the lag window and the $t = 8$ endpoint for each firm alongside the predicted distress probability and the realised outcome. A model that discriminates well should assign low $\hat{p}$ to the two controls and substantially higher $\hat{p}$ to the four distress cases. With only six firms, the case study is illustrative rather than a formal evaluation: it can show whether the model's predictions are directionally sensible but cannot support a quantitative assessment of predictive performance. It is also a reminder that the high aggregate AUC of Table 6.5 does not guarantee the reliability of any single firm-level prediction.

Table 6.6. Predicted distress probability at horizons $t \in \{8, 8.5, 9\}$ for the six case-study firms.

Ticker	Firm	Realised event	Lag window	$t = 8$ endpoint	$\hat{p}_{t=8}$	$\hat{p}_{t=8.5}$	$\hat{p}_{t=9}$
AAPL	Apple	Censored	2015-Q2–2020-Q4	2023-Q2	0.018	0.019	0.028
MSFT	Microsoft	Censored	2015-Q2–2020-Q4	2023-Q2	0.019	0.021	0.031
ENRNQ	Enron	2002-01-11	1996-Q1–2001-Q3	2004-Q1	0.149	0.140	0.181
KODK	Eastman Kodak	2012-01-18	2006-Q2–2011-Q4	2014-Q2	0.500	0.516	0.497
CCTYQ	Circuit City	2008-11-07	2003-Q1–2008-Q3	2011-Q1	0.664	0.663	0.606
GM	General Motors	2009-06-01	2003-Q3–2009-Q1	2011-Q3	0.608	0.629	0.678

Table 6.6 shows a clean directional separation: the two survivors receive $\hat{p} \approx 0.02$ at every horizon, while three of the four distress cases receive $\hat{p}$ in the range 0.50 to 0.68. The predictions are stable across horizons, with no firm's $\hat{p}$ varying by more than ~ 0.07 across $t \in \{8, 8.5, 9\}$, consistent with the cross-horizon flatness of the aggregate AUCs reported in Table 6.5.

Enron is a notable exception: $\hat{p}_{t=8} \approx 0.15$, well below the other three distress cases and only modestly above the survivor baseline. Enron's 2002 collapse was driven by off-balance-sheet entities and creative accounting that left the reported financial ratios in the 1996-2001 lag window looking healthy. A model trained on standardised financial covariates cannot, by construction, detect fraud-driven distress whose signature does not appear in those covariates; the low $\hat{p}$ for Enron therefore aligns with what one would expect of the architecture rather than indicating a model failure.

This contrast also bears on how a model of this kind is best used. For an individual investor selecting single positions, a model that misranks a firm such as Enron offers limited protection: one firm-level error can dominate a concentrated portfolio, and the case study provides no assurance that any single prediction is correct. The natural use case is instead institutional. A bank estimating default probabilities across a large book of exposures benefits from the property the case study confirms: individual prediction errors are largely idiosyncratic and partially wash out in aggregate, while the population-level discrimination captured by the AUC carries through to portfolio-level quantities such as expected loss, provisioning, and regulatory capital.

Chapter 7

Conclusion

7.1 Summary of Findings

This thesis asked whether the linearity and additivity restrictions of the sg-LASSO-MIDAS baseline of [Miao et al. \(2025\)](#) could be relaxed in a way that improved out-of-sample default prediction or yielded interpretive payoff that the linear baseline could not provide. Six variants developed in Chapter 5 systematically relax each restriction, all estimated under the same IPCW-weighted protocol so that any performance difference is attributable to the architectural extensions rather than to the censoring treatment.

The predictive verdict on the Chinese manufacturing panel (Section 6.3.1) is clean: no neural variant produces a statistically meaningful AUC improvement over the sg-LASSO-MIDAS baseline at any of the three horizons, and the NAM-MIDAS-family variants pay a substantial iteration-level variance premium relative to the linear model. The pattern of no AUC gain combined with the variance behaviour is consistent with the additional flexibility of f_k over the linear coefficient β_k fitting noise rather than adding signal at the available sample size.

Three diagnostic analyses (Sections 6.3.2, 6.3.4 and 6.3.5), each anchored to one of the architectural design choices, refine this Chinese-panel headline. The Residual variant’s mixing weights γ_k concentrate near the Gegenbauer end of the mixture across all covariates and horizons, settling on a small, diffuse neural correction rather than discovering covariate-specific structure. The factor block of FA-NAM-MIDAS, while carrying a substantial share of the logit contribution, does not improve mean AUC when isolated through FA-MIDAS; here it functions as a regulariser of the unstable per-covariate shape functions rather than as a carrier of non-linear signal. The hyperparameter profile reinforces this reading: the random search favours small factor counts and generously parameterised aggregation networks, placing the architecture’s useful flexibility in the per-covariate aggregation stage rather than in the per-covariate predictor.

What these findings imply for the individual shape functions depends on their identifiability. Section 5.11 catalogues three sources of non-identification of the individual shape functions f_k : a level-shift ambiguity, a slope-redistribution ambiguity between near-collinear covariates, and a within-covariate collinearity ambiguity for multi-input shape functions. Section 6.3.6 identifies a fourth, exact structural source: a rotational symmetry within each covariate's L -dimensional internal feature space. This is the only source the Procrustes alignment removes; after alignment, the population per-covariate shapes are statistically indistinguishable from horizontal lines. This finding is consistent with the absence of an AUC gain and with the FA-MIDAS-based reading of the factor block as a regulariser of the additive predictor rather than a carrier of new non-linear signal.

The variance premium and the flat population shapes on the Chinese panel motivated further investigation of whether the absence of a gain reflects a finite-sample limitation or a structural feature of the architecture. Section 6.4 tests this reading on a parallel US panel from a different economy, with a substantially larger sample size, a higher event rate, and broader sectoral scope. The predictive picture reverses sharply: FA-NAM-MIDAS delivers a uniform AUC improvement over sg-LASSO-MIDAS at every horizon with non-overlapping 95% confidence intervals (Table 6.5), while sg-LASSO-MIDAS itself achieves mean AUCs similar to those on the Chinese panel despite the richer information set. The additional information therefore produces no measurable AUC gain for the linear model but a substantial gain for FA-NAM-MIDAS, and the iteration variance premium also narrows, vanishing almost entirely at the shortest horizon.

The Procrustes-aligned shape-stability analysis of Section 6.4.3 qualifies this reading. Scale does make the architecture's flexibility productive, as the finite-sample hypothesis predicts, but it does so unevenly rather than across the board. At the shortest horizon the per-covariate shape functions f_k stay population-flat and the AUC gain is carried by the factor block g_0 alone; at the longer horizons the gain is shared, with a sparse handful of individual covariate shapes capturing genuine structure while the rest remain flat. The productive flexibility unlocked by scale is therefore concentrated rather than evenly spread: it sits in the factor channel and in a sparse subset of per-covariate shapes.

The case study of Section 6.4.4 provides a per-firm reading of the same architecture. Passed through the trained FA-NAM-MIDAS ensemble, the two surviving controls (Apple, Microsoft) receive predicted distress probabilities of about 0.02, and three of the four distress cases (Circuit City, Kodak, General Motors) receive probabilities between 0.50 and 0.68, a clean directional separation. Enron is the exception, receiving about 0.15: its collapse was driven by accounting fraud whose signature does not appear in the standardised financial covariates the model is trained on, so the low probability reflects a structural limit of covariate-based prediction rather than a model failure.

Read together, the two panels yield a more nuanced verdict than either alone would support. Had the US extension reproduced the absence of a gain on the Chinese panel, the verdict would have landed on the structural reading: that lag structures in corporate financial ratios do not, in general, carry the kind of non-linearity that the additive neural relaxation is designed to recover. The empirical reversal rejects that structural reading and confirms the finite-sample one, which positions the two approaches against each other along a sample-size dimension. For panels at the scale of the Chinese setting, sg-LASSO-MIDAS remains the preferred choice: it captures the available linear signal efficiently, is unfazed by the variance that per-covariate flexibility introduces, and is not outperformed in mean AUC by any of the neural extensions. For panels at the scale of the US setting, FA-NAM-MIDAS becomes preferable: the architectural flexibility now has enough signal to draw on, and the per-covariate shape functions and factor channel contribute genuine non-linear information that the linear baseline cannot match. The diagnostic tools developed in this thesis generalise across both settings and provide the population-level machinery for interpreting whichever model the data scale supports.

7.2 Future Research

The US replication of Section 6.4 addresses one dimension of external validity, and a second concerns the temporal stability of the lag dynamics. The observation window of both panels closes at 2020, before the pandemic-induced regime change in corporate financials. The lag dynamics on which the architectural family is trained may not remain representative of the post-pandemic distribution, and the predictive performance reported here cannot be assumed to carry over to the present without verification. Extending the observation window past 2020 to include the pandemic and post-pandemic periods would test whether both sg-LASSO-MIDAS and FA-NAM-MIDAS reproduce the headline AUCs on the more recent distribution. A sharper test of regime robustness is to train both models on the pre-pandemic window and evaluate them on distress horizons that fall within the pandemic-era regime; the gap between in-sample and out-of-regime performance would quantify how robust the additive neural family is to a macroeconomic regime change of the kind the pandemic period represents.

The mean AUCs of FA-NAM-MIDAS are stable across the three evaluation horizons on both panels (Sections 6.3.1 and 6.4.2), with the three US values lying within 0.001 of each other. The flatness suggests that the architectural family may retain predictive power at horizons longer than those tested here. A natural extension is to evaluate the same family at longer prediction windows, where the linear baseline's predictive content typically degrades as the relevance of recent lag information fades. The horizon at which the two models begin to

diverge would characterise the persistence advantage of the per-covariate aggregations and the factor channel, and would test whether the cross-horizon flatness observed in the present window is a structural property of the architecture or an artefact of the close spacing of the three reported horizons.

The per-covariate shape functions raise a separate cross-horizon question. The shape signal-to-noise analysis of Section 6.4.3 found that the number of covariates whose population shapes show non-trivial structure grows with the horizon, from none at $t = 8$ to a sparse handful at $t = 9$. Whether this trend continues at still longer horizons is open; if it does, longer-horizon evaluation would progressively reveal more of the per-covariate structure the additive design was built to capture.

A related question concerns sample size rather than horizon. In the US shape-stability analysis, the covariates that resolved population-level structure tended also to carry the largest Procrustes residuals, so that emerging signal and restart instability appeared together. A few otherwise flat covariates show similar residual outliers without yet clearing the SNR threshold. Section 6.4.3 reads these as restart-to-restart noise-fitting, but they are equally consistent with genuine structure that the present sample is too small to resolve. A still larger panel would separate the two readings, and if the latter holds, would bring more covariates above the threshold.

A natural extension of the parallel replication design is joint cross-panel modelling. Training a shared model on both panels under sector and economy fixed effects would test whether the additive neural family extracts common non-linear structure across markets. Training on one panel and evaluating on the other would isolate genuine transfer of the per-covariate shape functions f_k from artefacts of fitting a market-specific decision boundary, and would also test whether the small-panel finite-sample limitation can be circumvented by borrowing strength from larger panels rather than by waiting for the smaller market to accumulate more events.

The PCA projection P is fixed at the start of training and held constant thereafter, following Fan and Gu (2024) (Section 5.6.4). Joint estimation of P together with the factor network g_0 would be more flexible but considerably more demanding in optimisation complexity. On the Chinese panel, the FA-MIDAS-based reading of the factor block as a regulariser rather than a non-linear-discovery channel (Section 6.3.1) makes a joint-trained variant unlikely to add value. The US panel of Section 6.4, where FA-NAM-MIDAS substantially outperforms the linear baseline, is the natural setting for testing whether a learned P can extract additional non-linear signal beyond what the pre-trained PCA projection captures.

7.3 Methodological Contributions

This thesis contributes a family of architectures, the machinery to interpret them, and an empirical design to test them. The architectural contribution is a family of neural extensions of the sg-LASSO-MIDAS baseline, all estimated under a shared IPCW-weighted protocol. The interpretive contribution is a catalogue of the identifiability ambiguities of additive neural predictors, together with an orthogonal Procrustes alignment protocol that removes the rotational one, so the otherwise unstable per-covariate shape functions can be aggregated into population-level objects and compared. The empirical contribution is the application of the full protocol to two panels of very different size, which shows whether the result on the smaller Chinese panel reflects a structural limit of the additive neural family or simply its sample size. The R code accompanying the thesis is openly available at <https://github.com/Emil-Refn/fa-nam-midas>, covering panel construction, model estimation, and the shape-stability analysis.

Appendix A

Empirical Results

A.1 Covariate Importance Ranking

Table A.1. Cross-horizon top-ranked covariates by the FA-NAM-MIDAS feature importance $\mathbb{E}|f_k(h_{i,k})|$ (left) and the variance-weighted PCA loading mass $L_k = \sum_j \|P_{k,j}\|^2 \sigma_j^2$ (right).

Rank	NAM importance $\mathbb{E} f_k $		PCA loading mass L_k	
	Covariate	Value	Covariate	Value
1	(23) Interest-bearing debt ratio	0.308	(10) EBITDA / liabilities	0.149
2	(21) Equity / liabilities	0.279	(05) Current ratio	0.139
3	(09) Tangible / net debt	0.269	(12) ROA	0.133
4	(15) EBIT / op. income	0.262	(13) Net profit / assets	0.129
5	(31) Op. cash-flow / op. income	0.261	(18) EBIT / assets	0.126
6			(21) Equity / liabilities	0.121
7			(08) Tangible / liabilities	0.118

A.2 Learned Mixing Weights γ_k

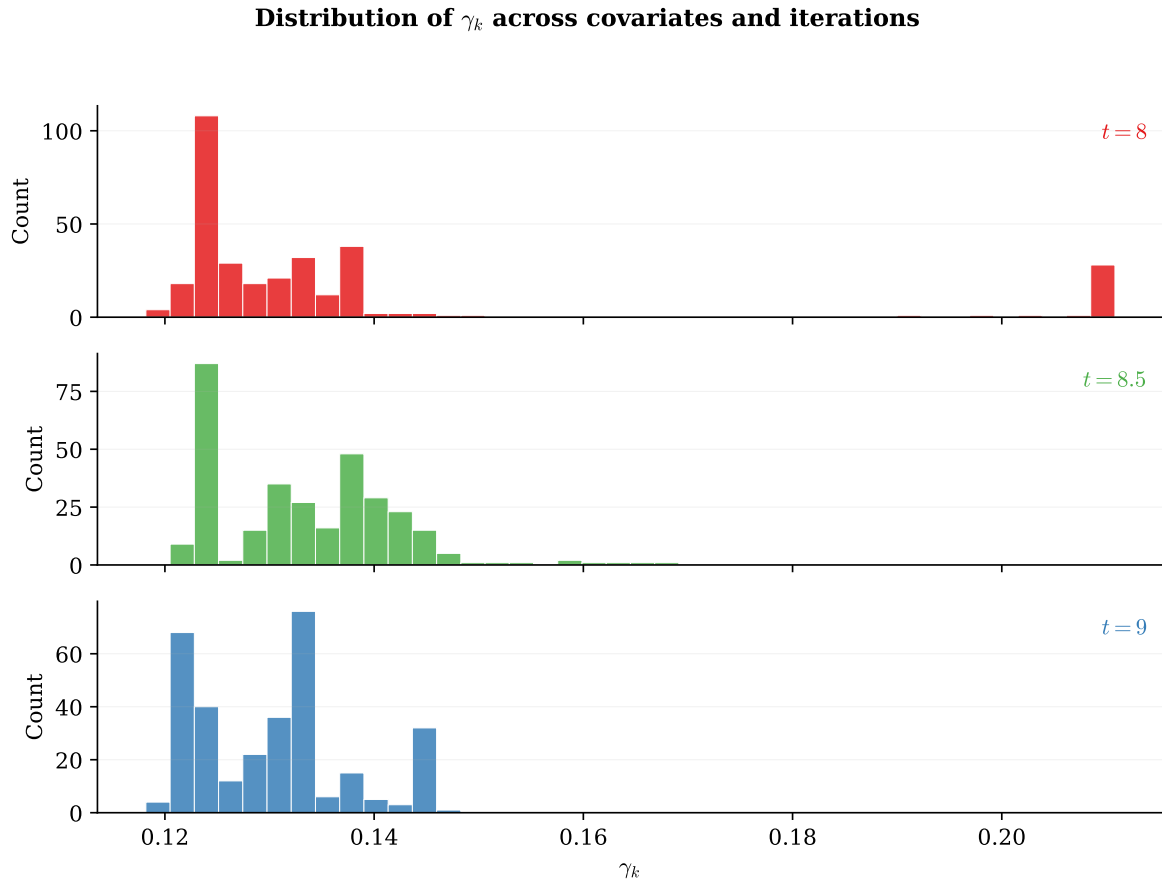


Figure A.1. Distribution of the per-covariate mixing weight $\bar{\gamma}_k$ across covariates, separated by forecast horizon. The mixing weights concentrate tightly near 0.13, uniformly across covariates and horizons. Higher values correspond to a larger neural contribution relative to the Gegenbauer aggregation.

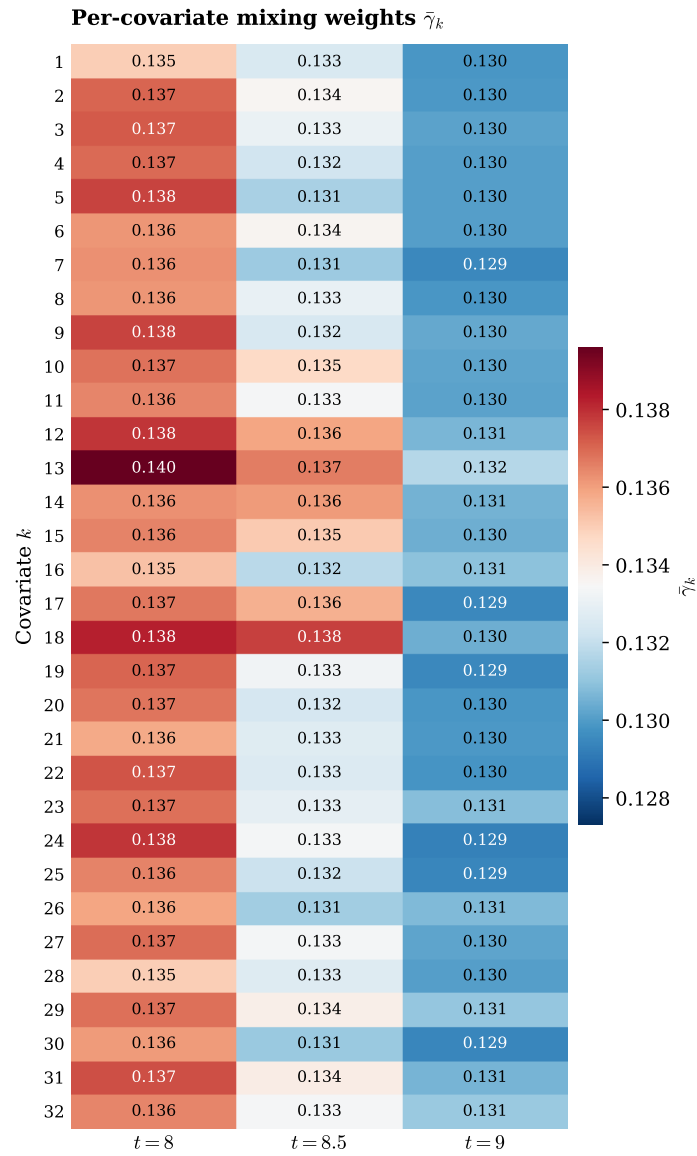


Figure A.2. Heatmap of $\bar{\gamma}_k$ across the 32 covariates (rows) and the three forecast horizons (columns). The colour scale is deliberately clipped to $[0.12, 0.14]$ to highlight the cross-covariate variation; no covariate at any horizon exceeds $\gamma_k = 0.5$. The pattern confirms that the neural aggregation is a small, diffuse correction rather than a mechanism that the model concentrates on a few covariates.

A.3 Factor Decomposition

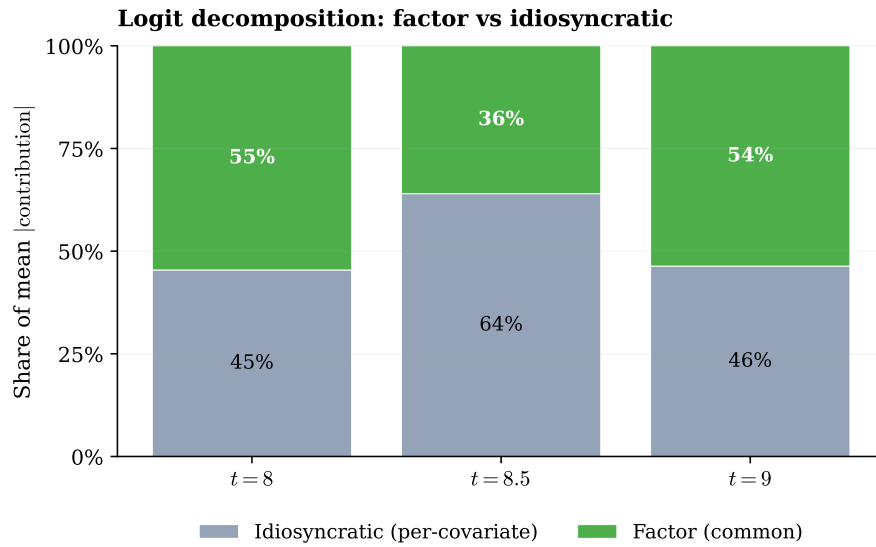


Figure A.3. Share of the mean absolute logit contribution carried by the factor component $g_0(F_i)$ in the FA-NAM-MIDAS estimator, by forecast horizon. The factor block carries roughly 55% at $t = 8$, 36% at $t = 8.5$, and 54% at $t = 9$, with the per-covariate shape functions carrying the remainder.

A.4 Shape Function Projections

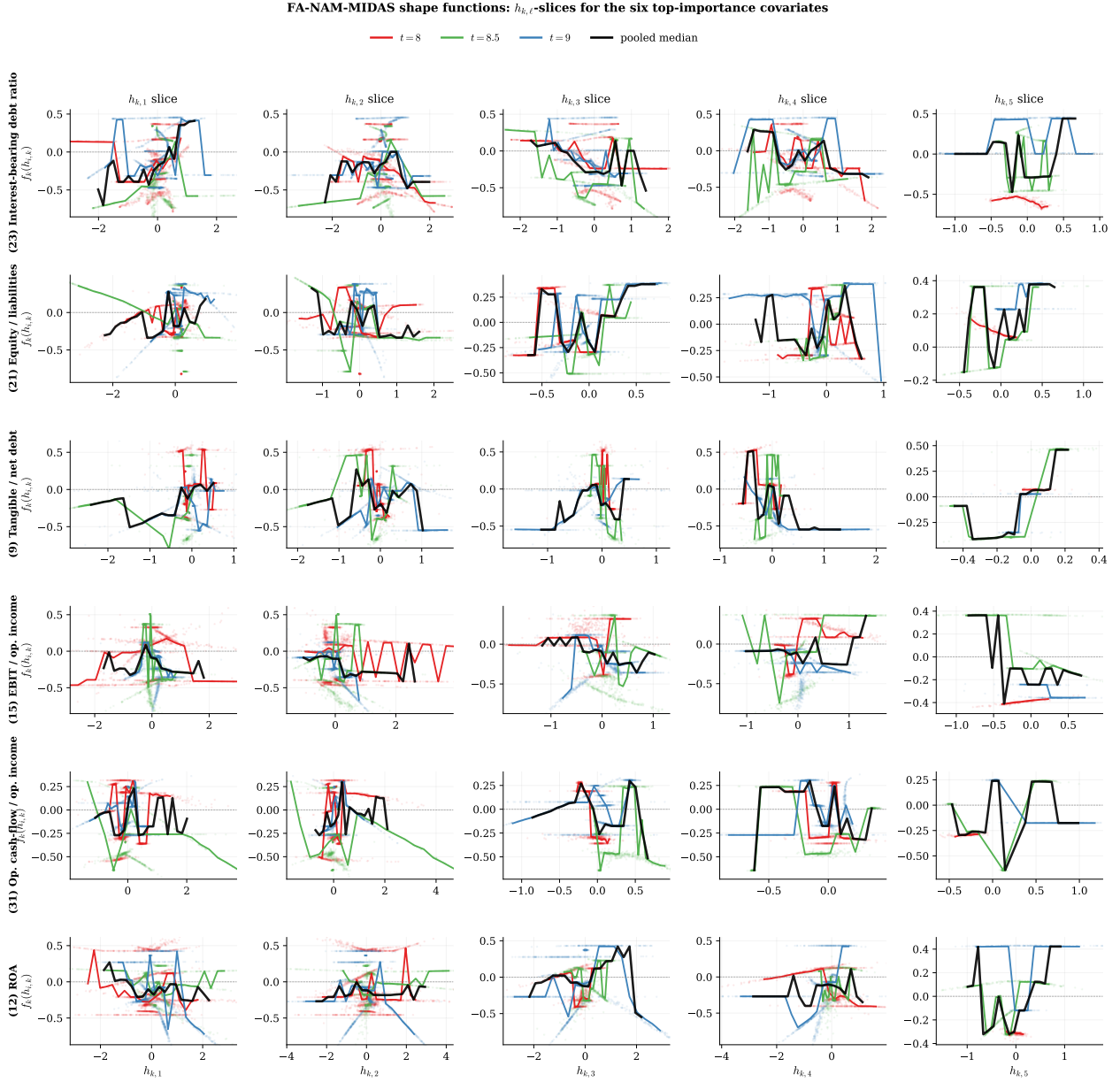


Figure A.4. $h_{k,\ell}$ -slice scatter of the FA-NAM-MIDAS shape functions for the six covariates with the highest cross-horizon NAM importance. Each row is one covariate; the five columns show $f_k(h_{i,k})$ plotted against $h_{k,1}, \dots, h_{k,5}$ in turn, with iterations pooled across horizons. No iteration centring is applied, so vertically stratified bands within a colour reflect the level-shift ambiguity of Equation (5.25) on the logit scale. Coloured curves are per-horizon binned medians; the heavy black curve is the pooled median. Columns $\ell = 4, 5$ pool only iterations with $L \geq \ell$.

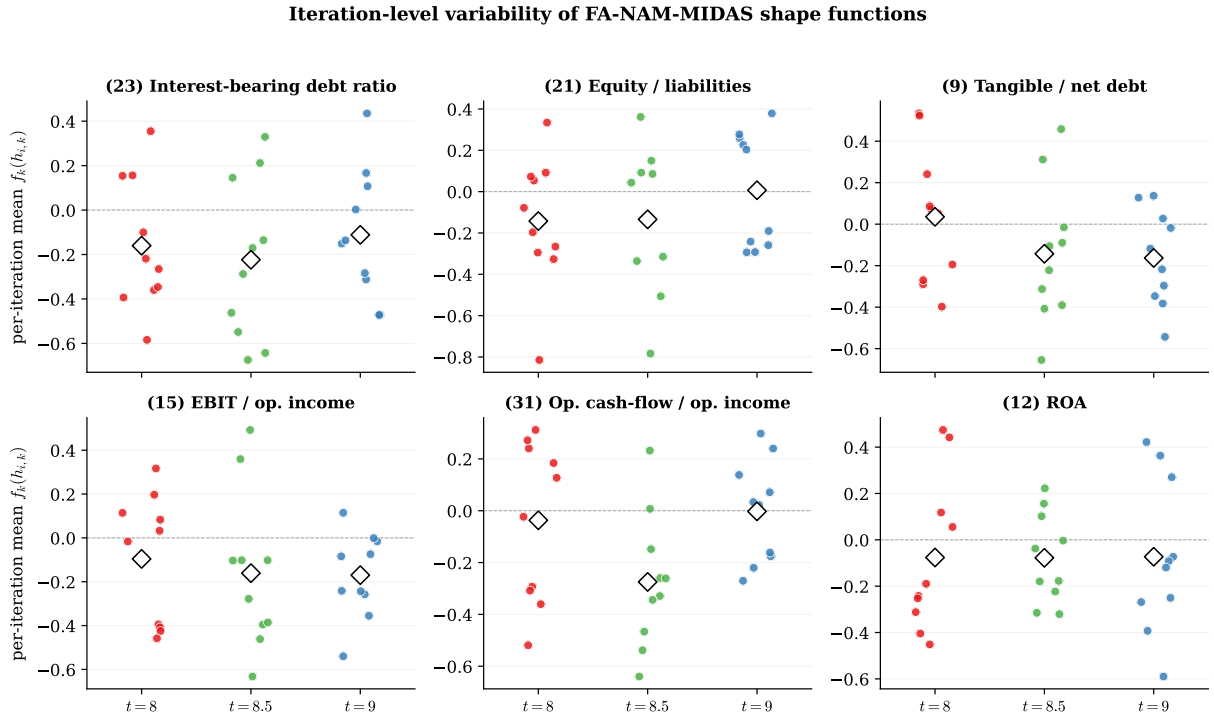


Figure A.5. Iteration-level variability of FA-NAM-MIDAS shape functions for the six top-importance covariates. Each coloured dot is the per-test-set mean of $f_k(h_{i,k})$ in one iteration; the white diamond marks the cross-iteration mean. Vertical spread within a horizon column measures iteration-level instability of f_k , the basis for the magnitudes reported in Section 6.3.3.

A.5 Per-Iteration Hyperparameter Selections and AUCs

This section reports the random-search hyperparameter selections and the realised out-of-sample AUC for each of the thirty train/test iterations (ten per horizon). Tables A.2 and A.3 list the NAM-side and FA-NAM-side configurations chosen by cross-validation, and Table A.4 reports the AUC realised by each of the five model variants on the held-out test sample for the same iteration. For compactness the tables report the architecture and penalty hyperparameters; the optimiser hyperparameters λ_{wd} , c_{clip} , and the early-stopping patience are searched over the grids of Section 6.2.3 but omitted here.

Table A.2. Per-iteration NAM-side hyperparameter selections from random-search cross-validation (Full NAM-MIDAS as scoring architecture). The selected configuration is reused by the Full, Ablation, and Residual variants for the final ensemble fit.

t	iter	L	H_{agg}	n_{agg}	H_{pred}	n_{pred}	λ_1	λ_2	p_{drop}	p_{group}	λ_{out}
8	1	2	8	2	32	2	0.001	0.010	0.10	0.50	0.100
8	2	3	8	1	4	1	0.100	0.010	0.10	0.10	0.001
8	3	5	128	4	4	1	0.000	0.000	0.40	0.00	0.010
8	4	5	32	1	16	1	0.010	0.001	0.10	0.20	0.100
8	5	2	128	2	16	3	0.000	0.500	0.00	0.00	0.010
8	6	5	32	2	32	3	0.001	0.000	0.10	0.00	0.010
8	7	4	8	3	32	4	0.010	0.000	0.20	0.30	0.100
8	8	2	128	3	4	2	0.100	0.000	0.40	0.00	0.010
8	9	4	128	4	4	1	0.000	0.000	0.40	0.00	0.010
8	10	3	64	3	16	3	0.000	0.100	0.20	0.10	0.000
8.5	1	5	32	1	16	1	0.001	0.001	0.20	0.20	0.001
8.5	2	5	32	1	16	1	0.001	0.001	0.20	0.20	0.001
8.5	3	3	128	1	32	4	0.000	0.000	0.00	0.00	0.010
8.5	4	5	128	4	4	1	0.000	0.000	0.40	0.00	0.010
8.5	5	2	8	1	32	2	0.100	0.500	0.50	0.00	0.100
8.5	6	4	128	2	16	1	0.100	0.001	0.00	0.50	0.100
8.5	7	2	32	4	4	1	0.100	0.100	0.40	0.30	0.010
8.5	8	5	64	2	8	1	0.001	0.000	0.10	0.00	0.010
8.5	9	2	32	3	4	3	0.001	0.001	0.20	0.20	0.100
8.5	10	4	128	4	4	1	0.000	0.000	0.40	0.00	0.010
9	1	3	128	4	16	2	0.100	0.000	0.40	0.00	0.100
9	2	2	32	4	16	1	0.001	0.001	0.20	0.20	0.001
9	3	5	32	2	16	1	0.001	0.001	0.20	0.20	0.100
9	4	2	64	2	16	4	0.100	0.100	0.10	0.00	0.001
9	5	4	16	4	8	1	0.500	0.100	0.40	0.10	0.000
9	6	3	128	1	16	4	0.100	0.000	0.00	0.00	0.010
9	7	3	128	3	4	2	0.000	0.000	0.40	0.00	0.010
9	8	3	64	1	32	1	0.001	0.100	0.50	0.50	0.000
9	9	5	128	1	32	3	0.000	0.000	0.00	0.00	0.010
9	10	2	64	4	32	1	0.000	0.000	0.00	0.30	0.000

Table A.3. Per-iteration FA-NAM-side hyperparameter selections from random-search cross-validation, governing the factor block of FA-NAM-MIDAS.

t	iter	L	r	H_{factor}	n_{factor}
8	1	5	4	32	1
8	2	2	2	8	2
8	3	2	5	32	2
8	4	4	3	64	1
8	5	2	5	32	1
8	6	2	4	64	1
8	7	4	2	32	2
8	8	4	3	64	1
8	9	2	5	32	1
8	10	4	3	8	1
8.5	1	2	5	64	1
8.5	2	5	5	32	1
8.5	3	2	2	8	2
8.5	4	5	3	64	2
8.5	5	5	3	32	1
8.5	6	4	3	8	1
8.5	7	2	2	8	2
8.5	8	2	4	8	2
8.5	9	5	3	32	2
8.5	10	3	3	8	1
9	1	4	3	8	1
9	2	4	3	64	2
9	3	5	5	32	1
9	4	4	2	8	1
9	5	4	3	8	1
9	6	5	2	32	1
9	7	3	2	32	1
9	8	3	3	16	1
9	9	2	3	16	1
9	10	4	2	32	2

Table A.4. Per-iteration out-of-sample time-dependent AUC for each model variant on the held-out test sample. Iterations are paired across variants: each row reports the AUC obtained on the same train/test split.

t	iter	sg-LASSO	Full	Ablation	Residual	FA-NAM
8	1	0.918	0.883	0.882	0.937	0.952
8	2	0.802	0.853	0.607	0.467	0.783
8	3	0.885	0.962	0.648	0.930	0.884
8	4	0.750	0.807	0.670	0.854	0.853
8	5	0.789	0.859	0.569	0.884	0.783
8	6	0.911	0.781	0.857	0.854	0.930
8	7	0.746	0.736	0.725	0.801	0.782
8	8	0.732	0.706	0.647	0.751	0.713
8	9	0.846	0.834	0.661	0.638	0.731
8	10	0.800	0.807	0.882	0.828	0.772
8.5	1	0.862	0.880	0.782	0.772	0.792
8.5	2	0.860	0.549	0.915	0.781	0.942
8.5	3	0.718	0.754	0.182	0.444	0.835
8.5	4	0.844	0.916	0.699	0.852	0.823
8.5	5	0.846	0.833	0.786	0.846	0.845
8.5	6	0.800	0.745	0.685	0.762	0.723
8.5	7	0.819	0.852	0.474	0.745	0.649
8.5	8	0.862	0.883	0.931	0.755	0.759
8.5	9	0.757	0.761	0.429	0.704	0.791
8.5	10	0.753	0.743	0.690	0.592	0.776
9	1	0.760	0.758	0.864	0.716	0.630
9	2	0.867	0.861	0.592	0.736	0.891
9	3	0.855	0.754	0.750	0.758	0.760
9	4	0.668	0.781	0.683	0.719	0.747
9	5	0.782	0.835	0.257	0.799	0.862
9	6	0.839	0.580	0.231	0.806	0.807
9	7	0.817	0.767	0.665	0.751	0.810
9	8	0.746	0.672	0.422	0.709	0.668
9	9	0.827	0.484	0.552	0.588	0.900
9	10	0.889	0.789	0.631	0.479	0.845

Appendix B

Covariate Definitions

Table B.1. The 32 financial ratios used as covariates. The Chinese panel of Chapter 6 indexes ratios by k with the labels in the second column. The third column gives the construction of the corresponding US ratio for the panel of Section 6.4.1, built from Compustat North America Fundamentals Quarterly fields; the suffix $_{4Q}$ denotes a trailing four-quarter sum applied to flow variables.

k	Chinese ratio	US formula
1	Accounts receivable turnover	$\text{Sales}_{4Q} / \text{Receivables}$
2	Current assets turnover	$\text{Sales}_{4Q} / \text{Current Assets}$
3	Fixed assets turnover	$\text{Sales}_{4Q} / \text{Net PP\&E}$
4	Total assets turnover	$\text{Sales}_{4Q} / \text{Total Assets}$
5	Current ratio	$\text{Current Assets} / \text{Current Liabilities}$
6	Quick ratio	$(\text{Current Assets} - \text{Inventory}) / \text{Current Liabilities}$
7	Equity ratio	$\text{Equity} / \text{Total Assets}$
8	Tangible assets / liabilities	$(\text{Total Assets} - \text{Intangibles}) / \text{Total Liabilities}$
9	Tangible assets / net debt	$(\text{Total Assets} - \text{Intangibles}) / (\text{Short-term Debt} + \text{Long-term Debt} - \text{Cash})$
10	EBITDA / liabilities	$\text{EBITDA}_{4Q} / \text{Total Liabilities}$
11	Cash flow / debt	$\text{Operating Cash Flow}_{4Q} / (\text{Short-term Debt} + \text{Long-term Debt})$
12	Return on assets (ROA)	$\text{Net Income}_{4Q} / \text{Total Assets}$
13	Net profit / assets	$\text{Net Income}_{4Q} / \text{Total Assets}$
14	Net profit / operating income	$\text{Net Income}_{4Q} / \text{Sales}_{4Q}$
15	EBIT / operating income	$\text{EBIT}_{4Q} / \text{Sales}_{4Q}$
16	Working capital / assets	$(\text{Current Assets} - \text{Current Liabilities}) / \text{Total Assets}$
17	Retained earnings / assets	$\text{Retained Earnings} / \text{Total Assets}$
18	EBIT / assets	$\text{EBIT}_{4Q} / \text{Total Assets}$
19	Market value of equity / liabilities	$\text{Market Value of Equity} / \text{Total Liabilities}$
20	Operating income / assets	$\text{Sales}_{4Q} / \text{Total Assets}$
21	Equity / liabilities	$\text{Equity} / \text{Total Liabilities}$
22	Debt ratio	$\text{Total Liabilities} / \text{Total Assets}$
23	Interest-bearing debt ratio	$(\text{Short-term Debt} + \text{Long-term Debt}) / \text{Total Assets}$
24	Equity multiplier	$\text{Total Assets} / \text{Equity}$
25	Current assets / total assets	$\text{Current Assets} / \text{Total Assets}$
26	Current liabilities / total liabilities	$\text{Current Liabilities} / \text{Total Liabilities}$
27	Operating cash flow per share	$\text{Operating Cash Flow} / \text{Shares Outstanding}$
28	Operating income per share	$\text{EBIT} / \text{Shares Outstanding}$
29	Profit before tax / dividend	$(\text{Net Income} + \text{Taxes}) / (\text{Dividends per Share} \cdot \text{Shares Outstanding})$
30	Net assets per share	$\text{Equity} / \text{Shares Outstanding}$
31	Operating cash flow / operating income	$\text{Operating Cash Flow}_{4Q} / \text{Sales}_{4Q}$
32	Net operating cash flow / operating income	$\text{Operating Cash Flow} / \text{Sales}$

Appendix C

Singular Value Decomposition and the Procrustes Solution

The orthogonal Procrustes problem of Section 6.3.6 is solved in closed form through the singular value decomposition (SVD) of the cross-product matrix $M = (H_k^{(b)})^\top H_k^{(b^*)} \in \mathbb{R}^{L \times L}$. This appendix records the construction of the SVD and verifies that the Procrustes rotation reduces to $Q^* = UV^\top$.

C.1 The Singular Value Decomposition

For any real matrix $M \in \mathbb{R}^{L \times L}$ there exists a decomposition

$$M = USV^\top,$$

where $U, V \in \mathbb{R}^{L \times L}$ are orthogonal matrices ($U^\top U = V^\top V = I$) and $S = \text{diag}(\sigma_1, \dots, \sigma_L)$ is a diagonal matrix of non-negative singular values arranged in decreasing order $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_L \geq 0$. The columns $u_1, \dots, u_L$ of U are the left singular vectors and form an orthonormal basis of the codomain of M ; the columns $v_1, \dots, v_L$ of V are the right singular vectors and form an orthonormal basis of its domain. They are paired by the singular values through $Mv_i = \sigma_i u_i$. Standard library routines compute the SVD via the bidiagonalisation-and-eigenproblem algorithm of [Golub and Reinsch \(1970\)](#); for the small $L \times L$ matrices arising in Section 6.3.6 the cost is negligible.

C.2 Derivation of the Procrustes Rotation

The orthogonal Procrustes problem seeks $Q \in O(L)$ minimising $\|HQ - H^*\|_F^2$, where $H = H_k^{(b)}$ and $H^* = H_k^{(b^*)}$. Expanding the squared Frobenius norm,

$$\|HQ - H^*\|_F^2 = \|H\|_F^2 + \|H^*\|_F^2 - 2 \operatorname{tr}(Q^\top H^\top H^*),$$

the first two terms do not depend on Q , so the minimisation reduces to maximising the trace term $\operatorname{tr}(Q^\top M)$ over orthogonal Q , with $M = H^\top H^*$. Substituting the SVD $M = USV^\top$ and using the cyclic property of the trace,

$$\operatorname{tr}(Q^\top M) = \operatorname{tr}(Q^\top USV^\top) = \operatorname{tr}(V^\top Q^\top U S) = \operatorname{tr}(ZS),$$

where $Z = V^\top Q^\top U$ is itself orthogonal as the product of orthogonal matrices. The diagonal entries of any orthogonal matrix satisfy $|Z_{ii}| \leq 1$, so

$$\operatorname{tr}(ZS) = \sum_{i=1}^L Z_{ii} \sigma_i \leq \sum_{i=1}^L \sigma_i,$$

with equality if and only if $Z = I$. The condition $V^\top Q^\top U = I$ yields $Q^* = UV^\top$, which is the closed-form Procrustes solution of [Schönemann \(1966\)](#).

The singular values σ_i do not appear in Q^* themselves; they only determine the residual Frobenius distance after alignment, which evaluates to

$$\|HQ^* - H^*\|_F^2 = \|H\|_F^2 + \|H^*\|_F^2 - 2 \sum_{i=1}^L \sigma_i.$$

Larger singular values therefore correspond to stronger directional agreement between H and H^* and a smaller residual after rotation.

C.3 Numerical Workflow

The full procedure for aligning restart b to the reference b^* at covariate k proceeds in five steps:

- (1) Apply the trained aggregator $g_k^{(b)}$ to the full panel of n standardised lag matrices to obtain $H_k^{(b)} \in \mathbb{R}^{n \times L}$, and similarly $H_k^{(b^*)}$ from the reference restart.
- (2) Compute the cross-product $M = (H_k^{(b)})^\top H_k^{(b^*)} \in \mathbb{R}^{L \times L}$.
- (3) Compute the SVD $M = U_{k,b} S_{k,b} V_{k,b}^\top$.

(4) Form the rotation $Q_k^{(b)} = U_{k,b} V_{k,b}^\top$.

(5) Apply the rotation to the shape function via $\tilde{f}_k^{(b)}(h) = f_k^{(b)}((Q_k^{(b)})^\top h)$.

The procedure is repeated for every restart b and every covariate k , after which the population-aligned shape $\bar{f}_k$ is the cross-restart average of $\tilde{f}_k^{(b)}$.

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